Between Weak and Contaminated: Identifying Monetary Policy Shocks with Imperfect Instruments[1]

Étienne Latulippe
University of British Columbia

This version: November 2026

[Click here for the most recent version]

Abstract

No available instrument for monetary policy shocks is both strong and clean. Orthogonalising high-frequency surprises on pre-announcement information makes them cleaner but weakens their first-stage strength, and that weakness magnifies any residual contamination. Instruments that cover the whole policy decision, rather than the narrow window around the announcement, are strong but exposed to other contamination. I group instruments built from the same construction into families, whose members share their contamination but differ in relevance. Demeaning within each family removes the shared contamination, and pooling across families strengthens the first stage. In U.S. data, the pooled instrument is much stronger than the orthogonalised surprise and identifies a gradual decline in output and prices and a widening of credit spreads.

Keywords: monetary policy shocks; high-frequency identification; orthogonalisation; weak instruments; instrument families; instrument pooling; proxy-SVAR.

JEL classification codes: E43, E52, E58, C32, C36

[1]First version: August 2026. This version: 11 November 2026. Email: elatulip@student.ubc.ca. I am grateful to Paul Beaudry, Giovanni Gallipoli, and Viktoriya Hnatkovska for their guidance throughout this project, to Luis Hernandez-Roman, Henry Siu, and Kevin Song for comments, and to seminar and conference participants at the Vancouver School of Economics, the Canadian Economics Association meetings (2026), the Student T2M Conference at HEC Montréal (2026) and Banco de México (2026) for discussions. I acknowledge financial support from the Fonds de recherche du Québec – Société et culture (FRQSC) and the Stone PhD Fellowship. All errors are mine.

1 Introduction

What are the effects of monetary policy on the economy? The question is central to macroeconomics, yet estimates of how output and prices respond to a monetary tightening still vary widely across studies (Ramey, 2016; Brennan et al., 2025). The difficulty lies in identifying the causal effect of policy. Changes in the policy rate largely reflect the central bank's systematic response to economic conditions. The effect of policy can only be read from its unanticipated part, the structural monetary policy shock, which is not observed. A common approach estimates a structural vector autoregression (VAR) with external instruments, variables that supply identifying variation because they move with the policy shock and not with the other forces driving the economy (Mertens and Ravn, 2013; Stock and Watson, 2018). A useful instrument must be strong, moving the policy rate enough to identify its effects precisely. It must also be valid, or clean, free of the contamination that other news about the economy brings.

For monetary policy, the literature has built several candidate instruments, each with its own strengths and weaknesses. The most widely used are high-frequency surprises, the changes in interest-rate futures in a narrow window around Federal Open Market Committee (FOMC) announcements (Kuttner, 2001; Gürkaynak et al., 2005; Gertler and Karadi, 2015; Nakamura and Steinsson, 2018). Recently, these surprises have been shown to be predictable from information available before the announcement (Miranda-Agrippino and Ricco, 2021; Bauer and Swanson, 2023b). Two views compete. In one, the announcement reveals private information the Fed holds about the economy (Nakamura and Steinsson, 2018; Miranda-Agrippino and Ricco, 2021). In the other, the Fed responds to news that the market had not yet anticipated and fully priced (Schmeling et al., 2022; Bauer and Swanson, 2023a). In both, the predictability is a sign that the surprise responds to news, not only to policy.

The standard remedy, orthogonalisation, projects the surprise on pre-announcement information and keeps the residual (Bauer and Swanson, 2023b). The remedy is necessary for the surprise to be orthogonal to the news that precedes the announcement, but it weakens the first-stage strength of the surprise (Bauer and Swanson, 2023b; Gürkaynak et al., 2026; Amodeo, 2026). The impulse responses identified with the orthogonalised surprise are then large and imprecise. The price level falls in the very month of the announcement, at odds with evidence that prices respond to monetary policy only gradually (Aruoba and Drechsel, 2024a; Buda et al., 2025). Other instruments have been proposed for the same shock, such as surprises cleaned of the information the Fed reveals (Jarociński and Karadi, 2020; Miranda-Agrippino and Ricco, 2021), narrative residuals of the Fed's decisions (Romer and Romer, 2004) and Cholesky-based innovations of a VAR (Christiano et al., 1999). Nothing guarantees that any of them is both strong and clean of non-policy news.

This paper combines several imperfect instruments for the same shock, none of which is assumed to be clean. The method applies to any shock with several candidate instruments. This paper applies it to measure the effects of monetary policy. Every instrument is first orthogonalised on the same pre-announcement information, to remove its predictable component, following Bauer and Swanson (2023b). The instruments are then grouped into families and pooled. Demeaning within families removes the contamination that the members of a family share, and pooling restores the first-stage strength that orthogonalisation removes.

The paper begins by showing that orthogonalisation is necessary but not sufficient. It removes the contamination that the controls span. Any news they do not span survives in the instrument. The contamination the instrument carries therefore always shrinks, but the resulting bias may

not. Orthogonalisation also weakens the first-stage strength, as the policy rate reacts to the same pre-announcement news. A weaker first stage magnifies the contamination that survives. When orthogonalisation removes a larger share of the first-stage strength than of the contamination, the bias in the estimated impulse responses rises. The strength lost depends on how predictable the instrument is and on how closely its predictable component moves with the policy rate. Two equally predictable instruments can therefore lose very different shares of their first-stage strength. The high-frequency surprise illustrates the mechanism. Orthogonalisation removes most of its first-stage strength. The contamination that survives can then be magnified more than twofold.

The paper therefore turns to instruments that bring strength back to the identification. These instruments widen the window. Instead of the change in expectations in a narrow window around the announcement, they use the variation in the policy rate over the whole decision. A wider window also captures the part of the decision that the market prices before the announcement (Lucca and Moench, 2015; Bernile et al., 2016; Cieslak et al., 2019). The first, the Romer–Romer instrument, follows Romer and Romer (2004). It removes from the intended change in the federal funds rate the Fed's information at the time of the decision, measured by its Greenbook forecasts. The second, the Cholesky-based instrument, expresses the recursive restrictions of a VAR as an instrument (Sims, 1980; Bernanke and Blinder, 1992; Christiano et al., 1999). It removes the month's output and price innovations from the policy-rate innovation. Both instruments are partly predictable from information available before the announcement. They are therefore orthogonalised on the same controls as the surprise. Whereas the first-stage strength of the high-frequency surprise falls sharply after orthogonalisation, the wide-window instruments preserve more of it, because their predictable component moves less closely with the policy rate.

The wider window comes at a cost. The wide-window instruments gain strength, but they admit news that the narrow window excludes, such as data releases and financial developments later in the day or the month. Orthogonalised instruments may also retain pre-announcement news that the controls do not span. The method proposed in this paper accepts this trade-off between strength and cleanliness. It removes the part of the residual contamination that similarly built instruments share.

Simple averaging of the instruments does not suffice. Averaging raises the first-stage strength and averages out the contamination specific to each instrument, as with factor-based instruments (Bai and Ng, 2010). The contamination that the instruments share remains no matter how many instruments are added. A pool that weights the instruments by their strength fares no better. It leans on the strongest instrument and inherits its contamination, and it gives narrow confidence bands around biased impulse responses.

The shared contamination must therefore be removed before the instruments are pooled. To that end, this paper groups the instruments into families. A family is a set of instruments built such that they share their exposure to non-policy news. A family typically gathers instruments built from the same source, over the same window and with the same orthogonalisation. The members of a family may differ, however, in their relevance for the policy shock. Demeaning the identifying moments, the covariances of each instrument with the VAR innovations, within each family removes the contamination the family shares. The relevance deviations that remain identify the impact direction of the policy shock, the direction in which it moves the endogenous variables on impact.

The within estimator solves a minimum-distance problem (Newey and McFadden, 1994). It selects the impact direction, the relevance of each instrument and the common contamination of each family that best fit the estimated moments. Each family's contamination shifts its moments

away from the impact direction, and demeaning removes these shifts. The demeaned moments then vary along the impact direction, in proportion to the relevance deviations. The estimate is therefore the leading eigenvector of the demeaned moments, the direction along which they vary most. The direction is identified when the members of each family share their contamination and relevance varies within at least one family. In a finite panel of instruments, relevance must also vary more within families than the contamination specific to individual instruments.

Two tests evaluate the family assignment. The first asks whether the demeaned moments carry a common direction, which requires relevance to vary within families. The second, an overidentification test, asks whether every instrument points to the same direction once the contamination of its family is removed.

The same identification can be written as a constrained pooling of the instruments. The pooling weights maximise the strength of a single combined instrument, subject to the constraint that it cancels the contamination the families share. The weights are free within families, and the family totals are set to satisfy this constraint. Used as a single external instrument, the constrained pooled instrument gives the same impulse responses as the within estimator. The family averages add strength without reintroducing the shared contamination only when the families outnumber the non-policy shocks in the VAR.

The monetary policy instruments fall into different families because of how they are constructed. Two families are built from high-frequency surprises around FOMC announcements, one from near-dated futures and one from the factors of Swanson (2021). One family is built from Romer–Romer instruments and one from Cholesky-based instruments. Applied to these orthogonalised U.S. instruments, the pooled instrument is strong where the orthogonalised surprise is weak. It identifies a standard monetary contraction. Output falls gradually, with its peak decline after fifteen months. Consumer prices fall little on impact and decline gradually, and the credit spread widens on impact.

An instrument pooled without demeaning is strong, but it leans on the strongest family, inherits its contamination and points to a much milder contraction. Comparing each family with the pooled estimate shows what demeaning removes. The wide-window families point to a milder contraction than the baseline, while the high-frequency family points to a stronger one, as its weak first stage magnifies its moderate contamination.

The robustness exercises change one choice at a time. The first set of exercises changes the panel of instruments. A fifth family of information-robust instruments, built from surprises cleaned of the information the Fed reveals, is added to the four baseline families. Adding instruments within families and repartitioning the instruments spread the identifying variation across more families, such that no single family dominates. In every exercise, the impulse responses remain close to the baseline.

The second set of exercises changes the VAR specification. The baseline estimate is robust to the shadow rate of Wu and Xia (2016) as the policy indicator and to other price measures. Larger systems trace the broader transmission of monetary policy, one added variable at a time. Each added variable brings a non-policy shock, and the method then requires an additional family of instruments. Housing starts, unemployment, stock prices and commodity prices respond as expected after a monetary tightening (Bernanke and Gertler, 1995; Christiano et al., 1999; Bernanke and Kuttner, 2005; Anzuini et al., 2013). With the instrument held fixed, the larger systems barely change the impulse responses of output and prices. Finally, the pooled instrument is a single series, and it can be reused in other frameworks. In local projections (Jordà, 2005) and in an

internal-instrument VAR (Plagborg-Møller and Wolf, 2021), the impulse responses of output and the credit spread have the same sign as in the baseline.

The paper contributes to three strands of the literature. The first contribution is to the literature on identification with many, weak or invalid instruments. Recent work relaxes the assumption of a single strong and valid instrument. It allows the instruments to be weak but keeps them valid (Angelini et al., 2024), bounds their exogeneity (Conley et al., 2012; Huang and Neri, 2026), or selects the valid ones as the largest group that agrees on the effect (Windmeijer et al., 2019, 2021; Lin et al., 2024). In this paper, the instruments are not assumed to be valid. The shock is identified from the structure of the contamination, which is shared within families, rather than from the validity of any instrument. The family assumption parallels the condition under which invalid instruments still identify an effect, when their direct effects are unrelated to their relevance (Kolesár et al., 2015). The method also delivers more than a direction. Constrained pooling turns the panel into the strongest single instrument that cancels the contamination the families share, and two tests evaluate the family assignment, one for a common direction and one for the overidentifying restrictions of the family structure. Since the instruments enter a single pooled series, the many-instrument bias of overidentified estimators (Bekker, 1994; Newey and Windmeijer, 2009; Mikusheva and Sun, 2022) does not arise. Earlier work combines several instruments for one shock, but treats each as valid. Proxy-SVARs use several external instruments for one shock (Stock and Watson, 2012; Lakdawala, 2019; Angelini and Fanelli, 2019), and cross-sectional instrumental variables average the first-stage predictions of many valid instruments (Kuersteiner and Okui, 2010). The method here instead extracts a principal component from the demeaned identifying moments, without assuming any instrument valid. The closest paper is Francis et al. (2026). They also combine several instruments into a composite, but as part of a general identification scheme, which maximises the correlation between each shock and its instrument and nests recursive identification. This paper instead builds the pooled instrument to cancel the contamination that families of instruments share, and uses it to identify a single shock.

The second contribution is to the literature on high-frequency identification and the predictability of the surprises. Beyond the information and Fed-response views, the predictability has been traced to the Fed's response to financial conditions (Chen, 2026), to shifts in the systematic component of policy (Hack et al., 2024), and to information effects whose importance varies over time (Hoesch et al., 2023). The remedies separate the policy shock from the information it reveals (Jarociński and Karadi, 2020; Miranda-Agrippino and Ricco, 2021) or project the surprise on pre-announcement controls, to remove the Fed's response to news that precedes the announcement (Bauer and Swanson, 2023a,b). The order of the projection and the frequency aggregation also matter (Amodeo, 2026). Bauer and Swanson (2023b) and Gürkaynak et al. (2026) note that the orthogonalisation weakens the instrument, and they treat this as a classical weak-instrument problem, to be met with more monetary events or with robust inference (Montiel Olea and Pflueger, 2013; Andrews et al., 2019; Montiel Olea et al., 2021; Jentsch and Lunsford, 2022). This paper extends the orthogonalisation to the wide-window instruments, built from the whole monetary policy decision. It also shows that the weakness and the bias cannot be treated separately. Orthogonalisation always shrinks the contamination the instrument carries. However, it can raise the bias by weakening the first stage more than it removes contamination and by moving contamination onto shocks the instrument did not carry. The orthogonalised impulse responses can then be more biased than those of the raw surprise, a case of weakness meeting a failure of exogeneity. The loss of strength is observable from the predictability of the instrument and the correlation of its predictable part with

the policy rate.

The third contribution is empirical. It provides new estimates of the effects of U.S. monetary policy on output and prices. The literature identifies the policy shock in several ways. Some schemes impose sign restrictions (Uhlig, 2005) or narrative restrictions (Antolín-Díaz and Rubio-Ramírez, 2018). Others restrict the systematic component of policy (Arias et al., 2019) or exploit heteroskedasticity (Rigobon, 2003). This paper instead combines high-frequency, narrative and Cholesky-based instruments into a single estimate, which does not rest on any one of them. With the pooled instrument, prices decline only gradually, in line with evidence that they respond to monetary policy with a lag (Aruoba and Drechsel, 2024a; Buda et al., 2025).

The remainder of the paper is organised as follows. Section 2 shows that orthogonalisation removes the predictability of the high-frequency surprise, and documents the implications for the estimated impulse responses. Section 3 derives what orthogonalisation removes and why a weaker first stage magnifies the contamination that survives it. Section 4 orthogonalises the wide-window instruments and shows that they keep more of their first-stage strength. Section 5 develops the method and two tests of the family assignment, and Section 6 applies it to U.S. monetary policy and shows what the demeaning step removes. Section 7 shows that the baseline estimate is robust to changes in the panel of instruments, in its partition and in the VAR specification. Section 8 concludes.

2 High-frequency identification and its orthogonalisation

The high-frequency surprise is the most widely used instrument for monetary policy shocks, and the starting point of this paper. It is predictable from information available before the announcement. The standard remedy orthogonalises it on pre-announcement controls. Once orthogonalised, the surprise has a much weaker first stage, and the impulse responses it identifies are imprecise and large. Prices fall in the month of the announcement, and the funds rate soon reverses. These five features are hard to reconcile with a valid instrument. They are consistent with two costs of a weaker first stage, imprecision and bias. The bias comes from the contamination that survives the orthogonalisation, magnified by the weaker first stage. Let the central bank's policy rule be

$$i_t = f_t(\Omega_t) + \varepsilon_t^p, \tag{1}$$

with i_t the policy rate, Ω_t the information the central bank responds to and f_t its systematic response. The policy shock ε_t^p is the part of the policy rate that is not a systematic response to economic conditions. This paper identifies it with external instruments (Mertens and Ravn, 2013; Stock and Watson, 2018). An instrument is a series z_t that is correlated with the policy shock and, in the ideal case, with no other structural shock, collected in $\boldsymbol{\varepsilon}_t^{-p}$. The two covariance conditions are summarised as

$$\mathbb{E}[z_t\, \varepsilon_t^p] = \phi \neq 0 \quad \text{(relevance)}, \qquad \mathbb{E}[z_t\, \boldsymbol{\varepsilon}_t^{-p}] = \mathbf{0} \quad \text{(exogeneity)}. \tag{2}$$

Relevance makes the instrument informative about the shock. Exogeneity ensures that what it identifies is the policy shock and nothing else. The relative covariances between the instrument and the VAR residuals identify the impact column b_p, the response of each variable to the policy

shock on impact, up to a normalisation.[2]

The high-frequency surprise is the change in short-term interest-rate futures in a narrow window around an FOMC announcement (Kuttner, 2001; Gürkaynak et al., 2005; Gertler and Karadi, 2015). For an announcement m, the surprise is the revision of the expected policy rate between the opening and the close of the window, $mps_m = \mathbb{E}_{m+\Delta}[i_m] - \mathbb{E}_{m-\Delta}[i_m]$, where $\mathbb{E}_{m-\Delta}$ and $\mathbb{E}_{m+\Delta}$ are the expectations priced just before the window opens and just after it closes.[3]

The narrow window isolates the announcement from other news, but it does not make the surprise unpredictable. The surprise still contains part of the Fed's systematic response to news the market had not yet priced. It is therefore predictable from information available before the announcement, and it must be orthogonalised on pre-announcement information.

2.1 Predictability and orthogonalisation

Two lines of work document the predictability of the surprise. Miranda-Agrippino and Ricco (2021) read it as an information effect, and project the surprise on the Fed's Greenbook forecasts and on its own lags. Bauer and Swanson (2023a,b) read it as the Fed's response to news the market had not yet fully priced, and project the surprise on publicly available data dated before the announcement. In their reading, the predictable part is the Fed's systematic response to that news, which the narrow window does not remove. I borrow the orthogonalisation of Bauer and Swanson (2023b) and apply the same procedure to every instrument in the paper. Every instrument is then orthogonalised on the same pre-announcement information, conditional on the choice of controls.

The orthogonalisation projects the surprise on a set $\mathbf{W}_m$ of six pre-announcement controls and keeps the residual.[4] Every control is measured before the announcement, and the market-based controls end the day before it. The projection can be applied announcement by announcement and then aggregated to the month (purge-then-aggregate, PTA; Amodeo, 2026), or to the monthly series (aggregate-then-purge, ATP; Bauer and Swanson, 2023b). The paper uses PTA for every instrument observed at the announcement level.[5] For a meeting m the orthogonalisation is

$$mps_m^{\perp} \;=\; mps_m - \text{Proj}\left(mps_m \mid \mathbf{W}_m\right),$$

where $\text{Proj}(y \mid \mathbf{X})$ denotes the linear projection of y on $\mathbf{X}$ and a constant. On the sample of this

[2] With u_t the vector of reduced-form VAR innovations and u_t^p the innovation in the policy rate, the identified impact vector, normalised such that the policy rate rises by one unit, is $\beta = b_p / b_{pp} = \mathbb{E}[u_t\, z_t] / \mathbb{E}[u_t^p\, z_t]$. It is the covariance of the instrument with each innovation relative to its covariance with the policy-rate innovation, which is the instrument's first stage. The response at horizon h is $\Phi_h \beta$, with Φ_h the moving-average coefficients of the VAR. Section 6.1 gives the empirical specification.

[3] The Bauer and Swanson (2023b) surprise is the first principal component of the changes in the first four quarterly Eurodollar futures (ED1 to ED4), which price the three-month rate over roughly the next year. i_m therefore stands for the expected path of the policy rate. The surprises are summed within the month, $z_t = \sum_{m \in t} mps_m$, to match the frequency of the VAR (Section 6.1).

[4] The six controls of Bauer and Swanson (2023b) are the nonfarm-payrolls surprise relative to the survey median, employment growth over the year to the latest release, the S&P 500 over the sixty-five trading days to the day before, the second principal component of the one- to ten-year yield curve over three months, commodity prices over three months, and the skewness implied by ten-year Treasury options over the prior month.

[5] Amodeo (2026) shows that orthogonalising monthly surprises (ATP) can bias the orthogonalised surprise in months with more than one announcement. Under ATP, the monthly sum is projected on controls dated before the first announcement of the month. The news that arrives between the two announcements is then left in the second surprise. Under PTA, each announcement is projected on its own controls.

paper, the six controls jointly predict the surprise with $R^2 = 0.17$ over 1988–2019 (Appendix Table 3, first column).

The orthogonalised surprise is, by construction, orthogonal to the chosen set of controls. Orthogonalisation removes the predictability that these controls span. The contamination that remains comes from news on the announcement day or later in the window, and from predictable news the controls do not span. The orthogonalised surprise is therefore clean only if the contamination lies entirely in the span of the controls (Section 3.2). The controls end the day before the announcement by design. Controls dated on announcement days would also span the policy shock, and the projection would remove it from the instrument.

Orthogonalisation also weakens the first-stage strength. The predictable part of the surprise consists of the systematic response to news that the market had not yet priced. Removing the predictable component therefore removes a portion of the first-stage strength. The loss depends less on how predictable an instrument is than on how closely its predictable part moves with the policy rate (Section 4.3).

2.2 What the orthogonalised surprise delivers

Orthogonalisation aims at a plausibly exogenous instrument, but it has side effects. Appendix Figure 8 shows the impulse responses identified with the raw Bauer and Swanson (2023b) surprise and with its orthogonalised version (PTA) as external instruments, each with its 68% confidence band, in a monthly VAR in industrial production, consumer prices, the excess bond premium (Gilchrist and Zakrajšek, 2012) and the federal funds rate (Section 6.1). Five empirical features stand out, summarised in the first two columns of Table 2.

(i) The orthogonalised instrument is weaker. Orthogonalisation weakens the first-stage strength of the surprise. Its first-stage F against the funds rate falls from 14.5 to 3.6. Predictable variation moves with the policy-rate innovation. Most of the Bauer and Swanson (2023b) controls are financial, and the predictable part is consistent with the Fed's response to financial conditions that the market had not fully priced (Chen, 2026). Removing the predictable part therefore also weakens the first-stage strength.

(ii) The confidence bands are wider. The weaker instrument identifies the impulse responses less precisely. At the peak decline in output, the 68% confidence band is four times as wide under the orthogonalised surprise as under the raw one.

(iii) The impulse responses are larger. The peak decline in output is 1.9% under the raw surprise and 6.6% under the orthogonalised one, and the credit spread rises on impact by 0.25 and 2.0 points. The true magnitudes are unknown, but these are large. On the same reduced form, they lie outside the range of five conventional identification schemes and of other high-frequency instruments (Online Appendix Figures OA.8 and OA.9).

(iv) Prices fall on impact. In the impulse responses identified with the orthogonalised surprise, the price level falls by 0.8% in the month of the announcement. A tightening usually reaches prices only gradually. It weakens aggregate demand, and firms' prices and workers' wages adjust to the resulting slack over time. An immediate fall in prices also sits uneasily with recent evidence on the timing of transmission, in which the aggregate price level takes several months at the earliest

to respond (Aruoba and Drechsel, 2024a) and consumer prices decline only gradually even as spending reacts within weeks (Buda et al., 2025).

(v) The funds rate reverses. After the normalised one-point increase on impact, the funds-rate impulse response identified with the orthogonalised surprise turns negative from the sixth month. It reaches -1.5 points after seventeen months, one and a half times the initial tightening. Under the raw surprise, the funds-rate impulse response remains above zero throughout the response horizon.

The orthogonalisation was meant to make the instrument credible. It also made the instrument weaker, and the impulse responses it identifies became larger as its first stage weakened. Features (i) and (ii) follow from the weaker first stage alone. Features (iii) to (v) are harder to attribute to weakness alone. A weaker first stage has two costs. The first is imprecision, the wider confidence bands of feature (ii). The second is bias. A weaker first stage magnifies any contamination that survives the orthogonalisation, and with it the bias of the identified impulse responses at every horizon (Section 3). Features (iii) to (v) are consistent with that bias. Section 6.2 revisits the five features under the pooled instrument.

3 The contamination–strength trade-off

An instrument that carries residual contamination biases the impulse responses it identifies. A relevant instrument attenuates that bias, and a weak first stage magnifies it. The first stage is the instrument's covariance with the policy-rate innovation. Contamination and strength must therefore be judged together. An instrument with less residual contamination can still be more biased than a more contaminated one if it is much weaker. This is the contamination–strength trade-off.

The trade-off holds for any instrument, and it matters most for orthogonalisation. Orthogonalising an instrument on pre-announcement controls removes predictable variation and reduces its contamination. It can also weaken its first-stage strength considerably. The bias in the impulse responses can then grow rather than shrink, the opposite of what the orthogonalisation intends. The section derives the bias, shows what orthogonalisation removes and how the bias that survives is magnified, and explains why weak-instrument-robust inference does not correct it.

3.1 Identification with a contaminated instrument

A single instrument may load on the non-policy shocks as well as on the policy shock. With it, the proxy-SVAR identifies the ratio of the instrument's covariance with each VAR innovation to its covariance with the policy-rate innovation. The bias follows from that ratio.

Let $y_t \in \mathbb{R}^n$ follow a stationary VAR with s lags, $y_t = c + \sum_{\ell=1}^{s} \Pi_\ell y_{t-\ell} + u_t$, with moving-average coefficients Φ_h ($\Phi_0 = I_n$). The response at horizon h to an impact vector b is $\Phi_h b$. The innovations are linked to orthogonal, unit-variance structural shocks by

$$u_t = B\varepsilon_t, \qquad \varepsilon_t = \begin{pmatrix} \varepsilon_t^{p} \\ \varepsilon_t^{-p} \end{pmatrix}, \qquad \mathbb{E}\left[\varepsilon_t \varepsilon_t'\right] = I_n, \qquad B = \left[\, b_p \;\; B_{-p} \,\right], \tag{3}$$

where ε_t^{p} is the policy shock, with impact vector b_p, and ε_t^{-p} the non-policy shocks, with impact

matrix B_{-p}. With s_p the vector that selects the policy rate, the policy-rate innovation is

$$u_t^p := s_p' u_t = b_{pp}\, \varepsilon_t^p + b_{p,-p}' \boldsymbol{\varepsilon}_t^{-p}, \tag{4}$$

where $b_{pp} := s_p' b_p > 0$ is the policy rate's response to its own shock and $b_{p,-p}' := s_p' B_{-p}$, the policy-rate row of B_{-p}, is its within-period response to the non-policy shocks. The impact of a policy shock that raises the policy rate by one unit is $\beta := b_p / b_{pp}$. Let a potentially contaminated instrument z_t be

$$z_t = \phi\, \varepsilon_t^p + \boldsymbol{\psi}' \boldsymbol{\varepsilon}_t^{-p} + \omega_t, \qquad \mathbb{E}[\boldsymbol{\varepsilon}_t\, \omega_t] = \mathbf{0}, \tag{5}$$

with relevance $\phi > 0$, *contamination loading* $\boldsymbol{\psi} \in \mathbb{R}^{n-1}$ on the non-policy shocks, and a remainder ω_t uncorrelated with the period's shocks. The identifying moment is $\mathbb{E}[u_t z_t] = \phi\, b_p + B_{-p}\boldsymbol{\psi}$. The first stage and the proxy-SVAR estimand are

$$\pi := s_p'\, \mathbb{E}[u_t z_t] = \phi\, b_{pp} + b_{p,-p}' \boldsymbol{\psi}, \qquad \tilde{\beta} := \frac{\mathbb{E}[u_t z_t]}{s_p'\, \mathbb{E}[u_t z_t]}. \tag{6}$$

The first stage has two sources. One is the policy shock the instrument contains, ϕb_{pp}. The other is the non-policy news it carries, $b_{p,-p}'\boldsymbol{\psi}$, which also moves the policy rate within the period. The proxy-SVAR estimator replaces the covariances in $\tilde{\beta}$ with their sample counterparts, and converges in probability to $\tilde{\beta}$ as the sample grows. Its bias is the gap between $\tilde{\beta}$ and β, which is (Appendix A.1)

$$\tilde{\beta} = \beta + \frac{d}{\pi}, \qquad d := \underbrace{B_{-p}\boldsymbol{\psi}}_{\text{contamination}} - \underbrace{\beta\, b_{p,-p}'\boldsymbol{\psi}}_{\text{induced policy response}}, \qquad s_p' d = 0. \tag{7}$$

A contaminated instrument therefore identifies the true impact vector plus a bias in the direction of d. At horizon h, the bias of the impulse response is $\Phi_h d / \pi$. The contamination term in equation (7) is the impact of the non-policy shocks on which the instrument loads. These shocks also move the policy rate within the period, and the proxy-SVAR attributes that movement to the policy shock. The induced policy response in equation (7) is the impact of a policy shock that moves the policy rate by the same amount. Hence, the contamination direction d is the contamination net of this induced policy response. It is linear in the loading, $d = M\boldsymbol{\psi}$, with $M := B_{-p} - \beta\, b_{p,-p}'$ the impact of each non-policy shock net of the policy response it induces. Since the impact response of the policy rate is normalised, the bias falls entirely on the variables other than the policy rate ($s_p' d = 0$).

The first stage scales the bias, and a weaker first stage magnifies it. Neither contamination nor strength is therefore enough on its own to judge an instrument. Take two instruments for the same policy shock, $z_t^{(1)}$ and $z_t^{(2)}$, the first less contaminated, $\|d^{(1)}\| < \|d^{(2)}\|$. The first is still the more biased if its first stage is weaker by a larger factor, $\pi^{(1)}/\pi^{(2)} < \|d^{(1)}\| / \|d^{(2)}\|$. The contamination is not observed, and the comparison cannot be made directly in the data. It shows that a cleaner instrument is not necessarily a better one. A cleaner but weaker instrument is often taken to cost only precision. With any residual contamination, it also costs bias. A small reduction in $\boldsymbol{\psi}$ bought with a large reduction in π moves the estimate further from the truth.

Orthogonalisation aims to remove the predictable variation in an instrument, the empirical footprint of its contamination. It also affects its first-stage strength π. Section 3.3 shows that it reduces the bias only if it removes a larger share of the contamination than of the first-stage

strength.

3.2 What orthogonalisation removes

Orthogonalisation removes the part of the instrument that the pre-announcement controls predict. The contamination that remains comes from news on the announcement day or later in the period, and from news the controls do not span. Part of the predictable variation is contamination that moves the policy rate. Removing it reduces the contamination and weakens the first stage at the same time. Whether the bias falls depends on which of the two falls by the larger share.

The controls $\mathbf{W}_t$ are dated before the announcement. They therefore respond neither to the policy shock it reveals nor to news that arrives on the announcement day or later in the period. Each control is a combination of the period's non-policy shocks that precede the announcement,

$$\mathbf{W}_t = A'\boldsymbol{\varepsilon}_t^{-p}, \tag{8}$$

with column j of A the loadings of control j on the non-policy shocks, and zero loadings on the shocks that arrive from the announcement on.

The orthogonalisation is the population regression of z_t on $\mathbf{W}_t$. It splits the instrument into the fitted value and the residual, $z_t = \hat{z}_t + z_t^{\perp}$, with $\hat{z}_t := \mathrm{Proj}(z_t \mid \mathbf{W}_t)$ the predictable part and $z_t^{\perp} := z_t - \hat{z}_t$ the orthogonalised instrument. Because the controls are orthogonal to the policy shock, the fitted value contains none of it, and the orthogonalised instrument keeps the relevance ϕ of the raw instrument. Orthogonalisation changes only the contamination loading, which becomes $\boldsymbol{\psi}^{\perp}$. The first stage, $\pi^{\perp} := \mathrm{Cov}(z_t^{\perp}, u_t^{p})$, and the contamination direction keep the form of equation (7), with $\boldsymbol{\psi}^{\perp}$ in place of $\boldsymbol{\psi}$,

$$\pi^{\perp} = \phi\, b_{pp} + b'_{p,-p}\boldsymbol{\psi}^{\perp}, \qquad d^{\perp} = B_{-p}\boldsymbol{\psi}^{\perp} - \beta\, b'_{p,-p}\boldsymbol{\psi}^{\perp}, \qquad \tilde{\beta}^{\perp} = \beta + \frac{d^{\perp}}{\pi^{\perp}}. \tag{9}$$

The first stage keeps the same two parts as the raw first stage. The policy shock the instrument contains, ϕb_{pp}, is unchanged. Any change in first-stage strength therefore comes from the contamination that moves the policy rate, $b'_{p,-p}\boldsymbol{\psi}^{\perp}$.

Proposition 1 describes what orthogonalisation removes and what it leaves. Let P be the projection onto the span of A, the combinations of non-policy shocks that the controls span. $P\boldsymbol{\psi}$ is the combination of the controls' loadings closest to the instrument's loading.

Proposition 1 (What orthogonalisation removes)**.** *With the controls of equation* (8)*:*

(i) *The contamination loading of the orthogonalised instrument is* $\boldsymbol{\psi}^{\perp} = (I - P)\boldsymbol{\psi}$.

(ii) *The loading shrinks by the variance of the predictable part,* $\|\boldsymbol{\psi}^{\perp}\|^2 = \|\boldsymbol{\psi}\|^2 - \mathrm{Var}(\hat{z}_t)$.

(iii) *The contamination cancels,* $\boldsymbol{\psi}^{\perp} = \mathbf{0}$*, if and only if the controls span it,* $\boldsymbol{\psi} = A\boldsymbol{w}$ *for some vector of weights on the controls* $\boldsymbol{w}$.

Details are in Appendix A.2. Part (i) shows that orthogonalisation removes $P\boldsymbol{\psi}$, the part of the contamination that the controls span, and nothing that arrives on the announcement day or later, since the controls have zero loadings on those shocks. Part (ii) shows that the loading always shrinks, strictly whenever the instrument is predictable. Part (iii) shows that exact cancellation is the exception. A control that moves with several shocks can be subtracted only whole. A stock

return moves with both demand and financial news, and subtracting it to remove demand news also subtracts financial news the instrument potentially did not carry. When the controls do not span the instrument's contamination, the remainder $(I - P)\boldsymbol{\psi}$ can load on shocks the instrument did not carry.

3.3 The magnification of the surviving bias

Proposition 1 shows that orthogonalisation always shrinks the contamination loading. The bias need not fall. The contamination direction weighs each shock by its effect on the economy, net of the induced policy response, and it can grow when the remainder moves to shocks with larger effects. The bias is the contamination direction magnified by a weaker first stage, and it can grow even when the contamination direction shrinks. Two shares summarise what orthogonalisation does to the bias, the share of first-stage strength it removes and the share of the contamination direction it removes.

Proposition 2 (Magnification)**.** *Let the instrument be contaminated, $\boldsymbol{\psi} \neq \mathbf{0}$, with positive first stages before and after orthogonalisation, $\pi, \pi^{\perp} > 0$.*

(i) *The share of the raw first-stage strength lost to orthogonalisation is*

$$\delta_{\pi} := \frac{\mathrm{Cov}(\hat{z}_t, u_t^p)}{\mathrm{Cov}(z_t, u_t^p)} = 1 - \frac{\pi^{\perp}}{\pi} = \frac{b'_{p,-p} P \boldsymbol{\psi}}{\pi}. \tag{10}$$

It is observable, and $\delta_{\pi} < 1$. It is positive when the predictable part moves with the policy rate as the instrument does, as for all three instruments of Section 4.3.

(ii) *The share of the contamination direction that orthogonalisation removes is*

$$\delta_d := 1 - \frac{\|d^{\perp}\|}{\|d\|} = 1 - \frac{\|M(I - P)\boldsymbol{\psi}\|}{\|M\boldsymbol{\psi}\|} \leq 1. \tag{11}$$

It is not observable, and $\delta_d = 1$ if the contamination cancels. It is negative when the contamination direction is longer after orthogonalisation.

(iii) *The factor by which orthogonalisation scales the bias is*

$$\varrho := \frac{\|\tilde{\beta}^{\perp} - \beta\|}{\|\tilde{\beta} - \beta\|} = \frac{\|d^{\perp}\| / \|d\|}{\pi^{\perp} / \pi} = \frac{1 - \delta_d}{1 - \delta_{\pi}}, \tag{12}$$

and

$$\varrho > 1 \iff \delta_d < \delta_{\pi}, \qquad \varrho \leq \frac{1}{1 - \delta_{\pi}} \ \text{ whenever } \ \delta_d \geq 0, \tag{13}$$

with equality in the bound if and only if $\delta_d = 0$.

Details are in Appendix A.3. Part (i) gives the cost of orthogonalisation. It weakens the first-stage strength only by removing contamination that moves the policy rate, $b'_{p,-p} P \boldsymbol{\psi}$. A positive loss is the expected case under the "Fed responds to news" reading (Bauer and Swanson, 2023a). Section 4.3 shows why equally predictable instruments lose very different shares of their first-stage strength. Part (ii) gives the benefit. Computing δ_d requires the loadings, which are not observed. Both

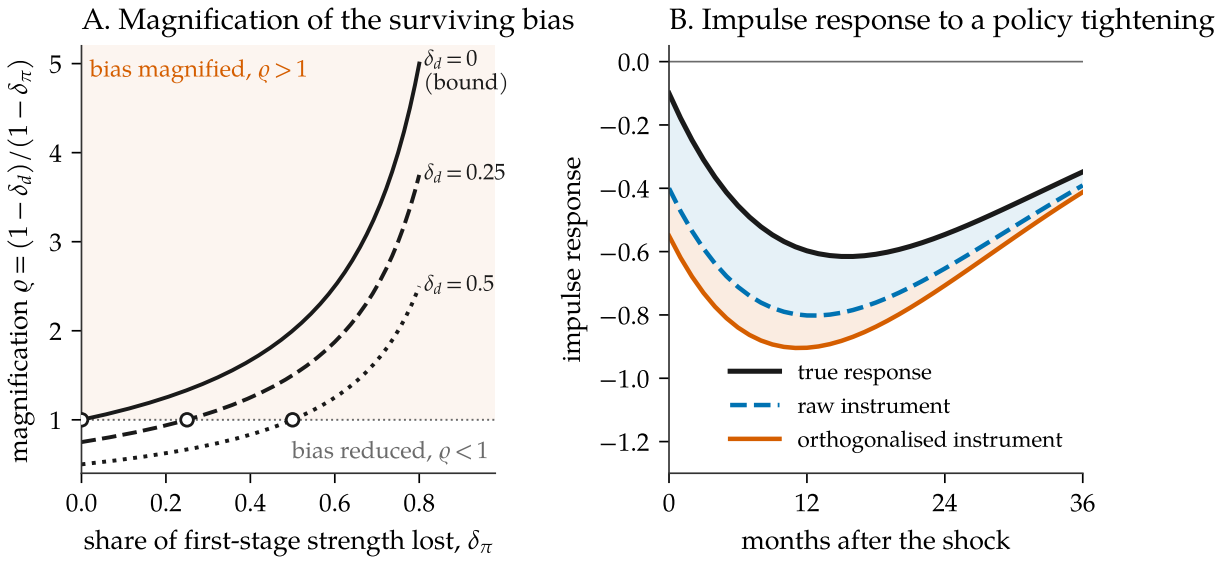


Figure 1: **The magnification of the surviving bias.** Panel A: the factor $\varrho = (1-\delta_d)/(1-\delta_\pi)$ of equation (12) by which orthogonalisation scales the bias, against the share of first-stage strength it removes, δ_π. Each curve holds the share of the contamination direction removed, δ_d, fixed, and crosses the dotted line $\varrho = 1$ at $\delta_\pi = \delta_d$ (circles). The curve $\delta_d = 0$ is the bound in equation (13). In the shaded region, $\varrho > 1$, orthogonalisation magnifies the bias. At a given δ_π, points between the curve $\delta_d = 0$ and the dotted line have $0 < \delta_d < \delta_\pi$, and points above the curve $\delta_d = 0$ have $\delta_d < 0$. Panel B: an illustrative example. Population impulse responses of a non-policy variable to a policy shock that raises the policy rate by one unit, in a bivariate VAR(1) in that variable and the policy rate. True response $\Phi_h\beta$; raw instrument $\Phi_h(\beta + d/\pi)$; orthogonalised instrument $\Phi_h(\beta + d^\perp/\pi^\perp)$, with $\delta_\pi = 0.5$, $\delta_d = 0.25$ and hence $\varrho = 1.5$. The contamination direction satisfies $s_p'd = 0$ and loads on the non-policy variable.

shares weigh the same removed contamination, $P\boldsymbol{\psi}$. The first stage weighs it by the policy rate's response to each shock, $b_{p,-p}$, in equation (10). The contamination direction weighs it by each shock's effect on the economy, M, in equation (11). Equation (12) has two components. The first component, $1/(1-\delta_\pi)$, is observable, and it is the magnification when orthogonalisation removes none of the contamination direction, $\delta_d = 0$. The second component, $1-\delta_d$, is the share of the contamination direction that survives. It depends on which non-policy news orthogonalisation removes. Removing contamination is therefore not enough. Magnification requires the controls to predict disproportionately the contamination that moves the policy rate. Orthogonalisation can also raise the bias by lengthening the contamination direction, $\delta_d < 0$, when the remainder moves to shocks with larger effects. Section 4.3 finds that the predictable part of the high-frequency surprise moves with the policy rate as closely as the surprise itself.

Figure 1 illustrates the magnification, with parameter values chosen for illustration, not estimated. In Panel A, each curve fixes the share of the contamination direction removed, δ_d, and the bias is magnified to the right of the point where the two shares are equal, $\delta_\pi = \delta_d$. The curve $\delta_d = 0$ is the bound $1/(1-\delta_\pi)$, which rises slowly at first and then steeply. For the orthogonalised surprise, $\delta_\pi \approx 0.54$ (Section 4.3), and any $\delta_d < 0.54$ magnifies the bias. Panel B shows what this does to the impulse response of a non-policy variable, in a stylised two-variable VAR. The raw

instrument overstates the decline of the variable. Orthogonalisation removes a quarter of the contamination but half of the first-stage strength, $\delta_d = 0.25$ and $\delta_\pi = 0.5$, and hence $\varrho = 1.5$. The orthogonalised impulse response lies further from the truth than the raw one and shows a larger decline. Appendix B.1 reports the magnification variable by variable and horizon by horizon.

An orthogonalised response larger than the raw one admits two readings. In the reading of Bauer and Swanson (2023b), orthogonalisation removes contamination that attenuated the raw estimate, $\delta_d > \delta_\pi$, and the orthogonalised estimate is closer to β. In the reading developed here, the contamination that survives is magnified by the weaker first stage. Two implications tell the readings apart. Under the magnification reading, the enlargement should be ordered by δ_π across instruments orthogonalised on the same controls, and an instrument that addresses the same contamination without weakening its first-stage strength should not enlarge the impulse responses. Section 4.4 examines both, using an identity that holds under both readings (Appendix B.1). A larger orthogonalised response is therefore not necessarily a better estimate of β. It may be residual contamination magnified by a weaker first stage.

Orthogonalisation leaves the surprise weak, and the standard response to a weak first stage is weak-instrument-robust inference.[6] An Anderson–Rubin-type confidence set has correct coverage for any instrument strength (Montiel Olea et al., 2021; Jentsch and Lunsford, 2022). It addresses weakness, not contamination (Andrews et al., 2019). Built on $z_t^\perp$, it covers the estimand of the contaminated instrument, $\beta + d^\perp/\pi^\perp$. It is therefore centred on the wrong value if some residual contamination survives, and it inherits the magnification of Proposition 2. Wider confidence bands repair neither the bias nor its magnification. Only an instrument that keeps its first-stage strength under the same orthogonalisation can reduce them. Section 4 turns to such instruments.

4 Widening the window

Neither version of the high-frequency surprise is satisfactory. Raw, it is predictable and contaminated. Orthogonalised, it becomes weak, and any contamination that survives is magnified by the weaker first stage. An ideal instrument would be both unpredictable from pre-announcement information and strongly correlated with the innovation in the policy rate.

The narrow window also limits how much of the policy shock the surprise contains. The surprise measures only the part of the decision revealed inside the window. By the rule in equation (1), the expectation at the opening of the window is $\mathbb{E}_{m-\Delta}[i_m] = \mathbb{E}_{m-\Delta}[f_m(\Omega_m)] + \mathbb{E}_{m-\Delta}[\varepsilon_m^p]$, and its second term may not be zero. On the eve of a meeting, and on the morning of the announcement, press reports and informal signals from policymakers can reveal part of the decision itself (Lucca and Moench, 2015; Cieslak et al., 2019). When the decision deviates from the rule, what is revealed in this way is part of the policy shock. It is not a response to the economy, and it moves the policy rate in the month of the decision. Because the market prices it before the window opens, it is already in the expectation at the opening of the window, and the surprise misses it. An instrument built on the decision itself keeps it, whenever the market learned it. If a share a of the shock's variance is revealed before the window opens, the surprise's relevance is only $1 - a$ times that of the decision. Widening the window therefore adds policy shock, not only contamination.

The window can also close too early. Markets take at least forty minutes to price FOMC statements, and up to an hour at long maturities. The conventional thirty-minute window can

[6] Weakness is a bias problem in its own right. Under weak identification, the IV estimate is biased toward its least-squares counterpart, which in a proxy-SVAR is the response with the policy rate ordered first in a Cholesky decomposition of the residual covariance matrix (Stock and Watson, 2018).

therefore miss part of the reaction, and the length of the window changes what the surprises capture (Tran, 2026).

Another remedy is to add events rather than widen the window. Bauer and Swanson (2023b); Swanson and Jayawickrema (2023) add Fed Chair speeches, press conferences and congressional testimony to the FOMC announcements, which raises the first-stage strength of the surprise. The added strength may come from contamination rather than from the policy shock. Bauer and Swanson (2023b) note that speeches and testimonies can carry larger information effects. A speech mostly conveys the Fed's view of the outlook and how it intends to respond to it. In terms of the rule in equation (1), this is news about the information the Fed responds to or about its systematic response, not a policy shock. The longer windows of these events can also let other news in.

Widening the window from the announcement to the decision addresses both shortfalls of the narrow window. The two instruments proposed below are reaction-function residuals, each the change in the policy rate that the policymaker's information set does not explain. Neither construction removes all of the pre-announcement information that predicts the high-frequency surprise. Both remain predictable from the controls of Bauer and Swanson (2023b) (Appendix Table 3). For consistency, they are orthogonalised on the same pre-announcement controls as the surprise.

Orthogonalisation deals with only one source of contamination, and widening the window adds a second. Figure 2 separates the two. The first is predictability. The controls are dated before each announcement and capture news the market had not yet priced. Orthogonalisation removes the part of that news the controls span. The second is news that arrives on the day of the decision or later in the window, such as data releases or financial developments. Orthogonalisation cannot remove it. Controls dated on the day of the announcement would also span the policy shock, and would remove it along with the contamination. Exposure to this news increases with the length of the window. It is limited for the high-frequency surprise and larger for the two wider-window instruments.

No instrument is therefore clean after orthogonalisation. The news that survives is not specific to each instrument, though. Instruments built from the same source over the same window share it. Section 5 groups such instruments into families and removes what they share, the part of the contamination that orthogonalisation cannot reach. The first instrument widens the window to the whole decision, with the narrative approach of Romer and Romer (2004). The second widens it to the whole month, with the VAR's own policy-rate innovation.

4.1 From the announcement to the decision: the Romer–Romer instrument

To widen the window to the whole monetary policy decision, I revisit the narrative instrument of Romer and Romer (2004). Measuring the whole decision rather than the announcement window aims to capture the part of the policy shock that the high-frequency surprise misses. The Romer–Romer construction suits that aim for two reasons. Because its left-hand side is the decision itself, it keeps the part of the shock revealed before the announcement. Because it removes the Fed's systematic response with the Fed's own forecasts, less of its relevance should rest on what the pre-announcement controls predict.

The Romer–Romer instrument, Z_m^{RR}, is the portion of the FOMC's intended change in the funds-rate target at meeting m, Δi_m, that the Fed's own information set does not explain. The information set contains the pre-meeting target level i_{m^-} and the real-time Greenbook forecasts

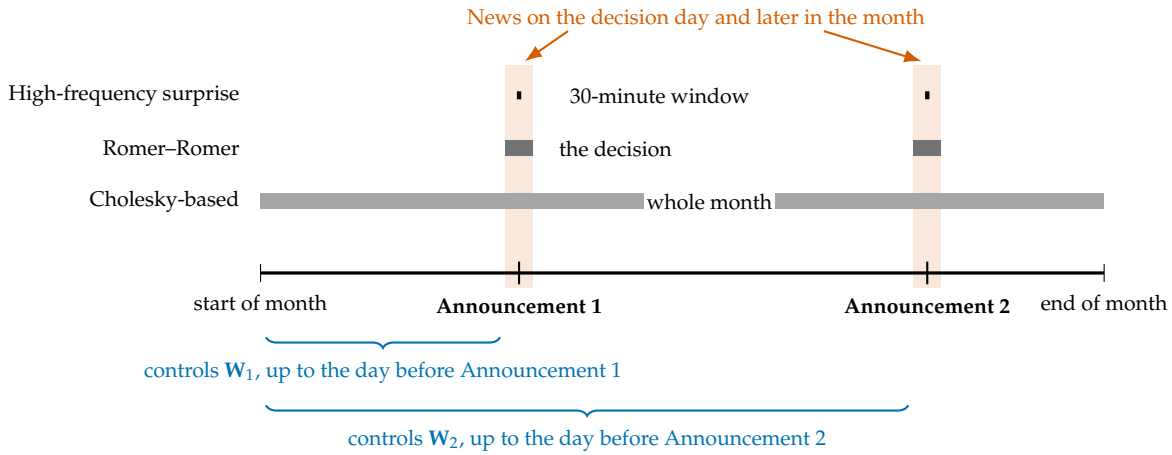


Figure 2: **Two sources of contamination within a month.** A month with two monetary policy announcements. Grey bars: the window over which each instrument measures the decision. The Romer–Romer instrument is the decision itself, net of the Fed's response to its Greenbook forecasts, whenever the market learned it. Blue braces: the pre-announcement controls used to orthogonalise each announcement (PTA), which end the day before it. Shaded bands: the day of each decision. Orange: news on the decision day and later in the month, inside the window of the wider instruments and outside every control set. Orthogonalisation removes the predictable part of the instrument, the news in the controls. It cannot remove news that arrives on the decision day or later in the window, which the wider instruments admit; the within transform of Section 5 removes the part that a family of instruments shares.

$\mathbf{G}_m$. Romer and Romer (2004) give the details of the construction, and the series is extended here to 2019. The orthogonalised instrument controls for the same information set and, jointly, for the pre-announcement controls $\mathbf{W}_m$ of Bauer and Swanson (2023b),

$$Z_m^{\mathrm{RR},\perp} = \Delta i_m - \mathrm{Proj}\left(\Delta i_m \mid i_{m^-}, \mathbf{G}_m, \mathbf{W}_m\right). \tag{14}$$

The projection can be computed jointly or in two steps.[7] The Romer–Romer instrument is predictable from the pre-announcement controls ($R^2 = 0.11$, Appendix Table 3), mostly through the yield-curve slope and Treasury skewness. The predictability has two potential sources.

The first is *timing*. The Greenbook closes about six days before the meeting. Information that arrives in between, such as an employment report or a move in financial markets, can influence the Committee's decision but does *not* enter the forecasts $\mathbf{G}_m$. If the Committee reacts to it at the meeting, the residual Z_m^{RR} still carries a systematic response, visible as predictability from the controls $\mathbf{W}_m$.

The second is that the two sets cover different information. The Greenbook forecasts output, inflation and unemployment, while four of the six controls are market prices, namely the S&P 500, the slope of the yield curve, commodity prices and Treasury skewness. The controls therefore load more on financial conditions, and they span different structural shocks from the Greenbook

[7] By the Frisch–Waugh–Lovell theorem, the joint projection is a two-step residual, namely the Romer–Romer instrument orthogonalised on the part of the controls that its own information set does not already explain, $Z_m^{\mathrm{RR},\perp} = Z_m^{\mathrm{RR}} - \mathrm{Proj}(Z_m^{\mathrm{RR}} \mid \tilde{\mathbf{W}}_m)$ with $\tilde{\mathbf{W}}_m = \mathbf{W}_m - \mathrm{Proj}(\mathbf{W}_m \mid i_{m^-}, \mathbf{G}_m)$. The same holds for the Cholesky-based instrument of Section 4.2, whose own information set is the month's output and price innovations, (u_t^o, u_t^c).

forecasts. The two information sets overlap, but neither contains the other.

The joint orthogonalisation in equation (14) removes both the systematic response that the Greenbook spans and the response to news that arrives after the Greenbook closes. After orthogonalisation, the instrument keeps most of its first-stage strength. Its first-stage F falls from 37.3 to 29.6 (Section 4.3). It is not clean. News that arrives after the controls are measured survives, as does any response that neither information set captures. This contamination differs from that of the high-frequency surprise, because the instrument is built from different information over a different window.

4.2 To the whole month: the Cholesky-based instrument

The Romer–Romer instrument widens the window from the announcement to the decision. The Cholesky-based instrument widens it further, to the whole month. It removes the central bank's systematic response within the VAR, rather than with the Fed's forecasts. Its purpose is to add first-stage strength to the identification. The cost is a window that no longer surrounds the announcement. The instrument admits contamination from news later in the month, which differs from the contamination of the high-frequency surprise and of the Romer–Romer instrument. Like the other instruments, it is orthogonalised on the same controls, such that it does not carry the response to pre-announcement information.

The construction has two steps. The first removes what past data predict. The VAR is estimated and its policy-rate residual is kept, the innovation u_t^p, the part of the month's policy rate not predicted by the lags of the system. It contains the whole decision, and every other shock that moves the policy rate within the month.

The second step removes the Fed's systematic reaction within the month. The policy rate may respond to output and prices in the same month. The instrument keeps only the variation in the policy rate beyond that reaction.

Order the VAR as output (o), prices (c, for the CPI), the policy rate (p) and the excess bond premium (f), with innovations u_t^i and impact responses b_{ij} of variable i to shock j, and let ε_t^f be the financial shock.

The timing restrictions of the recursive identification (Sims, 1980; Bernanke and Blinder, 1992; Christiano et al., 1999) can be written as an instrument. It is the policy-rate innovation residualised on the contemporaneous output and price innovations,

$$Z_t^{\text{CHOL}} = u_t^p - \text{Proj}\left(u_t^p \mid u_t^o, u_t^c\right) = b_{pp}\varepsilon_t^p + b_{pf}\varepsilon_t^f, \tag{15}$$

with the effective federal funds rate as the policy variable.

Two restrictions of the recursive ordering (o, c, p, f) give the second equality in equation (15). First, the policy shock does not move output and prices within the month, $b_{op} = b_{cp} = 0$. The innovations u_t^o and u_t^c are then orthogonal to ε_t^p, and residualising on them removes none of the policy shock. Second, the financial shock does not move output and prices within the month either, $b_{of} = b_{cf} = 0$. The projection then removes exactly the output and price shocks. The recursive ordering would also restrict the policy rate not to respond to the financial shock within the month, $b_{pf} = 0$. This restriction is relaxed. The Fed can respond to financial news within the month, as the predictability of the surprises from financial information suggests (Bauer and Swanson, 2023a,b). The instrument therefore carries the Fed's systematic response to the financial shock, $b_{pf}\varepsilon_t^f$. By

construction, it is uncorrelated with the month's output and price innovations. Used alone, it identifies no response of output or prices on impact. This is the recursive restriction, and it is valid if the policy shock does not move them within the month.

The Cholesky-based instrument remains predictable from the controls ($R^2 = 0.06$, $p = 0.017$, Appendix Table 3), through the payroll surprise, the slope of the yield curve and Treasury skewness. One source is the financial term in equation (15). If the Fed responds to financial conditions within the month, $b_{pf} \neq 0$, the timing restrictions leave that systematic response in the instrument. Most of the controls are financial market variables, and they predict this response. The instrument is therefore orthogonalised on the same controls,

$$Z_t^{\text{CHOL},\perp} = u_t^p - \text{Proj}\left(u_t^p \mid u_t^o, u_t^c, \mathbf{W}_t\right),$$

computed jointly or in two steps (footnote 7). The instrument is built from monthly VAR innovations. Its orthogonalisation therefore uses monthly controls, dated before the month's announcement, and aggregates before orthogonalising (ATP). It removes the part of $b_{pf}\varepsilon_t^f$ that the controls span, the financial exposure the ordering was meant to exclude. The part the controls do not span remains in the instrument.

The VAR innovations cover the whole month, including the days after the decision, and both projections inherit that window. The recursive restriction then requires that the policy shock not move output and prices before the end of the month. Such a monthly transmission lag is consistent with the gradual response of prices (Aruoba and Drechsel, 2024a). The controls, by contrast, end the day before the meeting. Variation in u_t^p dated after the announcement lies outside their span, and orthogonalisation cannot remove it. This variation contains the rest of the policy shock, which the instrument should keep. It also contains any other news that moves the policy rate later in the month. This news is the cost of the wider window, a contamination that the narrower instruments exclude.

Two further features of the orthogonalised instrument matter. First, $Z_t^{\text{CHOL},\perp}$ is defined only on 1988–2019, where the controls exist. On this subsample, the recursive zero restrictions no longer hold exactly, and the projection on the controls moves the instrument further away from them. The instrument therefore no longer forces the impact responses of output and prices to zero. In the estimates, they are small. Second, the monthly controls exist only in months with an FOMC announcement in the Bauer–Swanson sample. $Z_t^{\text{CHOL},\perp}$ is therefore zero in the other months, which carry the largest raw innovations of the window (Figure 10).

Both wider-window instruments keep the whole decision, at the cost of more contamination from news within their window. In exchange, they keep much of their first-stage strength once orthogonalised.

4.3 First-stage strength after orthogonalisation

Two questions follow. Do the wider-window instruments keep their first-stage strength after the orthogonalisation that removes predictability, where the high-frequency surprise does not? And why do the three instruments lose such different shares of it? All versions of an instrument are on the same months, and the raw wider-window instruments are re-estimated on the window of their orthogonalised versions, such that each comparison isolates the effect of the controls.

The pattern is the one Section 3 anticipates. Orthogonalisation weakens every instrument, but

by very different amounts (Table 1). The high-frequency surprise falls well below the conventional threshold for a strong first stage, while both wider-window instruments remain strong. The first-stage F varies with the policy variable and the way of orthogonalising, but the ordering of the instruments and the collapse of the orthogonalised surprise do not (Appendix Table 4).

All three instruments are predictable from the same controls. Predictability alone therefore does not explain the difference. By Proposition 2(i), the share of first-stage strength that the orthogonalisation removes, δ_π, is observable. Appendix B shows that it is the product of two observable numbers,

$$\delta_\pi = \sqrt{R^2}\,\frac{\hat{\rho}}{\rho}, \qquad \hat{\rho} := \mathrm{Corr}(\hat{z}_t, u_t^p), \quad \rho := \mathrm{Corr}(z_t, u_t^p), \tag{16}$$

where R^2 is the share of the instrument's variance that the orthogonalisation removes and $\hat{z}_t = z_t - z_t^\perp$ the predictable part. The first factor is how predictable the instrument is. The second, $\hat{\rho}/\rho$, is the correlation of the predictable part with the policy-rate innovation, relative to that of the instrument as a whole. The predictable part covaries with the policy-rate innovation only through the policy rate's response, within the month, to the news it carries. Predictable news to which the policy rate does not respond within the month lowers $\hat{\rho}/\rho$ (Appendix equation (33)).

Table 1: The loss of first-stage strength and the magnification bound. R^2: share of the monthly variance of the instrument that the orthogonalisation removes, $\mathrm{Var}(\hat{z}_t)/\mathrm{Var}(z_t)$ with $\hat{z}_t = z_t - z_t^\perp$. It exceeds the R^2 of the regressions in Appendix Table 3, which are at the frequency of the announcements and cover only the regression on the controls. δ_π: share of the first-stage covariance that the orthogonalisation removes (Proposition 2(i)). $\hat{\rho}/\rho = \delta_\pi/\sqrt{R^2}$, equation (16). First-stage F: HC0, each instrument alone for the funds-rate innovation in the baseline VAR, on the 383 months within 1988m1–2019m12 on which all its versions are observed. Orthogonalisation: PTA, PTA-FWL and ATP-FWL, as defined in Appendix Table 4. Magnification bound: $1/(1-\delta_\pi)$, valid when $\delta_d \geq 0$ (Proposition 2).

			Strength lost	First-stage F		Magnification
	R^2	$\hat{\rho}/\rho$	δ_π	raw	orthogonalised	bound
High-frequency surprise	0.288	1.00	0.54	14.5	3.6	2.15
Romer–Romer instrument	0.129	0.43	0.15	37.3	29.6	1.18
Cholesky-based instrument	0.344	0.61	0.36	966	534	1.55

Table 1 reports the two factors for each instrument. Were the predictable part of each instrument as correlated with the policy-rate innovation as the instrument itself, $\hat{\rho}/\rho = 1$, the three instruments would lose 54%, 36% and 59% of their first-stage strength. In practice, the surprise loses exactly that share, while the Romer–Romer and Cholesky-based instruments lose only 15% and 36%. The gap lies entirely in the second factor, $\hat{\rho}/\rho$.

The predictable part of the surprise moves with the policy rate as closely as the surprise itself, $\hat{\rho}/\rho = 1.00$. The surprise therefore loses as much strength as its predictability implies. The controls predict news the market had not yet priced, to which the Fed responds systematically. The Fed's systematic response moves the policy rate. Removing it takes more than half of the surprise's covariance with the policy-rate innovation, and three quarters of its first-stage F. More than half of the raw surprise's first stage therefore came from the contamination that orthogonalisation removes. The contamination that survives can then be magnified up to 2.15 times.

For the two wider-window instruments, $\hat{\rho}/\rho$ is well below one, 0.43 and 0.61. Their predictable parts are correlated with the policy-rate innovation less strongly than the instruments themselves. Their construction already removes most of the systematic response, with the Greenbook forecasts for the Romer–Romer instrument and with the month's output and price innovations for the Cholesky-based instrument. The controls may still predict news to which the policy rate does not respond within the month, which lowers $\hat{\rho}/\rho$. Orthogonalisation removes a much smaller share of their first-stage strength, and the magnification bound stays close to one for the Romer–Romer instrument and below 1.6 for the Cholesky-based instrument (Table 1). The loss of first-stage strength therefore depends less on how predictable an instrument is than on how its predictable component moves with the policy rate, the second factor in equation (16).

Keeping strength is not evidence that the orthogonalisation cleans an instrument. The controls end the day before the announcement, and they span only part of the news that precedes it. Orthogonalisation cannot remove contamination that arrives on the announcement day or later in the window, nor contamination the controls do not span, and it can move contamination onto shocks the instrument did not carry (Proposition 1). Strength instead limits how far the contamination that survives distorts the estimated impulse responses.

4.4 Strength limits the surviving bias

Neither wider-window instrument is clean. The Romer–Romer instrument carries any news that neither the Greenbook nor the controls span. The Cholesky-based instrument carries the financial exposure the controls do not span, and any news later in the month. Each is contaminated in a direction of its own, possibly more than the orthogonalised surprise. Both, however, keep their first-stage strength, far above that of the orthogonalised surprise (Table 1). By the bias identity in equation (7), that strength limits how much the contamination that survives distorts the impulse responses, and it is magnified far less.

Appendix Figures 11 and 12 report the impulse responses to the two instruments, raw and orthogonalised, and Appendix Figure 8 the corresponding figure for the surprise. Under either wider-window instrument, raw or orthogonalised, output falls gradually, by about a quarter of its decline under the orthogonalised surprise, and prices rise. This price puzzle is a well-documented symptom of contamination (Ramey, 2016), which orthogonalisation on pre-announcement controls cannot remove by construction.

Similar predictability need not imply a similar loss of strength. The high-frequency surprise and the Romer–Romer instrument are both predictable from the same controls, and their predictable parts identify nearly the same impulse responses. Output and prices rise, and credit spreads narrow. What differs is the share of each instrument's first stage that this predictable part carries, about half for the surprise and less than a sixth for the Romer–Romer instrument (Table 1). Orthogonalisation removes that share, δ_π, and magnifies the contamination that remains by $1/(1-\delta_\pi)$. It moves the impulse responses away from those identified with the predictable part, and the larger the share removed, the further they move. The move scales with $\delta_\pi/(1-\delta_\pi)$, which is 1.15 for the surprise and 0.18 for the Romer–Romer instrument (Appendix B.1). The output decline after two years under the surprise more than doubles. The size of the move therefore reflects, at least in part, the first-stage strength an instrument loses, and not only the contamination it sheds. The Cholesky-based instrument loses close to half of its first-stage F but remains strong, and its impulse responses barely move.

Neither orthogonalised instrument is clean, and the output decline is three to five times larger under the orthogonalised surprise than under the orthogonalised Romer–Romer instrument. This gap does not come from the news that orthogonalisation removes, which is similar for the two. It comes from the contamination that remains, which is not observed, and from how much each first stage magnifies it. By Proposition 2, the remaining contamination is magnified more for the orthogonalised surprise, which keeps less than half of its first-stage strength. The analysis therefore allows every instrument to remain contaminated.

The wider-window instruments rest on distinct information sets and windows, and their contamination differs. Section 6 exploits this structure. It groups instruments that share a contamination direction into families and removes what the members of each family share.

5 Identification from many instruments

Pooling the instruments raises the first-stage strength. A simple pool, however, does not make the instruments clean. No single instrument can be assumed valid, and the pool inherits the contamination of its members. Contamination that several instruments share does not average out as instruments are added. The estimator below therefore demeans before it pools. It groups instruments that share their exposure to non-policy shocks into families, removes the contamination each family shares, and only then combines them. Its target is the direction of the impact of the policy shock on the variables of the VAR, from which the causal impulse responses to that shock follow. The differences in relevance within families then identify the direction of the policy shock. The estimator is the leading eigenvector of the demeaned moments, and a single pooled instrument delivers its first stage.

5.1 The instrument panel

Each instrument considered so far may load on non-policy shocks. The instruments also differ in relevance for the policy shock, because they are built from different sources, windows and maturities. The estimator uses this dispersion in relevance rather than assuming it away, because relevance carries the information about b_p. Recall that the reduced form is the VAR,

$$y_t = c + \sum_{\ell=1}^{s} \Pi_\ell \, y_{t-\ell} + u_t, \qquad u_t = B\varepsilon_t, \qquad \mathbb{E}\left[\varepsilon_t \varepsilon_t'\right] = I_n, \qquad B = \left[\, b_p \;\; B_{-p} \,\right],$$

with unit-variance structural shocks and residual covariance matrix $\Sigma_u := \mathbb{E}[u_t u_t'] = BB'$. The estimator recovers the direction of b_p, and the impact vector $\beta = b_p / b_{pp}$ of equation (4) is obtained at the end by normalising the policy coordinate to one. The covariance of a single strong and valid instrument with the VAR innovations is proportional to the impact column b_p, and alone identifies the direction of b_p.

Suppose instead there are m candidate instruments, none of which is assumed to be valid. The direction of b_p must then be recovered from contaminated moments. Let each candidate instrument z_{jt} be

$$z_{jt} = \phi_j \, \varepsilon_t^{p} + \boldsymbol{\psi}_j' \boldsymbol{\varepsilon}_t^{-p} + \omega_{jt}, \qquad g_j := \mathrm{Cov}(u_t, z_{jt}) = \phi_j \, b_p + B_{-p} \boldsymbol{\psi}_j \in \mathbb{R}^n, \qquad j = 1, \ldots, m, \tag{17}$$

with relevance ϕ_j, contamination loading $\boldsymbol{\psi}_j \in \mathbb{R}^{n-1}$ and measurement noise ω_{jt} orthogonal to all

shocks. The identifying moment g_j is the covariance of the VAR innovations with instrument j. It has two components. One is the policy-shock impact vector scaled by relevance, $\phi_j b_p$. The other is the impact of the non-policy shocks the instrument loads on, $B_{-p}\boldsymbol{\psi}_j$. Stacking the moments into the matrix $G := [g_1, \ldots, g_m] \in \mathbb{R}^{n\times m}$, $\phi := (\phi_1, \ldots, \phi_m)'$ and $\Psi := [\boldsymbol{\psi}_1, \ldots, \boldsymbol{\psi}_m]' \in \mathbb{R}^{m\times(n-1)}$,

$$G = \underbrace{b_p \phi'}_{\text{relevance term}} + \underbrace{B_{-p}\Psi'}_{\text{contamination term}}. \tag{18}$$

Without a restriction on the contamination term, the moments cannot separate it from the relevance term (Section 5.4). The restriction used here is that instruments in the same family share their contamination.

5.2 Families and the within transform

A family is a group of instruments built similarly, from the same source and over the same window. Examples are high-frequency surprises at several maturities around the same announcements, or several orthogonalisations of the same narrative series. Instruments in a family are exposed to similar non-policy news. Therefore, they share the mean of their contamination, while their relevance to the shock of interest can differ within a family. The difference in relevance is what the estimator uses for identification.

The main assumption is that the instruments can be grouped into C such families. Instrument j belongs to family c, written $j \in c$, and $|c|$ is the size of the family.

Assumption 1 (Common within-family contamination)**.** *Instruments can be grouped into C families. Within family c,*

$$\boldsymbol{\psi}_j = \bar{\boldsymbol{\psi}}_c + \boldsymbol{\nu}_j, \qquad j \in c, \tag{19}$$

where the idiosyncratic contamination $\boldsymbol{\nu}_j$ satisfies two conditions: (a) Normalisation. $\mathbb{E}[\boldsymbol{\nu}_j \mid c] = \mathbf{0}$*, which defines $\bar{\boldsymbol{\psi}}_c$ as the family mean of the loadings. (b)* No relation to relevance. $\mathbb{E}[\boldsymbol{\nu}_j \mid \boldsymbol{\phi}_c, c] = \mathbb{E}[\boldsymbol{\nu}_j \mid c]$*, with $\boldsymbol{\phi}_c := \{\phi_k\}_{k\in c}$ the relevances of the family's members.*

Assumption 1 splits each instrument's contamination loading into two parts, a family-common part $\bar{\boldsymbol{\psi}}_c$, shared by all members of the family, and an idiosyncratic part $\boldsymbol{\nu}_j$. The common part contributes $\tau_c := B_{-p}\bar{\boldsymbol{\psi}}_c$ to the moment of every member, the family's common contamination. For example, high-frequency surprises measured over the same announcement window share the news in that window that the controls cannot span. Condition (a) separates the two parts. It sets the common part to the family mean of the loadings, and the idiosyncratic part to each member's deviation from that mean.

Condition (b) rules out a systematic link between relevance and idiosyncratic contamination within a family. On average, a more relevant member is not contaminated differently from a less relevant one.

Substituting the assumption, each moment is $g_j = \phi_j b_p + \tau_c + B_{-p}\boldsymbol{\nu}_j$ for $j \in c$. Demeaning within families, with $\tilde{g}_j := g_j - \bar{g}_c$, where $\bar{g}_c = |c|^{-1}\sum_{k\in c} g_k$,[8] and likewise $\tilde{\phi}_j$ and $\tilde{\boldsymbol{\nu}}_j$,

$$\tilde{g}_j = \tilde{\phi}_j b_p + B_{-p}\tilde{\boldsymbol{\nu}}_j. \tag{20}$$

[8] In the application, the family means weight each instrument by the precision of its moment (Section 6).

The common contamination τ_c cancels exactly within families. The demeaned moment keeps two terms, the policy-shock term $\tilde{\phi}_j b_p$ and the idiosyncratic contamination. The policy-shock term survives the within transform only if relevance differs within families. If all members of a family were equally relevant, demeaning would remove the policy-shock term along with the common contamination. The direction of b_p is the direction along which the demeaned moments move with the relevance deviations $\tilde{\phi}_j$. The total within-family dispersion of relevance adds up, over all instruments, the squared deviation of each instrument's relevance from its family's average,

$$\sigma_\phi^2 := \sum_{c=1}^{C} \sum_{j \in c} \tilde{\phi}_j^2 = \sum_c \sum_{j \in c} (\phi_j - \bar{\phi}_c)^2. \tag{21}$$

It measures the signal the target leaves in the demeaned moments and plays the role of a first stage for the direction. It grows with the number of families whose members differ in relevance. The direction is identified when this dispersion is positive and large relative to the idiosyncratic contamination.

Figure 3 is an illustrative example. It shows why the demeaning step is needed, with two families of six instruments and two variables. Panel (a) plots each instrument's moment, its covariance with the two VAR innovations. Each family lies along a line parallel to the true direction b_p, shifted away from the origin by its common contamination τ_c. Pooling the moments without demeaning mixes the two shifts with the target. The first principal component of the raw moments therefore points in neither direction, and a simple pool recovers the wrong direction. Panel (b) plots the same moments, on the same axes, after each family's mean has been subtracted. Demeaning subtracts each family's own mean, and both families now lie around the origin, along b_p. The first principal component of the demeaned moments is the true direction. It is recovered because relevance varies within each family. The moments of a family are dispersed along b_p, and that dispersion survives demeaning. Appendix D shows, in a simulated three-variable VAR, the impulse responses implied by each of these directions.

5.3 The estimator

The estimator, called the within estimator, works in two stages. It first whitens the moments. It then demeans them within families and finds the direction along which they vary most.

Whitening expresses each moment in terms of the structural shocks rather than the VAR innovations. The innovations are measured in the units of the variables, and they are correlated because each shock moves several variables at once. A covariance with the innovations therefore mixes the shocks. A non-policy shock can move the variables in proportions close to those of the policy shock, and its contamination then resembles relevance. Rescaling the innovations by the inverse square root of their covariance matrix, $\Sigma_u^{-1/2}$, removes both the units of the variables and the correlation between the innovations. Define the orthonormal matrix $\mathcal{Q} := \Sigma_u^{-1/2} B = [\, q_p \;\; \mathcal{Q}_{-p} \,]$, with q_p the whitened impact vector of the policy shock and $\mathcal{Q}_{-p}$ those of the non-policy shocks. Since $\Sigma_u = BB'$,

$$\mathcal{Q}'\mathcal{Q} = B'\Sigma_u^{-1}B = B'(BB')^{-1}B = I_n, \qquad \text{with} \qquad q_p'q_p = 1, \quad q_p'\mathcal{Q}_{-p} = \mathbf{0}'.$$

The whitened innovations, $\Sigma_u^{-1/2} u_t = \mathcal{Q}\varepsilon_t$, therefore have unit variance, are uncorrelated, and are

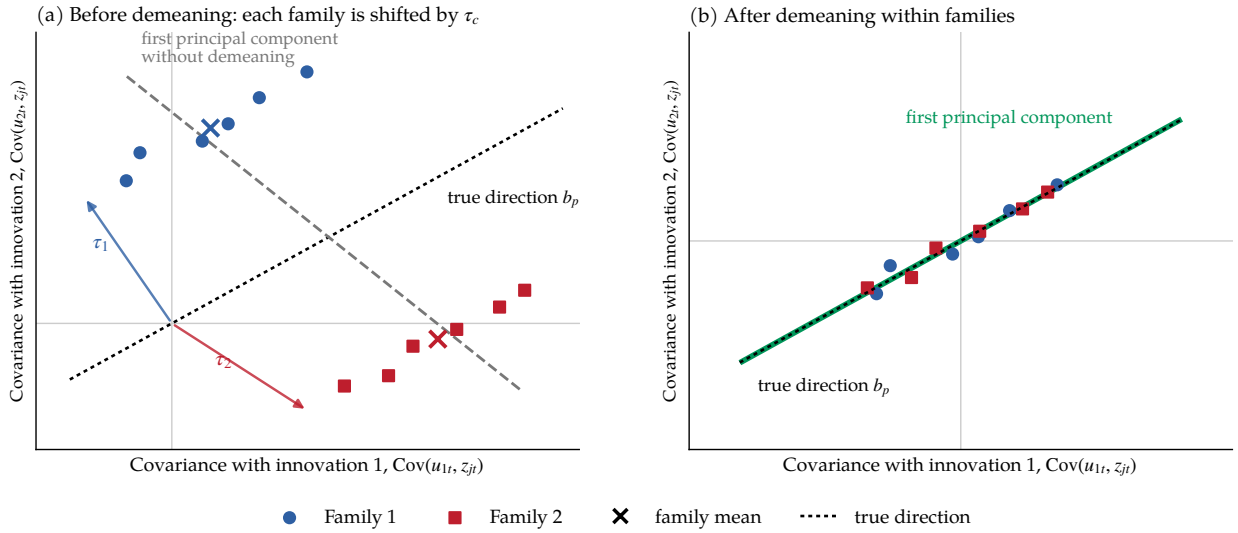


Figure 3: **Demeaning within families recovers the direction of the policy shock.** Simulated moments of two families of six instruments in a two-variable system, $g_j = \phi_j b_p + \tau_c + e_j$. Panel (a): the moments, with each family's mean (cross), the contamination τ_c that the family shares (arrows), the true direction b_p (dotted) and the first principal component of all the moments without demeaning (dashed). Panel (b): the same moments on the same axes, after each family's mean is subtracted, with their first principal component (solid) and the true direction (dotted). Relevance ϕ_j varies within each family. The idiosyncratic contamination e_j is drawn uncorrelated with relevance within each family, as Assumption 1(b) requires. The estimator also whitens the moments, a step the figure omits.

a rotation of the structural shocks. The whitened moment is the covariance of the instrument with them, $g_j^w := \mathrm{Cov}(\Sigma_u^{-1/2} u_t, z_{jt}) = \Sigma_u^{-1/2} g_j$. It collects the covariances of the instrument with the structural shocks, in rotated coordinates,

$$g_j^w = \phi_j q_p + \mathcal{Q}_{-p} \boldsymbol{\psi}_j. \tag{22}$$

The first term is the target direction q_p, the whitened impact of the policy shock, scaled by relevance. The second term is the contamination. The two terms are orthogonal, $q_p' \mathcal{Q}_{-p} = \mathbf{0}'$, and the size of each no longer depends on the units of the variables. After demeaning, the idiosyncratic contamination therefore adds noise to the estimated direction, but under condition (b) of Assumption 1 it does not shift the direction systematically towards the contamination.

The estimator is derived directly in these whitened coordinates. Let $\hat{g}_j^w := \hat{\Sigma}_u^{-1/2} \hat{g}_j$ denote the whitened estimated moments. By equation (22), they satisfy $\hat{g}_j^w = \phi_j q_p + \tau_c^w + e_j^w$ for $j \in c$, with $\tau_c^w := \mathcal{Q}_{-p} \bar{\boldsymbol{\psi}}_c$ the family's common contamination and e_j^w collecting the idiosyncratic contamination $\mathcal{Q}_{-p} \boldsymbol{\nu}_j$ and the sampling error. The estimator chooses the family effects, the relevances and the direction that bring this structure closest to the moments, by least squares:

$$Q(q, \{\phi_j\}, \{\tau_c^w\}) = \sum_{c=1}^{C} \sum_{j \in c} \left\| \hat{g}_j^w - \phi_j q - \tau_c^w \right\|^2. \tag{23}$$

Written in terms of the original moments, this is weighted least squares with weight $\hat{\Sigma}_u^{-1}$, the metric of generalised least squares.

For a given direction, the criterion is quadratic in the family effects $\{\tau_c^w\}$ and in the relevances $\{\phi_j\}$. Therefore, it can be minimised in three steps. The first two remove the family effects and the relevances in closed form. The last finds the direction as the leading eigenvector of a matrix.

Step 1: Remove the family contamination, for given $(q, \{\phi_j\})$. Only the terms of family c involve τ_c^w. The first-order condition for τ_c^w, with $\bar{g}_c^w$ and $\bar{\phi}_c$ the family means of $\hat{g}_j^w$ and ϕ_j, gives

$$\hat{\tau}_c^w(q, \phi) = \bar{g}_c^w - \bar{\phi}_c\, q, \qquad \text{hence} \qquad \hat{g}_j^w - \phi_j q - \hat{\tau}_c^w = \tilde{g}_j^w - \tilde{\phi}_j\, q, \tag{24}$$

with the within transform $\tilde{g}_j^w := \hat{g}_j^w - \bar{g}_c^w$. In the rest of this section, $\bar{g}_c^w$, $\tilde{g}_j^w$ and the matrix $\widetilde{G}^w$ below are computed from the estimated moments. Estimating the family effects removes each family's common contamination. However, it also removes the family's mean relevance $\bar{\phi}_c$, as the two enter the criterion only through the sum $\bar{\phi}_c q + \tau_c^w$. Only the relevance deviations $\tilde{\phi}_j = \phi_j - \bar{\phi}_c$ survive, and they identify the direction of the policy shock.

Step 2: Estimate the relevance deviations, for given q. Substituting equation (24) into equation (23) gives the within criterion,

$$Q^w(q, \{\tilde{\phi}_j\}) := \min_{\{\tau_c^w\}} Q = \sum_j \left\| \tilde{g}_j^w - \tilde{\phi}_j q \right\|^2,$$

which separates across j. For fixed q each term is a scalar least-squares problem. Its solution, $\hat{\tilde{\phi}}_j(q) = q' \tilde{g}_j^w / q'q$, is the projection coefficient of the demeaned moment on the candidate direction q. It is the relevance deviation that q implies for instrument j, its relevance relative to the average of its family. The minimised value of term j measures the part of $\tilde{g}_j^w$ that q does not explain. Summing these over instruments gives the concentrated criterion, a function of the candidate direction q alone,

$$Q^c(q) := \min_{\{\tilde{\phi}_j\}} Q^w = \sum_j \|\tilde{g}_j^w\|^2 - \frac{q' \widetilde{G}^w \widetilde{G}^{w\prime} q}{q'q}, \qquad \widetilde{G}^w := [\tilde{g}_1^w, \ldots, \tilde{g}_m^w]. \tag{25}$$

The first term does not depend on q. The second is a Rayleigh quotient, invariant to the scale and the sign of q. The estimator therefore identifies a direction only, and the normalisation of equation (4) sets its scale.

Step 3: Find the direction. Minimising Q^c is the same as maximising the Rayleigh quotient of $\hat{S} := \widetilde{G}^w \widetilde{G}^{w\prime}$, over $q \neq \mathbf{0}$. The maximum is the largest eigenvalue of $\hat{S}$, attained at the corresponding eigenvector. The eigenvector is unique up to sign when $\hat{\lambda}_1 > \hat{\lambda}_2$, the two largest eigenvalues of $\hat{S}$. Hence

$$\hat{q} = \arg\min_q Q^c(q) \quad \Longleftrightarrow \quad \hat{S}\,\hat{q} = \hat{\lambda}_1\,\hat{q}, \qquad Q^c(\hat{q}) = \sum_j \|\tilde{g}_j^w\|^2 - \hat{\lambda}_1. \tag{26}$$

The estimate $\hat{q}$ is the first principal component of the whitened demeaned moments. Undoing the whitening gives the impact vector in the original units, $\hat{b}_p = \hat{\Sigma}_u^{1/2} \hat{q}$, which is then normalised such that the policy rate rises by one unit, $\hat{\beta} = \hat{b}_p / (s_p' \hat{b}_p)$.

Substituting $\hat{q}$ back into the projection coefficient of Step 2 and into equation (24) returns the fitted relevance deviations $\hat{\tilde{\phi}}_j$ and family effects $\hat{\tau}_c^w$. Therefore, the three steps deliver every parameter of equation (23). Details are in Appendix C.2. The ratio of the two largest eigenvalues,

the eigengap $\hat{\lambda}_1/\hat{\lambda}_2$, measures how much more the demeaned moments vary along $\hat{q}$ than along any other axis. It describes how sharply the data determine the direction of the policy shock.

5.4 Identification

The estimator returns a direction, the proportions in which the policy shock moves the variables of the VAR on impact. Its target is the direction of the true impact vector b_p, all its entries up to a common scale. Whitening rescales b_p one-to-one, $b_p = \Sigma_u^{1/2} q_p$. The results below are stated for b_p.

Two questions decide whether the estimated direction is the target. The first is identification. With the moments known exactly, and the restriction of Assumption 1 holding as stated, do they determine b_p for some estimator? The second is recovery. With the finite panel of instruments at hand, does the leading eigenvector of the demeaned moments land on b_p? Identification asks what the moments and the restriction reveal. Recovery asks how this estimator performs in a given panel of instruments.

Which restriction identifies the target. The moments cannot separate the policy shock's impact from contamination that grows with relevance. Each moment is the sum of a relevance term and a contamination term, equation (18). Consider a shift vector $\alpha \in \mathbb{R}^{n-1}$, with one entry per non-policy shock, that changes the two terms by offsetting amounts. It changes the policy shock's impact from b_p to $b_p + B_{-p}\alpha$, a mix of its own impact and the impact of the non-policy shocks. It also reduces each instrument's contamination loading by $\phi_j \alpha$, in proportion to its relevance. In the moment of instrument j, the policy term gains $\phi_j B_{-p}\alpha$ and the contamination term loses the same amount. The two changes offset each other, and every instrument keeps the same moment (Lemma C.1). This relevance-proportional shift is the obstacle to identification. Each α gives a description of the data that fits the moments equally well, and the true description has $\alpha = \mathbf{0}$. Economically, suppose that the instruments that move the policy rate more also carry more non-policy news. Their contamination then grows with their relevance, and the effect of that news is attributed to the policy shock.

Demeaning removes only part of the shift α. The relevance can be written as $\phi_j = \bar{\phi}_c + \tilde{\phi}_j$. The shifted component $-\bar{\phi}_c \alpha$ is common to the family. It enters τ_c, and demeaning removes it. The component $-\tilde{\phi}_j \alpha$ survives demeaning. It is idiosyncratic contamination that moves with the relevance deviation. Condition (b) of Assumption 1 excludes it.

Proposition 3 (Identification)**.** *Let relevance vary within at least one family, $\sigma_\phi^2 > 0$ in equation (21).* (i) *Under Assumption 1, $\alpha = \mathbf{0}$ is the only relevance-proportional shift consistent with the restriction, and b_p is identified up to scale.* (ii) *Without condition (b), b_p is not identified. The shift α adds $-\tilde{\phi}_j \alpha$ to each idiosyncratic contamination, a term that moves with relevance. Every α then gives a description that fits the moments and satisfies condition (a) at any m.*

Details are in Appendix C.3. Part (i) shows that condition (b) identifies the target direction, and part (ii) shows why it cannot be dropped. A failure of condition (b) common to all families also fits the moments exactly (Lemma C.1). Identification therefore rests on the assumption that, within a family, contamination does not move with relevance.

Recovery in a finite panel. Condition (b) is a statement about expectations. It requires the idiosyncratic contamination to be unrelated to relevance on average, across panels of instruments

that could be drawn. Proposition 3 relies on this average. A given panel has a finite number of instruments, and their idiosyncratic contamination may not average out exactly. Recovery asks whether, in a given panel, the leading eigenvector of the demeaned moments still recovers b_p. The benchmark is exact block structure, with no idiosyncratic contamination, $\boldsymbol{\nu}_j = \mathbf{0}$ for every instrument. Every demeaned moment then lies on the line through b_p, and the leading eigenvector is b_p with any number of instruments.

Within a family, the estimator compares instruments that carry the same common contamination but move the policy rate by different amounts. The difference between their moments reflects the policy shock, scaled by the difference in their relevance. With idiosyncratic contamination, it also reflects the difference in what each instrument carries on its own. This difference matters in two ways. Where it happens to line up with the relevance deviations, it pulls the estimate away from b_p. Where it is widely dispersed, it creates rival directions that can replace b_p as the leading one. The matrix $N := \sum_j \tilde{\boldsymbol{\nu}}_j \tilde{\boldsymbol{\nu}}_j'$, of size $(n-1) \times (n-1)$, measures this dispersion in the panel, with $\tilde{\boldsymbol{\nu}}_j$ the idiosyncratic contamination demeaned within its family. Its largest eigenvalue, $\lambda_{\max}(N)$, is the dispersion along the direction in which it is largest. The ordering condition requires the dispersion of relevance to exceed it,

$$\underbrace{\sigma_\phi^2}_{\text{dispersion of relevance}} \quad > \quad \underbrace{\lambda_{\max}(N)}_{\text{largest dispersion of the idiosyncratic contamination}} . \tag{27}$$

Proposition 4 first lets the number of instruments grow, then turns to the finite panel at hand.

Proposition 4 (Recovery)**.** *Let relevance vary within at least one family, $\sigma_\phi^2 > 0$.*

(i) *Under Assumption 1, and with idiosyncratic contamination independent across instruments (Appendix C.4), as the number of instruments grows, b_p is the leading eigenvector if and only if the ordering condition, equation (27), holds in the limit. When it fails, the estimator returns a contamination direction.*

(ii) *Under the same conditions, in a finite panel, if the ordering condition holds, the leading eigenvector is close to b_p. Its deviation from b_p has mean zero over panels and shrinks as instruments are added.*

Details are in Appendix C.4. Part (i) gives the condition under which the estimator picks b_p rather than a contamination direction. Within families, the instruments must differ more in their relevance than in their idiosyncratic contamination. Part (ii) applies to the panel at hand. Its idiosyncratic contamination can happen to be larger for the more relevant instruments, and the estimate then moves toward the contamination. Condition (b) of Assumption 1 makes this deviation zero on average.

Hence, identification of the target rests on two conditions, common within-family contamination (Assumption 1) and relevance that varies within families ($\sigma_\phi^2 > 0$). Recovery of the target by the leading eigenvector in a finite panel adds a third, the ordering condition. Relevance that varies within families runs through all the results. It is the premise of Propositions 3 and 4, and partition condition (P1) of Proposition 5. The ordering condition strengthens it, from $\sigma_\phi^2 > 0$ to $\sigma_\phi^2 > \lambda_{\max}(N)$.

5.5 The family partition

The estimator needs a partition of the instruments into families, with enough instruments and enough families. Under Assumption 1, demeaning removes each family's common contamina-

tion for any partition. Each family spends one instrument on it, and only the remaining $m - C$ instruments carry information about the direction. Identification requires relevance to vary within families, $\sigma_\phi^2 > 0$. This needs at least one family with two instruments of unequal relevance, and hence $m - C \geq 1$. This is partition condition (P1). Testing the family structure needs one more instrument, $m - C \geq 2$.

Demeaning also discards each family's average relevance. Partition condition (P2) counts the families needed to use the family averages without bringing the common contamination back. A combination of family averages adds strength without bias only if the common contaminations of the families cancel in it. These contaminations lie in at most $n - 1$ directions, one for each non-policy shock. A nonzero combination of them cancels only if there are more families than the directions they occupy.

Proposition 5 (Conditions on the partition of instruments)**.** *Under Assumption 1, the partition must meet two conditions.*
Partition condition 1 (P1). Relevance varies within a family. *At least one family contains two instruments of unequal relevance, $\sigma_\phi^2 > 0$. Under Assumption 1, condition (P1) identifies the direction. It implies $m - C \geq 1$. When condition (P1) fails, the direction is not identified. In addition, if $m - C \geq 2$, the family structure places $(n-1)(m-C-1) > 0$ overidentifying restrictions on the moments.*
Partition condition 2 (P2). Enough families to use the levels. *Let*

$$r := \operatorname{rank}[\bar{\boldsymbol{\psi}}_1, \ldots, \bar{\boldsymbol{\psi}}_C] \;\leq\; \min(C, n-1) \tag{28}$$

be the number of distinct directions of the family contaminations. Condition (P2) is $C > r$. Since $r \leq n - 1$, a sufficient condition is

$$C \;>\; n - 1.$$

When condition (P2) fails, the direction is still identified, and the estimate remains free of the common contamination. Only the strength of the strongest cancelling combination falls, to the part carried by the relevance deviations, which is proportional to σ_ϕ^2.

Condition (P1) governs the direction, the condition $m - C \geq 2$ the overidentification test, and condition (P2) the strength. Condition (P2) also becomes harder to meet as the VAR grows. Each added variable adds a non-policy shock, and hence one more direction the family contaminations may occupy, while the number of families is fixed by the instruments available. It is assessed on the family-average contamination of the C families, estimated from the family means of the moments.

The partition is chosen on economic grounds, before the moments are formed. Families follow the construction of the instruments. Instruments measured at a different point in time are exposed to different news and form families of their own. With a long enough sample, the partition can also be estimated from the data. Once the target direction is removed, instruments of the same family cluster around their family's common contamination. An algorithm searches over these clusters, chooses the number of families and assigns each instrument to one (Online Appendix OA.3.1). In simulated samples, it recovers the number of families once the sample is long enough, and the assignment of every instrument under exact block structure. Details are in Online Appendix Section OA.4.

Testing the partition. Two tests evaluate the partition, in order. The first asks whether the partition carries a direction at all. The second asks whether the restrictions of the family structure

hold.

A common direction. Identification requires relevance to vary within families, partition condition (P1). The test of no common direction takes $H_0^{\mathrm{d}} : \sigma_\phi^2 = 0$ as its null hypothesis. The demeaned moments then contain only sampling noise, and the estimated direction is an arbitrary axis. The statistic measures how much the demeaned moments vary along the estimated direction. It is the largest eigenvalue of the weighted sum of their outer products. Its distribution under the null hypothesis is obtained by drawing moments with no common direction from their estimated sampling covariance. A rejection means that the partition carries a direction. Details are in Appendix C.5.

Exact block structure. Under exact block structure, the idiosyncratic contamination is zero. Once each family's common contamination is removed, every instrument points to the same direction. The number of restrictions this places on the moments follows from counting. The moments supply nm moment conditions, n for each instrument. The unknowns are the direction of b_p, with $n-1$ free entries once its scale is fixed, the m relevances, and the C family contaminations τ_c. Each τ_c is a combination of the $n-1$ columns of B_{-p}, and hence also has $n-1$ free entries. The moments therefore exceed the unknowns by

$$nm - (n-1) - m - (n-1)C = (n-1)(m-C-1).$$

Each family spends one instrument on its common contamination, and one more instrument is spent on the direction. Each remaining instrument supplies $n-1$ overidentifying restrictions, one for each direction orthogonal to the target.

The overidentification test evaluates these restrictions. Its null hypothesis is exact block structure, $H_0^{\mathrm{o}} : \boldsymbol{\nu}_j = \mathbf{0}$ for every j. Once each family's common contamination is removed, do the moments of all instruments line up along one direction and differ only in their relevance? Let $\hat{X} := \hat{G}^{w\prime}(I_n - \hat{q}\hat{q}')$, with $\hat{G}^w := [\hat{g}_1^w, \ldots, \hat{g}_m^w]$, collect the whitened moments of the m instruments, one row per instrument, with the estimated direction removed. The test fits them by generalised least squares on a family effect and a common multiple of the estimated relevances $\hat{\phi}$, and adds up the weighted squared residuals,

$$J_T = \min_{\Gamma} \operatorname{tr}\left[\left(\hat{X} - K_R\Gamma\right)' \hat{V}^{-1}\left(\hat{X} - K_R\Gamma\right)\right], \tag{29}$$

with $K_R := [\iota_1, \ldots, \iota_C, \hat{\phi}]$ the indicators of the families' members and the estimated relevances, Γ the family effects and the common multiple, and $\hat{V}$ the sampling covariance of the moments. The covariance keeps its off-diagonal elements, since instruments with a common construction are correlated. Under H_0^{o}, J_T is asymptotically χ^2 with $(n-1)(m-C-1)$ degrees of freedom. Details are in Appendix C.5. A rejection means that, once each family's common contamination is removed, the moments do not line up along one direction. The null hypothesis is stronger than Assumption 1, which allows idiosyncratic contamination unrelated to relevance. The test can therefore reject while the direction is identified. The test is informative once the first test rejects, since its limit requires the direction to be identified. Section 6 reports both tests.

5.6 Pooling across families

The within estimator identifies the direction from the within-family moments. Pooling reaches the same direction another way. It combines the instruments into a single instrument whose

contamination cancels. This pooled instrument gives the same impulse responses as the within estimator. It also delivers an instrument series. Therefore, its first-stage strength can be reported.

How the instruments are combined matters. When instruments share contamination within families, a simple average does not identify the direction. It adds strength, but it keeps the common contamination of each family in full, however many instruments it averages (Appendix C.6). Panel (a) of Figure 3 illustrates the consequence. Without demeaning, the leading direction of the moments differs from the target. Constrained pooling chooses the weights to remove the common contamination.

Constrained pooling. Constrained pooling seeks the strongest combination of the instruments whose contamination is zero. A combination $w'z_t = \sum_j w_j z_{jt}$ inherits the three parts of its members, equation (17), each combined with the same weights,

$$w'z_t = \underbrace{(w'\phi)}_{\text{relevance}}\,\varepsilon_t^p + \underbrace{(\Psi'w)'}_{\text{contamination}}\,\boldsymbol{\varepsilon}_t^{-p} + \underbrace{w'\omega_t}_{\text{noise}}, \qquad \operatorname{Var}(w'\omega_t) = w'\Sigma_\omega w.$$

First-stage strength is measured by the concentration parameter $\mu^2(w) = T(w'\phi)^2/(w'\Sigma_\omega w)$, relevance squared over noise. For a cancelling combination, its population first-stage F increases with $\mu^2(w)$.[9] The program maximises concentration subject to cancelling its contamination,

$$w^\star = \arg\max_w \frac{(w'\phi)^2}{w'\Sigma_\omega w} \quad \text{s.t.} \quad \Psi'w = \mathbf{0}. \tag{30}$$

The constraint restricts the contamination of the combination to zero. Column k of Ψ gives each instrument's loading on non-policy shock k. Therefore, $\Psi'w$ is the loading of the combination on each shock. The weights that satisfy the constraint form the *kernel* of Ψ', $\ker(\Psi') := \{w \in \mathbb{R}^m : \Psi'w = \mathbf{0}\}$, the set of cancelling combinations. Under exact block structure, every member of family c carries the family's contamination $\bar{\boldsymbol{\psi}}_c$. The contamination of a combination is then $\sum_c a_c \bar{\boldsymbol{\psi}}_c$, with $a_c := \sum_{j\in c} w_j$ the total weight on family c. A combination that puts weight on the family levels cancels this sum only if the family contaminations offset each other, which requires condition (P2). Under Assumption 1, the same combinations cancel the common contamination and leave the idiosyncratic contamination.

With homoskedastic independent noise, $\Sigma_\omega = \sigma_\omega^2 I$, and the normalisation $w'\phi = 1$, the solution is

$$w^\star = \frac{P_\perp \phi}{\|P_\perp \phi\|^2}, \qquad \mu^2(w^\star) = T\,\|P_\perp \phi\|^2/\sigma_\omega^2, \tag{31}$$

where $P_\perp := I - \Psi(\Psi'\Psi)^+\Psi'$, and $\mu^2(w^\star)$ defines the strength of the strongest cancelling combination. Appendix C.7 derives the solution for a general noise covariance Σ_ω.

Without the constraint, the program would weight each instrument by its relevance, $w \propto \phi$. The strengths of the instruments would then add, $\mu^2 = \sum_j \mu_j^2$ with independent noise (Appendix C.8), but common contamination would remain. This pool leans on the most relevant instruments and inherits their contamination. The constrained weights $P_\perp \phi$ instead keep only the relevance that the

[9] For a cancelling combination, $F(w) = T\kappa r(w)/\big(1 + (1-\kappa)r(w)\big)$, with $r(w) := \mu^2(w)/T$ and κ the share of the variance of the policy-rate innovation that the policy shock explains.

contamination loadings do not explain. Constrained pooling therefore adds first-stage strength while cancelling the common contamination.

In practice, Ψ is unobserved. Once the direction is estimated, the residual of each whitened moment, $\hat{g}_j^w - (\hat{q}'\hat{g}_j^w)\,\hat{q}$, provides an estimate of the contamination of instrument j. The feasible weights are the estimated relevances with their projection on these residuals removed, giving the pooled instrument $z_t^\star := \sum_j \hat{w}_j z_{jt}$. Since the feasible weights cancel the estimated contamination, the moments of the pooled instrument point exactly along $\hat{q}$. Its impact vector is therefore the within direction. It gives the same impulse responses as the within estimator, the minimum-distance estimator of equation (23). The pooled instrument adds an instrument series and its first stage. Details are in Appendix C.1.

6 Empirical application to high-frequency monetary surprises

This section applies the within estimator to U.S. monetary policy. Each instrument is orthogonalised on pre-announcement controls, grouped into families, demeaned within families and pooled. The panel holds twelve U.S. monetary policy instruments in four families of three. The pooled instrument is strong. Its first stage is above the conventional threshold for a single instrument (Stock and Yogo, 2005). The main difference from the orthogonalised surprise is in prices. The orthogonalised surprise implies an immediate and significant fall in prices, which is hard to reconcile with sticky prices. Under the pooled instrument, the fall in prices on impact shrinks and is no longer significantly different from zero. The impulse responses of output and the credit spread are also smaller in magnitude. The estimated effects of monetary policy are therefore attenuated, and the confidence bands are narrower.

6.1 Data, the instrument panel, and the partition

The baseline model is a monthly VAR with six lags and a constant,

$$y_t = c + \sum_{\ell=1}^{6} \Pi_\ell\, y_{t-\ell} + u_t, \qquad \mathbb{E}[u_t u_t'] = \Sigma_u, \qquad u_t = B\varepsilon_t,$$

with $y_t = (\mathrm{IP}_t, \mathrm{CPI}_t, \mathrm{EBP}_t, i_t)'$: the logs of industrial production and of the consumer price index, the excess bond premium of Gilchrist and Zakrajšek (2012), and the federal funds rate as policy variable. The VAR is estimated from January 1973 to December 2019. The identifying moments $\hat{g}_j$ are formed on each instrument's own support between January 1988 and December 2019, the period over which the Bauer and Swanson (2023b) controls are available.

Impulse responses are normalised to raise the federal funds rate by one percentage point on impact, and are reported over 36 months. First-stage strength is the heteroskedasticity-robust F of the regression of the policy-rate innovation on the instrument. The estimates use the algorithm of Appendix C.1 with precision weights.[10] Confidence bands are 68% median-recentred percentile intervals from a residual moving-block bootstrap (Jentsch and Lunsford, 2022) with 1,000 draws (Online Appendix equation (OA.6)). Each draw resamples the VAR residuals and the instruments

[10] Each instrument is weighted by $1/\hat{v}_j$, the sampling variance of its moment, and contributes on its own support. The weights make the estimate invariant to the units of each family, and they give more influence to the moments measured precisely.

jointly and rebuilds the data from the estimated VAR. The VAR coefficients, Σ_u, the moments, the precision weights and the direction are then re-estimated. Only the partition is held fixed.

Instrument families. The baseline panel holds twelve instruments (Appendix Table 5) in four families of three. Every instrument is orthogonalised on the six Bauer and Swanson (2023b) controls, at the announcement level and then summed within the month (PTA), jointly with the instrument's own first stage (PTA-FWL), or after monthly aggregation (ATP), alone or jointly with the instrument's own first stage (ATP-FWL).

- *High-frequency surprises.* The changes in the three-month-ahead federal funds future and in the four-quarter-ahead Eurodollar future around FOMC announcements (FF4 and ED4, from the Acosta et al. (2025) database, starting in February 1994), and the Bauer and Swanson (2023b) surprise, each orthogonalised at the announcement level (PTA).
- *Swanson factors.* Swanson (2021) extracts three factors from the high-frequency responses of futures and Treasury yields to FOMC announcements and rotates them to separate a change in the federal funds rate, forward guidance and asset purchases. The family uses the factors of Swanson (2024), each orthogonalised at the announcement level (PTA).
- *Romer–Romer.* The Romer and Romer (2004) instrument (Section 4.1), in the same three orthogonalisations.
- *Cholesky-based instruments.* The Cholesky innovations of Section 4.2 for the federal funds rate and the one- and two-year Treasury yields, orthogonalised after monthly aggregation (ATP-FWL).

The assignment is economic. Each family shares a source, a window and the controls its members are orthogonalised on. The controls end the day before each announcement and span only part of the news that precedes it. Orthogonalisation therefore leaves the contamination that arrives on the announcement day or later in the window, and the contamination the controls do not span. After orthogonalisation, a family shares the news inside its window. The window is the announcement window for the high-frequency and Swanson families, the whole decision on the day of the meeting for the Romer–Romer family, and the month for the Cholesky-based instruments. Its members differ in relevance through their maturity, their factor, or the way they are orthogonalised. In addition, since each member is orthogonalised on the same controls, the pooled instrument is also unpredictable from the spanned variation (Appendix Table 3).

6.2 The baseline impulse responses

Table 2 compares the baseline estimate, the impulse responses identified by the pooled instrument, with those of the orthogonalised surprise alone and of the raw surprise, on the five features of Section 2.2. Pooling moderates each of them (Appendix Figure 9).

The first stage rises to $F = 32.6$, nine times that of the orthogonalised surprise, and the confidence bands narrow by more than half. Output falls by 0.5% on impact. Its peak decline, 4.0% after fifteen months, lies between those of the raw surprise (1.9%) and the orthogonalised one (6.6%). Prices fall gradually, by 0.9% after two years, and their fall on impact is no longer significant. The excess bond premium rises by 0.9 points on impact, less than half its rise under the orthogonalised surprise, and the funds rate turns negative later and less deeply.

Table 2: The five features of Section 2.2. Raw and Orthogonalised (PTA): the Bauer and Swanson (2023b) surprise, raw and orthogonalised announcement by announcement; Pooled baseline estimate. Impulse responses to a +1 pp funds-rate shock; confidence bands are 68% moving-block bootstrap intervals.

		Raw	Orthogonalised (PTA)	Pooled
(i) First stage	F, funds rate	14.5	3.6	32.6
(ii) Precision	width of the 68% confidence band, output at its peak decline	2.5	10.6	4.3
	width of the 68% confidence band, prices at 24 months	1.2	6.1	2.6
(iii) Magnitude	peak output decline, % (month)	−1.9 (20)	−6.6 (13)	−4.0 (15)
	credit spread on impact, pp	+0.25	+1.97	+0.88
(iv) Prices on impact	price level in the month of the shock, %	−0.15	−0.83	−0.22
(v) Funds-rate path	first month below zero	–	6	9
	lowest value, pp (month)	+0.17 (36)	−1.46 (17)	−0.39 (20)

Pooling moderates the five features of the orthogonalised surprise for the reason of Section 3. The orthogonalised surprise has a weak first stage, and its impulse responses magnify any contamination that survives the orthogonalisation. The pooled instrument is nine times stronger, and the contamination that survives in it is magnified far less. The impact responses show the difference most, as the fall in prices and the jump in the credit spread both shrink. The credit spread still rises on impact. A monetary tightening lowers the value of borrowers' collateral and of lenders' capital, and the premium that lenders demand rises (Gertler and Karadi, 2015). The smaller rise in the premium is consistent with the milder reversal of the funds rate. The Fed eases in response to tighter credit conditions (Caldara and Herbst, 2019). The funds rate crosses zero at nine months, after the premium has peaked on impact and before output falls most, at fifteen months.

These estimates reflect what each family brings to the pool. Strength comes from the level of relevance, and identification from its dispersion within families. A family can supply much of one and little of the other. Most instruments are weak on their own.[11] The Cholesky-based instruments are by far the strongest, as they are functions of the VAR's own innovations, and the Romer–Romer instruments are moderately strong (left panel of Appendix Figure 13). In the pooled instrument, the Cholesky-based and Romer–Romer families each supply a third of the first stage, and the Swanson family only an eighth.

A family's share of the identification is its contribution to the variation that identifies the shock. The Swanson family supplies 71% of it, against 28% for the Cholesky-based family (right panel of Appendix Figure 13). Its three factors separate the surprise in the funds rate from forward guidance and asset purchases. These components move the funds rate by very different amounts, and this dispersion in relevance identifies the shock. Relevance varies little within the Romer–Romer family, which holds three highly correlated orthogonalisations of one series, and within the high-frequency family, whose surprises move the funds rate by similar amounts. On its own, the Swanson family identifies a sharp contraction, and the Cholesky-based family a mild one with higher prices. The baseline lies between the two.

[11] Five of the twelve instruments have a first stage below 10 on their own. Pooling keeps all instruments, as weak ones can still provide identifying variation. In the pool, a weak member enters a single instrument with a small weight. It therefore does not create the many-weak-instruments bias of two-stage least squares (Staiger and Stock, 1997). Dropping the weak members would remove the forward-guidance and asset-purchase factors of the Swanson family, and with them most of the dispersion in relevance that identifies the shock (Appendix Table 6).

Diagnostics. The two tests of Section 5.5 evaluate the partition, in order. The test of no common direction rejects its null hypothesis. Under the null, the demeaned moments vary along the estimated direction as much as in the data in only 0.8% of draws ($p = 0.008$). Relevance therefore varies within families, and the partition carries a direction. The overidentification test then does not reject exact block structure, with $J_T = 14.5$ on $(n-1)(m-C-1) = 21$ degrees of freedom ($p = 0.85$). The data are therefore consistent with instruments that, within a family, differ only in how strongly they move the policy rate, and with families that point to the same policy shock. The ordering condition, equation (27), also holds. The within-family dispersion of relevance is 22.6, against 1.3 for the largest dispersion of the idiosyncratic contamination (Online Appendix OA.2.3). The leading eigenvector therefore recovers the policy shock rather than a contamination direction. Together, the three results support the choice of families. The partition carries a direction, its restrictions are not rejected, and the idiosyncratic contamination is too small to tilt the estimate. None of them can detect a relevance-correlated component common to all families (Lemma C.1).

6.3 Why pooling needs demeaning

Pooling alone is not enough. A pool without demeaning weights the instruments by their strength (Section 5.6). It leans on the strongest instruments and inherits their contamination. The baseline therefore combines two steps, and each accomplishes a different task. Pooling supplies strength, and strength narrows the confidence bands. Demeaning removes contamination. A pool that is strong but not demeaned therefore gives narrow confidence bands around biased impulse responses.

Pooling the twelve instruments without demeaning illustrates this risk (Appendix Table 6 and Figure 4). The pooled instrument is very strong ($F = 146.9$), but it leans on its strongest family, the Cholesky-based instruments, and inherits their contamination. Output falls by only 2.0% at its peak decline, and the funds rate never turns negative. These impulse responses are close to those the Cholesky-based family identifies on its own (Appendix Figure 12). Demeaning costs most of the strength, down to $F = 32.6$, and the peak decline in output doubles, to 4.0%. The contamination removed is the news inside each family's window, which orthogonalisation on pre-announcement controls cannot reach.

6.4 What demeaning removes

Demeaning removes each family's contamination from the pooled estimate. Comparing each family with the pooled estimate illustrates how differently that contamination would move each impulse response, family by family and variable by variable.

It does so by subtracting each family's mean moment from its members. Under Assumption 1, the mean moment of family c is $\bar{g}_c = \bar{\phi}_c\, b_p + \bar{\tau}_c$, the family's mean relevance times the impact of the policy shock, and its mean contamination $\bar{\tau}_c$, the common contamination τ_c together with the mean idiosyncratic contamination of its members. To isolate what demeaning removes, each family's average instrument $\bar{z}_{ct}$, the precision-weighted mean of its members' instruments, is used as the only instrument, and its impulse responses are compared with the pooled estimate. The estimated impact vector is $\tilde{\beta}_c := \hat{\bar{g}}_c / \hat{\pi}_c$, with $\hat{\bar{g}}_c$ the sample mean moment and $\hat{\pi}_c$ its funds-rate coordinate,

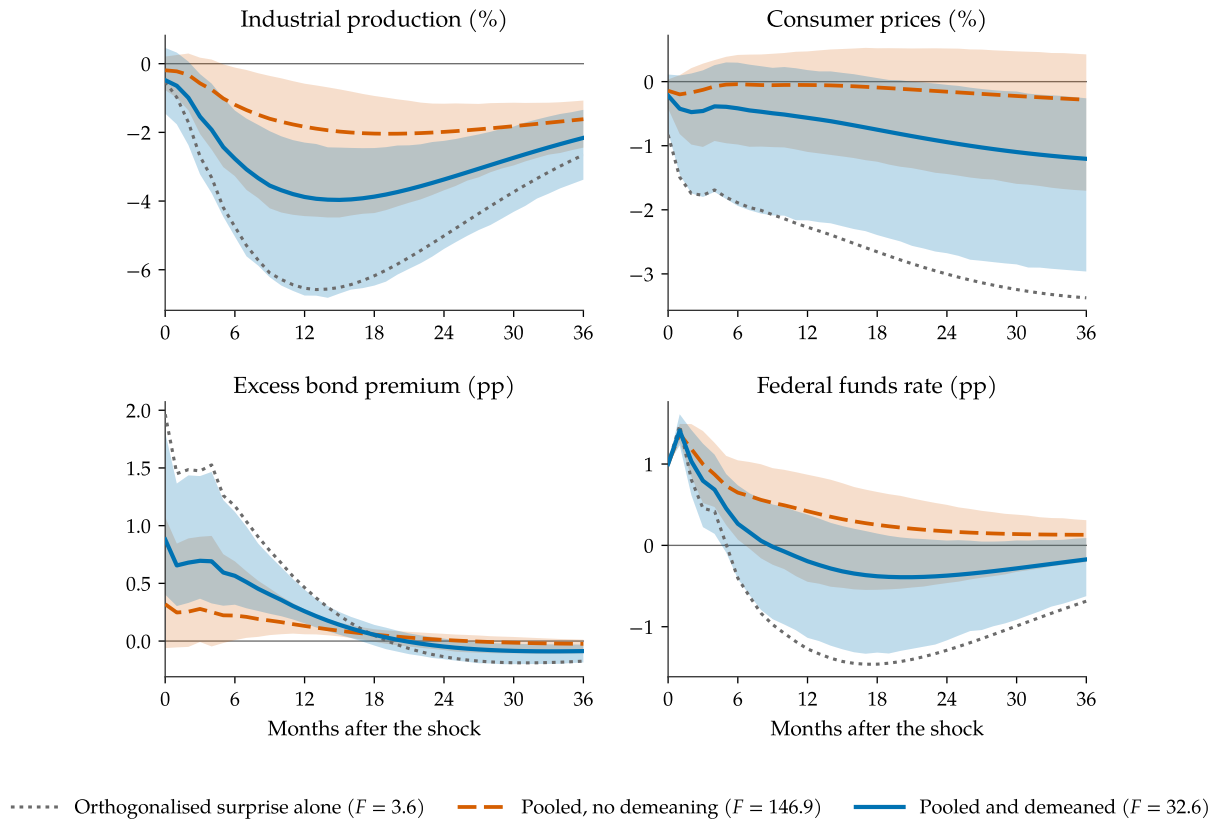


Figure 4: **What each step contributes.** The orthogonalised Bauer and Swanson (2023b) surprise alone (dotted, $F = 3.6$), the twelve instruments pooled without demeaning (dashed, $F = 146.9$), and the baseline (solid, $F = 32.6$). 68% confidence bands.

the family's first stage. Relative to the pooled estimate $\hat{\beta}$,

$$\tilde{\beta}_c - \hat{\beta} = \frac{\hat{\tilde{g}}_c - \hat{\pi}_c \hat{\beta}}{\hat{\pi}_c} \xrightarrow{p} \frac{\bar{\tau}_c - \bar{\tau}_{c,p}\beta}{\bar{\phi}_c^p + \bar{\tau}_{c,p}}, \tag{32}$$

when the pooled estimate is consistent for β, with $\bar{\phi}_c^p$ the family's mean relevance in units of the policy rate, the movement of the funds rate that the average instrument captures through the policy shock, and $\bar{\tau}_{c,p}$ the funds-rate coordinate of $\bar{\tau}_c$. The equality holds exactly in sample. Equation (32) is equation (7) applied to the family's average instrument. The numerator is the family's contamination direction, its contamination net of the policy response that the contamination induces. For each variable, it measures how much the news shared by the family moves the variable beyond the effect implied by its movement of the funds rate. The denominator is the family's first stage. Figure 5 plots the shift $\tilde{\beta}_c - \hat{\beta}$, mapped into impulse responses on impact and after twelve months. Each bar is measured relative to the pooled estimate, not to the true impulse responses. Reading it as a bias relies on the pooled estimate being consistent.

The families fall into two groups (Figure 5). The two wide-window families, Romer–Romer

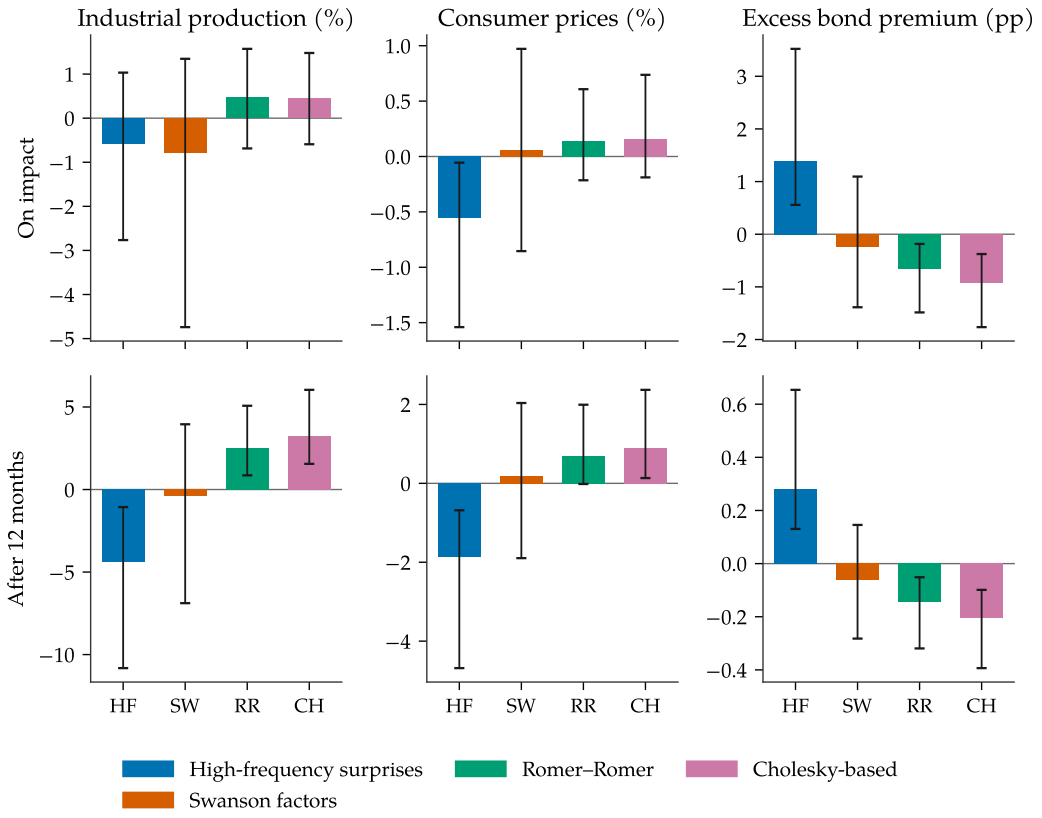


Figure 5: **The family-common component removed by demeaning.** For each family, the impulse response to a $+1$ percentage-point funds-rate shock identified with the family's average instrument as the only instrument, minus the pooled response, the sample equality in equation (32). Top row: on impact; bottom row: after twelve months. Whiskers: 68% median-recentred percentile intervals, 1,000 draws, with the VAR, the moments, the weights, and the pooled direction re-estimated in each draw. The family and the pooled impulse responses are differenced within each draw. The whiskers therefore account for the correlation between the two estimates. Each bar is a ratio with the family's first stage in the denominator, and the whiskers are widest for the two families with a weak first stage. Appendix E.1 splits each bar into the family's contamination and its first stage. HF: high-frequency surprises; SW: Swanson factors; RR: Romer–Romer; CH: Cholesky-based.

and Cholesky-based, point toward a *milder* contraction. Their average instruments would have raised the impulse responses of output and prices and lowered that of the credit spread. Their instruments are measured over longer windows, which also contain news other than the policy decision, and the instruments attribute part of that news to the policy shock. The high-frequency family points the other way, toward a *stronger* contraction. Its mean contamination also raises the credit-spread response on impact. This is consistent with news in the announcement window that moves interest rates and credit spreads together.

Equation (32) explains the size of the bars. Each bar is the family's contamination, magnified by the inverse of its first stage. The bars are comparable across families, as $\tilde{\beta}_c$ is normalised on the funds rate, and Appendix E.1 splits each bar into these two sources. The two wide-window families

carry the most contamination. The Cholesky-based family carries the largest, about four times that of the high-frequency family for output after twelve months. This is the cost that Section 4 anticipated. Widening the window restores first-stage strength, but it admits news that a narrow window excludes (Figure 2). The first stage works the other way. A weaker first stage magnifies the contamination of a family, as Section 3.3 shows for a single instrument. The high-frequency family carries moderate contamination, but its weak first stage magnifies it about nine times. The strong first stage of the Cholesky-based family magnifies its larger contamination by a factor of only 1.7 (Appendix Table 7). The Romer–Romer family lies in between. Magnification acts only on the contamination a family carries. The Swanson family, also built from surprises in the announcement window, has the weakest first stage of the four families, but its contamination is small, and so are its bars.

The two groups point in opposite directions, which is why the pooled estimate sits between the single instruments (Appendix Table 6). The contributions do not sum to the gap between the pools with and without demeaning in Figure 4, for two reasons. First, the pool without demeaning does not average the four families. It weights them by their strength and leans on the Cholesky-based family, whose bar points toward a milder contraction. Second, each bar is magnified by its family's own first stage, while the pool without demeaning has a single, combined first stage.

7 Robustness of the baseline estimate

The baseline rests on choices made in the empirical application, and the data cannot fully validate them. They concern which instruments enter, how they are grouped into families, how the VAR is specified, and which estimator is used. The exercises below change one choice at a time and keep the baseline families, weights, and windows unless stated otherwise.

7.1 Alternative panels of instruments

The identification rests on the panel of instruments and on its partition into families. If the baseline reflected one particular choice of instruments, a different panel would give a different estimate. The exercises below add a family, diversify the families and repartition the instruments.

A fifth family. An information-robust family (IR) can be added as a fifth family. Its three members start from a rate surprise in the announcement window and remove the information that the announcement reveals about the economy. The Miranda-Agrippino and Ricco (2021) surprise does so with the Fed's own forecasts. It is the residual of the three-month federal funds future surprise (FF4) on the Greenbook forecasts and their revisions. The other two members use the poor man's sign restriction of Jarociński and Karadi (2020). A monetary tightening should lower stock prices. When the rate surprise and the S&P 500 surprise in the same window move in the same direction, the announcement is read as news about the economy, and the surprise is set to zero. Every member is first orthogonalised at the announcement level on the Bauer and Swanson (2023b) controls (PTA). The sign restriction is then applied to the orthogonalised FF4 surprise and to the orthogonalised Bauer and Swanson (2023b) surprise. In both, 35% of the orthogonalised announcements still move with stock prices and are set to zero. The IR members are built from announcement-window surprises, like the high-frequency family. Removing the information effect, however, changes the news they carry, and with it their contamination.

Adding the IR family moves the estimate little. The first stage is $F = 26.8$, and output falls by 4.3% at its peak decline, against 4.0% in the baseline. The overidentification test does not reject exact block structure ($p = 0.72$). The IR family therefore points to the same policy shock as the baseline families.

A fifth family also allows each family to be dropped in turn, while partition condition (P2) still holds with the remaining four. The direction and the pooled instrument are re-estimated each time (Appendix Table 8 and Figure 14). The Swanson family sets the depth of the contraction. Without it, output falls by about half as much, and prices barely move. The Cholesky-based family supplies strength and tempers the magnitude. Without it, the pooled instrument is weak ($F = 7.5$), and the impulse responses are larger. The high-frequency and Romer–Romer families confirm the direction. Without either, the impulse responses remain close to those of the five-family estimate. These two families supply overidentifying restrictions and the number of families that condition (P2) requires.

Diversified families. In the baseline, the Swanson family carries most of the identifying variation. Adding instruments within families can reduce this dependence. The goal is to introduce dispersion in relevance within each family. Each family must still group instruments exposed to the same non-policy news, and the demeaning step still removes their common exposure. Two families are diversified (Appendix Table 10).

- *Maturity ladder.* The two-, five- and ten-year Treasury surprises in the announcement window, added to the high-frequency family. They share the family's window, and their relevance falls with maturity.
- *Narrative family* (NR). In the baseline, the Romer–Romer family holds three orthogonalisations of the same instrument (PTA, PTA-FWL and ATP). Their relevance along the pooled direction is nearly the same. The family carries 31% of the first-stage strength but adds little dispersion in relevance. NR keeps the Romer–Romer instrument orthogonalised before aggregation (PTA) and replaces the two other versions by two specifications of the Aruoba and Drechsel (2024b) shock. Like the Romer–Romer instrument, each is the residual of a regression of the change in the intended funds rate on the Greenbook forecasts. The regression also includes sentiment indicators extracted from Fed staff documents. The two specifications differ in whether the regressors enter with lags. The Aruoba and Drechsel (2024b) shocks cover only scheduled meetings from 1982 to 2008, a shorter support than the baseline window.

All new instruments are orthogonalised on the same controls as the baseline instruments. Each new family adds dispersion in relevance. The Treasury surprises share the announcement window of the near-dated futures and are exposed to the same news, but longer maturities respond less to the policy rate. Demeaning within the family removes the common news and keeps these differences in relevance, which point to the policy direction without the Swanson factors. The Aruoba and Drechsel (2024b) shocks remove more of the predictable part of the change in the funds rate. Their residuals therefore move the funds rate by different amounts from the Romer–Romer instrument, and demeaning within the family keeps these differences in relevance. With the two diversified families and the IR family together, the Swanson family's share of the identifying variation falls from 71% to about half. The impulse responses remain close to the baseline (Appendix Figure 15). Dropping the Swanson family now moves the estimate much less than in the baseline panel, because the other families carry identifying variation of their own. The dependence on the Swanson family can thus be reduced without altering the baseline estimates.

Repartitioning the instruments. The two previous exercises add instruments to the panel. The instruments can also be repartitioned into different families, which shows whether the results depend on the baseline partition. The new partition keeps the instruments of the previous exercises and forms five families, each holding instruments built from the same data over the same window. The five families are the near-term futures (the baseline high-frequency family), the Treasury curve (the two-, five- and ten-year Treasury surprises), the IR family, the narrative instruments (the Romer–Romer instrument and the two Aruoba and Drechsel (2024b) shocks) and the Cholesky-based instruments. The first three share the announcement window, the narrative instruments the decision day and the Cholesky-based instruments the month. The Swanson factors no longer form a family of their own. The target, path and asset-purchase factors measure the surprise in the current rate, the expected path of policy over the next year or so and the long end of the curve, respectively. The first two therefore join the near-term futures, and the third joins the Treasury curve.

The impulse responses nearly coincide with the baseline (Appendix Figure 16). The peak decline in output and the fall in prices after two years differ from the baseline by less than a tenth and a fifth of a percentage point, respectively, and both lie well inside the baseline confidence bands. The new partition passes both tests ($p = 0.008$ for no common direction, $p = 0.64$ for block structure). The target factor remains the main source of identifying variation. Within the near-term family, it moves the funds rate far more than the futures surprises do, and this dispersion in relevance identifies the shock. The asset-purchase factor contributes less than in the baseline, because its relevance deviation is now measured against the Treasury surprises. The role of the Swanson factors in the identification thus falls, as a by-product of the new partition. Without any factor, the near-dated futures and the Treasury surprises differ too little in relevance to identify the direction, and the estimate is close to the estimate of the Cholesky-based family alone. Moving the path factor to the Treasury curve leaves the estimate unchanged.

The estimate survives the addition of a family, the diversification of the families, and a different partition. In all three exercises, the impulse responses remain broadly consistent with the baseline.

7.2 The VAR specification

The exercises below return to the baseline panel of four families and change the composition of the VAR. They replace the federal funds rate by the shadow rate as the policy indicator, replace the price level with inflation measures and core prices, and add variables one at a time to trace the transmission of a monetary policy tightening.

Policy indicator and price measures. From December 2008 to December 2015 the federal funds rate was at its effective lower bound, and these months contribute little identifying variation. Policy still acted through forward guidance and asset purchases, which moved the yield curve. The shadow rate of Wu and Xia (2016) is constructed from the yield curve and can turn negative at the lower bound. Used as the policy indicator, it keeps moving in those months, which then add identifying variation. The results remain consistent with the baseline (Appendix Figure 17). Output and prices fall, and the excess bond premium rises, with impulse responses slightly larger than under the funds rate. The pooled instrument is weaker but remains strong ($F = 20.3$).

The orthogonalised surprise implies an immediate fall in the price level, at odds with the gradual transmission of monetary policy (feature (iv) of Section 2.2). Appendix Figure 18 replaces the log

level of the consumer price index in turn with year-over-year and month-over-month inflation, and with core CPI, which excludes food and energy. Under every measure, prices respond sooner and by more under the orthogonalised surprise than under the pooled instrument. This holds for headline and core prices, in levels and in inflation. The pooled instrument implies smaller price responses.

Transmission to the rest of the economy. A larger VAR traces the transmission of the policy shock to the rest of the economy. In the within estimator, each additional variable brings a non-policy shock of its own, which may contaminate the instruments. Demeaning removes the contamination that each family shares from the new shock. However, each added variable makes condition (P2) harder to meet (Section 5.5). With the IR family of Section 7.1 as a fifth family, condition (P2) holds in a five-variable VAR ($C = 5 > n - 1 = 4$). Each larger system therefore adds one variable at a time to the baseline four. Each panel of Figure 6 is a separate five-variable VAR, the baseline four variables and the one shown, estimated with the baseline weights and windows on the baseline sample, January 1973 to December 2019. The four variables shown are the unemployment rate, a commodity-price index, the S&P 500 and housing starts.

The impulse responses follow the conventional transmission of a monetary tightening. Financial markets react first, and the S&P 500 falls on impact (Bernanke and Kuttner, 2005). Housing, the most interest-sensitive component of demand, contracts next, ahead of output (Bernanke and Gertler, 1995). Unemployment rises, and commodity prices fall more gradually, as demand weakens (Christiano et al., 1999; Anzuini et al., 2013). Monetary policy affects real activity and prices only with a lag. In all four systems, the impulse responses of output and prices are close to the baseline, and exact block structure is not rejected (p-values between 0.55 and 0.96).

The first-stage strength varies across systems, from 11.2 to 21.0, for two reasons. First, the policy-rate innovation changes, because it is the part of the funds rate that the lags of all the VAR's variables cannot predict. Second, the weights of the pooled instrument change, because they are set to cancel the contamination from the new non-policy shock. When the instruments move with that shock on impact, canceling it also removes part of their relevance. Stock prices react within minutes of the announcement. The instruments therefore move with the stock-market shock on impact, and the VAR that includes the S&P 500 has the weakest first stage (Figure 6). Appendix E.2 separates the change in the impulse responses into the change of VAR, with the instrument held fixed, and the change of instrument, with the VAR held fixed. With the instrument held fixed, the impulse responses of output and prices stay close to the baseline. Re-pooling the instrument in the larger VAR weakens the first stage and widens the confidence bands, most for the S&P 500 (Appendix Figure 20).

Appendix Figure 19 extends the exercise to eight further variables, in the same five-variable systems. Financial conditions tighten on impact, while demand, the labour market and credit contract more gradually, consistent with the lagged effects of monetary policy on real activity. In every system, output and prices fall, and exact block structure is not rejected (p between 0.30 and 0.89). Appendix Table 9 reports these systems and the other candidate variables screened.

7.3 Reusing the pooled instrument in other frameworks

Constrained pooling delivers a series, the pooled instrument $z_t^\star$, which can be reused in any framework that takes an instrument. Two such frameworks are an internal-instrument VAR

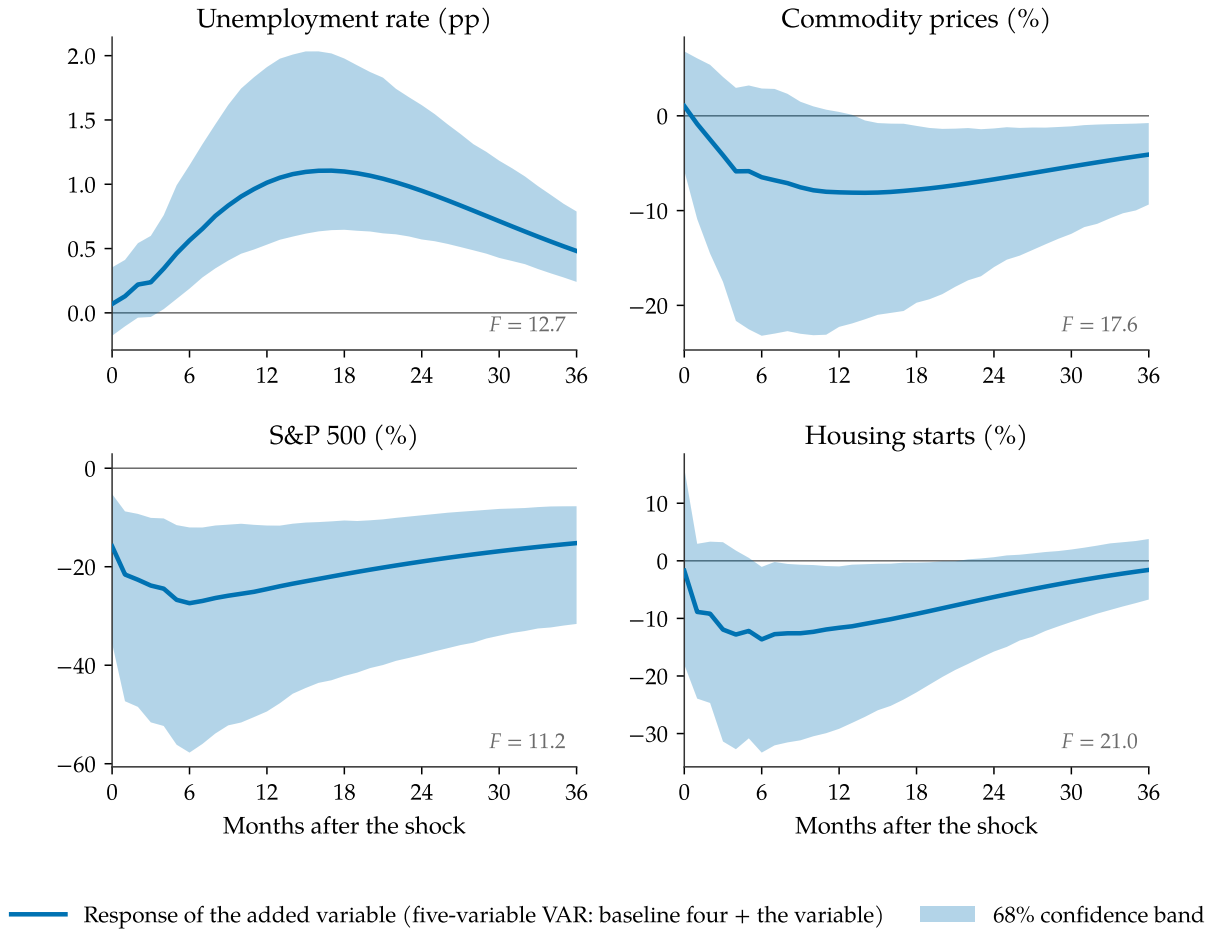


Figure 6: **Larger systems, one added variable at a time.** Each panel is estimated separately, as a five-variable VAR in the baseline four variables plus the variable shown, with the baseline families and the IR family as a fifth, and the baseline precision weights, windows and bootstrap. The panel plots only the impulse response of the added variable to a +1 percentage-point funds-rate shock, with its 68% confidence band.

(Plagborg-Møller and Wolf, 2021) and local projections (Jordà, 2005). In the internal-instrument VAR, the pooled instrument is ordered first, and the shock is identified from a Cholesky decomposition of the extended covariance matrix. In local projections, each outcome variable is regressed, horizon by horizon, on the policy rate instrumented by the pooled instrument. Appendix E.3 gives the specifications of both.

The pooled instrument exists only on the moment window, from January 1988 to December 2019. Both frameworks use it as a regressor, and their sample is restricted to that window. For a consistent comparison, the external-instrument VAR is re-estimated on the same window, with the baseline variables, lags and normalisation. Only the framework then changes across the three. Appendix Figures 21 and 22 show the three estimates, with the baseline, estimated from 1973, for reference.

The pooled instrument remains strong in all three frameworks, and the estimates agree on

impact. Output falls, and the excess bond premium rises by similar amounts. The transmission that follows differs in timing and depth. The external-instrument VAR extrapolates the dynamics from its six lags, which smooths the impulse responses at longer horizons. Local projections estimate each horizon separately, without this extrapolation. Their impulse responses are therefore more irregular and less precise, and their output decline is earlier and deeper. The internal-instrument VAR gives a shallower decline in output than the external estimate. Prices differ. They fall at every horizon in the external-instrument VAR but rise after impact in the internal-instrument VAR and the local projections.

8 Conclusion

No available instrument for monetary policy is known to be both strong and valid. The identification method proposed in this paper does not require one. It groups the instruments into families whose members share their exposure to non-policy news but differ in how strongly they move the policy rate. Demeaning the identifying moments within families removes the contamination each family shares. The relevance deviations that remain identify the direction of the policy shock. Pooling across families restores the first-stage strength that each instrument may lack.

The method addresses the trade-off that orthogonalisation creates. Orthogonalising an instrument on pre-announcement information is necessary to remove its predictable variation. However, it weakens the first-stage strength, and it cannot remove news that the controls do not span. Any contamination that survives orthogonalisation is then magnified by the weaker first stage. The high-frequency surprise is the extreme case. It loses most of its first-stage strength, and its impulse responses are large in magnitude. The paper extends the orthogonalisation to instruments built from wider windows around the policy decision, the Romer–Romer instrument and the Cholesky-based instrument. They are partly predictable, but they retain most of their first-stage strength under the same orthogonalisation. They may, however, admit news that the narrow window excludes. Orthogonalisation and the within estimator are therefore complements. The first removes the variation that was predictable before the announcement. The second removes the news that a family shares within its window, and restores the strength that the first takes away.

On twelve orthogonalised U.S. instruments in four families, the pooled instrument is strong where the orthogonalised surprise is weak, and it identifies a standard monetary contraction. Output falls gradually, consumer prices decline slowly and the credit spread widens. The estimate survives the addition of a family, the diversification of the families and a different partition of the instruments. The pooled instrument can also be reused in other frameworks, such as an internal-instrument VAR and local projections.

The method is not specific to monetary policy. It suits any shock that several imperfect instruments identify. Oil supply shocks are a natural candidate. They are instrumented by high-frequency surprises around OPEC announcements (Känzig, 2021), by measures of exogenous supply disruptions (Kilian, 2008), and by oil price increases above their recent peak (Hamilton, 2003; Ramey and Vine, 2011). Fiscal policy offers other applications. Government spending shocks are instrumented by narrative military news (Ramey, 2011; Ramey and Zubairy, 2018) and by the excess returns of defence contractors (Fisher and Peters, 2010), and tax shocks by narrative tax changes (Romer and Romer, 2010; Mertens and Ravn, 2013).

References

Acosta, Miguel, Andrea Ajello, Michael D. Bauer, Francesca Loria, and Silvia Miranda-Agrippino. Financial Market Effects of FOMC Communication: Evidence from a New Event-Study Database. Working Paper 2025-30, Federal Reserve Bank of San Francisco, 2025. Revise and resubmit, Journal of Monetary Economics.

Amodeo, Francesco. The Ordering of Orthogonalisation and Aggregation in Monetary Policy Surprises. Working paper, 2026.

Andrews, Isaiah, James H. Stock, and Liyang Sun. Weak Instruments in Instrumental Variables Regression: Theory and Practice. *Annual Review of Economics*, 11:727–753, 2019.

Angelini, Giovanni, and Luca Fanelli. Exogenous Uncertainty and the Identification of Structural Vector Autoregressions with External Instruments. *Journal of Applied Econometrics*, 34(6):951–971, 2019.

Angelini, Giovanni, Giuseppe Cavaliere, and Luca Fanelli. An Identification and Testing Strategy for Proxy-SVARs with Weak Proxies. *Journal of Econometrics*, 238(2):105604, 2024.

Antolín-Díaz, Juan, and Juan F. Rubio-Ramírez. Narrative Sign Restrictions for SVARs. *American Economic Review*, 108(10):2802–2829, 2018.

Anzuini, Alessio, Marco J. Lombardi, and Patrizio Pagano. The Impact of Monetary Policy Shocks on Commodity Prices. *International Journal of Central Banking*, 9(3):125–150, 2013.

Arias, Jonas E., Dario Caldara, and Juan F. Rubio-Ramírez. The Systematic Component of Monetary Policy in SVARs: An Agnostic Identification Procedure. *Journal of Monetary Economics*, 101:1–13, 2019.

Aruoba, S. Borağan, and Thomas Drechsel. The Long and Variable Lags of Monetary Policy: Evidence from Disaggregated Price Indices. *Journal of Monetary Economics*, 148:103635, 2024a.

Aruoba, S. Borağan, and Thomas Drechsel. Identifying Monetary Policy Shocks: A Natural Language Approach. NBER Working Paper 32417, National Bureau of Economic Research, 2024b.

Bai, Jushan, and Serena Ng. Instrumental Variable Estimation in a Data Rich Environment. *Econometric Theory*, 26(6):1577–1606, 2010.

Bauer, Michael D., and Eric T. Swanson. An Alternative Explanation for the "Fed Information Effect". *American Economic Review*, 113(3):664–700, 2023a.

Bauer, Michael D., and Eric T. Swanson. A Reassessment of Monetary Policy Surprises and High-Frequency Identification. *NBER Macroeconomics Annual*, 37:87–155, 2023b.

Bekker, Paul A. Alternative Approximations to the Distributions of Instrumental Variable Estimators. *Econometrica*, 62(3):657–681, 1994.

Bernanke, Ben S., and Alan S. Blinder. The Federal Funds Rate and the Channels of Monetary Transmission. *American Economic Review*, 82(4):901–921, 1992.

Bernanke, Ben S., and Mark Gertler. Inside the Black Box: The Credit Channel of Monetary Policy Transmission. *Journal of Economic Perspectives*, 9(4):27–48, 1995.

Bernanke, Ben S., and Kenneth N. Kuttner. What Explains the Stock Market's Reaction to Federal Reserve Policy? *Journal of Finance*, 60(3):1221–1257, 2005.

Bernile, Gennaro, Jianfeng Hu, and Yuehua Tang. Can Information Be Locked Up? Informed Trading Ahead of Macro-News Announcements. *Journal of Financial Economics*, 121(3):496–520, 2016.

Brennan, Connor M., Margaret M. Jacobson, Christian Matthes, and Todd B. Walker. Monetary Policy Shocks: Data or Methods? *B.E. Journal of Macroeconomics*, 2025. doi:10.1515/bejm-2024-0175. Earlier circulated as Finance and Economics Discussion Series 2024-011.

Buda, Gergely, Vasco M. Carvalho, Giancarlo Corsetti, João B. Duarte, Stephen Hansen, Afonso S. Moura, Álvaro Ortiz, Tomás Rodrigo, José V. Rodríguez Mora, and Guilherme Alves da Silva. The Short Lags of Monetary Policy. CEPR Discussion Paper 18022, Centre for Economic Policy Research, 2025. Revised 2025; first circulated in 2023 as "Short and Variable Lags".

Caldara, Dario, and Edward Herbst. Monetary Policy, Real Activity, and Credit Spreads: Evidence from Bayesian Proxy SVARs. *American Economic Journal: Macroeconomics*, 11(1):157–192, 2019.

Chen, Zhengyang. Demystifying Monetary Policy Surprises: Fed Response to Financial Conditions and Wait and See for New Economic Data. *Journal of Macroeconomics*, 87:103736, 2026.

Christiano, Lawrence J., Martin Eichenbaum, and Charles L. Evans. Monetary Policy Shocks: What Have We Learned and to What End? In Taylor, John B., and Michael Woodford, editors, *Handbook of Macroeconomics*, volume 1, chapter 2, pages 65–148. Elsevier, 1999.

Cieslak, Anna, Adair Morse, and Annette Vissing-Jorgensen. Stock Returns over the FOMC Cycle. *Journal of Finance*, 74(5):2201–2248, 2019.

Conley, Timothy G., Christian B. Hansen, and Peter E. Rossi. Plausibly Exogenous. *Review of Economics and Statistics*, 94(1):260–272, 2012.

Fisher, Jonas D. M., and Ryan Peters. Using Stock Returns to Identify Government Spending Shocks. *Economic Journal*, 120(544):F414–F436, 2010.

Francis, Neville, Peter Reinhard Hansen, and Chen Tong. Principled Identification of Structural Dynamic Models. arXiv:2512.17005, version of April 30, 2026, 2026.

Gertler, Mark, and Peter Karadi. Monetary Policy Surprises, Credit Costs, and Economic Activity. *American Economic Journal: Macroeconomics*, 7(1):44–76, 2015.

Gilchrist, Simon, and Egon Zakrajšek. Credit Spreads and Business Cycle Fluctuations. *American Economic Review*, 102(4):1692–1720, 2012.

Gürkaynak, Refet S., Brian Sack, and Eric T. Swanson. Do Actions Speak Louder Than Words? The Response of Asset Prices to Monetary Policy Actions and Statements. *International Journal of Central Banking*, 1(1):55–93, 2005.

Gürkaynak, Refet S., Eric T. Swanson, and Jonathan H. Wright. High-Frequency Identification. Working paper, August 2026, 2026.

Hack, Lukas, Klodiana Istrefi, and Matthias Meier. The Systematic Origins of Monetary Policy Shocks. CEPR Discussion Paper 19063, Centre for Economic Policy Research, 2024. Also Banque de France Working Paper 973.

Hamilton, James D. What Is an Oil Shock? *Journal of Econometrics*, 113(2):363–398, 2003.

Hoesch, Lukas, Barbara Rossi, and Tatevik Sekhposyan. Has the Information Channel of Monetary Policy Disappeared? Revisiting the Empirical Evidence. *American Economic Journal: Macroeconomics*, 15(4):355–387, 2023.

Huang, Jiaming, and Luca Neri. Beyond Validity: SVAR Identification Through the Proxy Zoo. Working Paper 2601.11195, arXiv, 2026.

Jarociński, Marek, and Peter Karadi. Deconstructing Monetary Policy Surprises—The Role of Information Shocks. *American Economic Journal: Macroeconomics*, 12(2):1–43, 2020. Updated Fed shocks: `github.com/marekjarocinski/jkshocks_update_fed`.

Jentsch, Carsten, and Kurt G. Lunsford. Asymptotically Valid Bootstrap Inference for Proxy SVARs. *Journal of Business & Economic Statistics*, 40(4):1876–1891, 2022.

Jordà, Òscar. Estimation and Inference of Impulse Responses by Local Projections. *American Economic Review*, 95(1):161–182, 2005.

Känzig, Diego R. The Macroeconomic Effects of Oil Supply News: Evidence from OPEC Announcements. *American Economic Review*, 111(4):1092–1125, 2021.

Kilian, Lutz. Exogenous Oil Supply Shocks: How Big Are They and How Much Do They Matter for the U.S. Economy? *Review of Economics and Statistics*, 90(2):216–240, 2008.

Kolesár, Michal, Raj Chetty, John Friedman, Edward Glaeser, and Guido W. Imbens. Identification and Inference with Many Invalid Instruments. *Journal of Business & Economic Statistics*, 33(4): 474–484, 2015.

Kuersteiner, Guido, and Ryo Okui. Constructing Optimal Instruments by First-Stage Prediction Averaging. *Econometrica*, 78(2):697–718, 2010.

Kuttner, Kenneth N. Monetary Policy Surprises and Interest Rates: Evidence from the Fed Funds Futures Market. *Journal of Monetary Economics*, 47(3):523–544, 2001.

Lakdawala, Aeimit. Decomposing the Effects of Monetary Policy Using an External Instruments SVAR. *Journal of Applied Econometrics*, 34(6):934–950, 2019.

Lin, Yiqi, Frank Windmeijer, Xinyuan Song, and Qingliang Fan. On the Instrumental Variable Estimation with Many Weak and Invalid Instruments. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 86(4):1068–1088, 2024.

Lucca, David O., and Emanuel Moench. The Pre-FOMC Announcement Drift. *Journal of Finance*, 70 (1):329–371, 2015.

Mertens, Karel, and Morten O. Ravn. The Dynamic Effects of Personal and Corporate Income Tax Changes in the United States. *American Economic Review*, 103(4):1212–1247, 2013.

Mikusheva, Anna, and Liyang Sun. Inference with Many Weak Instruments. *Review of Economic Studies*, 89(5):2663–2686, 2022.

Miranda-Agrippino, Silvia, and Giovanni Ricco. The Transmission of Monetary Policy Shocks. *American Economic Journal: Macroeconomics*, 13(3):74–107, 2021.

Montiel Olea, José Luis, and Carolin Pflueger. A Robust Test for Weak Instruments. *Journal of Business & Economic Statistics*, 31(3):358–369, 2013.

Montiel Olea, José Luis, James H. Stock, and Mark W. Watson. Inference in Structural Vector Autoregressions Identified with an External Instrument. *Journal of Econometrics*, 225(1):74–87, 2021.

Nakamura, Emi, and Jón Steinsson. High-Frequency Identification of Monetary Non-Neutrality: The Information Effect. *Quarterly Journal of Economics*, 133(3):1283–1330, 2018.

Newey, Whitney K., and Daniel McFadden. Large Sample Estimation and Hypothesis Testing. In *Handbook of Econometrics*, volume 4, pages 2111–2245. Elsevier, 1994.

Newey, Whitney K., and Frank Windmeijer. Generalized Method of Moments with Many Weak Moment Conditions. *Econometrica*, 77(3):687–719, 2009.

Plagborg-Møller, Mikkel, and Christian K. Wolf. Local Projections and VARs Estimate the Same Impulse Responses. *Econometrica*, 89(2):955–980, 2021.

Ramey, Valerie A. Identifying Government Spending Shocks: It's All in the Timing. *Quarterly Journal of Economics*, 126(1):1–50, 2011.

Ramey, Valerie A. Macroeconomic Shocks and Their Propagation. In *Handbook of Macroeconomics*, volume 2, pages 71–162. Elsevier, 2016.

Ramey, Valerie A., and Daniel J. Vine. Oil, Automobiles, and the U.S. Economy: How Much Have Things Really Changed? *NBER Macroeconomics Annual*, 25:333–367, 2011.

Ramey, Valerie A., and Sarah Zubairy. Government Spending Multipliers in Good Times and in Bad: Evidence from US Historical Data. *Journal of Political Economy*, 126(2):850–901, 2018.

Rigobon, Roberto. Identification Through Heteroskedasticity. *Review of Economics and Statistics*, 85 (4):777–792, 2003.

Romer, Christina D., and David H. Romer. A New Measure of Monetary Shocks: Derivation and Implications. *American Economic Review*, 94(4):1055–1084, 2004.

Romer, Christina D., and David H. Romer. The Macroeconomic Effects of Tax Changes: Estimates Based on a New Measure of Fiscal Shocks. *American Economic Review*, 100(3):763–801, 2010.

Schmeling, Maik, Andreas Schrimpf, and Sigurd A. M. Steffensen. Monetary Policy Expectation Errors. *Journal of Financial Economics*, 146(3):841–858, 2022.

Sims, Christopher A. Macroeconomics and Reality. *Econometrica*, 48(1):1–48, 1980.

Staiger, Douglas, and James H. Stock. Instrumental Variables Regression with Weak Instruments. *Econometrica*, 65(3):557–586, 1997.

Stock, James H., and Mark W. Watson. Disentangling the Channels of the 2007–09 Recession. *Brookings Papers on Economic Activity*, Spring:81–135, 2012.

Stock, James H., and Mark W. Watson. Identification and Estimation of Dynamic Causal Effects in Macroeconomics Using External Instruments. *Economic Journal*, 128(610):917–948, 2018.

Stock, James H., and Motohiro Yogo. Testing for Weak Instruments in Linear IV Regression. In *Identification and Inference for Econometric Models: Essays in Honor of Thomas Rothenberg*, pages 80–108. Cambridge University Press, Cambridge, 2005.

Swanson, Eric T. Measuring the Effects of Federal Reserve Forward Guidance and Asset Purchases on Financial Markets. *Journal of Monetary Economics*, 118:32–53, 2021.

Swanson, Eric T. The Macroeconomic Effects of the Federal Reserve's Conventional and Unconventional Monetary Policies. *IMF Economic Review*, 72(3):1152–1184, 2024.

Swanson, Eric T., and Vishuddhi Jayawickrema. Speeches by the Fed Chair Are More Important than FOMC Announcements: An Improved High-Frequency Measure of U.S. Monetary Policy Shocks. Working paper, University of California, Irvine, 2023.

Tran, Paul L. How Long Do Markets Need to Fully React to Monetary Policy Announcements? Job market paper, University of Texas at Austin, 2026.

Uhlig, Harald. What Are the Effects of Monetary Policy on Output? Results from an Agnostic Identification Procedure. *Journal of Monetary Economics*, 52(2):381–419, 2005.

Windmeijer, Frank, Helmut Farbmacher, Neil Davies, and George Davey Smith. On the Use of the Lasso for Instrumental Variables Estimation with Some Invalid Instruments. *Journal of the American Statistical Association*, 114(527):1339–1350, 2019.

Windmeijer, Frank, Xiaoran Liang, Fernando P. Hartwig, and Jack Bowden. The Confidence Interval Method for Selecting Valid Instrumental Variables. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 83(4):752–776, 2021.

Wu, Jing Cynthia, and Fan Dora Xia. Measuring the Macroeconomic Impact of Monetary Policy at the Zero Lower Bound. *Journal of Money, Credit and Banking*, 48(2–3):253–291, 2016.

Appendix

A The bias of an orthogonalised instrument

Orthogonalisation removes the contamination that the controls span, and leaves the rest. Appendix A.2 derives the loadings of Proposition 1, and Appendix A.3 the magnification of Proposition 2.

A.1 The bias identity

Derivation of the bias identity in equation (7). With the shocks orthogonal and of unit variance, and ω_t uncorrelated with them, the identifying moment is $\mathbb{E}[u_t z_t] = \phi\, b_p + B_{-p}\boldsymbol{\psi}$, and its policy-rate entry is $\pi = \phi\, b_{pp} + b'_{p,-p}\boldsymbol{\psi}$. Since $b_p = b_{pp}\beta$, the policy term is $\phi\, b_p = \phi b_{pp}\,\beta = (\pi - b'_{p,-p}\boldsymbol{\psi})\beta$. Substituting it in the moment gives $\mathbb{E}[u_t z_t] = \pi\beta + B_{-p}\boldsymbol{\psi} - \beta\, b'_{p,-p}\boldsymbol{\psi} = \pi\beta + d$. Dividing by π gives $\tilde{\beta} = \beta + d/\pi$. Since $s'_p\beta = 1$, $s'_p d = b'_{p,-p}\boldsymbol{\psi} - b'_{p,-p}\boldsymbol{\psi} = 0$. With $M := B_{-p} - \beta\, b'_{p,-p}$ as in Section 3, the contamination direction is linear in the loading, $d = M\boldsymbol{\psi}$. □

A.2 The loadings after orthogonalisation

Proof of Proposition 1 (what orthogonalisation removes). (i) The loadings follow in three steps.

1. *Loadings are covariances with the shocks.* The shocks are orthogonal with unit variance, and ω_t is uncorrelated with them. By equation (5), $\phi = \mathbb{E}[\varepsilon_t^p z_t]$ and $\boldsymbol{\psi} = \mathbb{E}[\boldsymbol{\varepsilon}_t^{-p} z_t]$. The loadings of $z_t^\perp$ are its covariances with the shocks in the same way.

2. *Covariances of the controls.* By equation (8), $\mathbb{E}[\boldsymbol{\varepsilon}_t^{-p}\mathbf{W}'_t] = A$, $\mathbb{E}[\varepsilon_t^p \mathbf{W}'_t] = \mathbf{0}$ and $\mathrm{Var}(\mathbf{W}_t) = A'A$. Multiplying equations (8) and (5), $\mathbb{E}[\mathbf{W}_t z_t] = A'\boldsymbol{\psi}$. The regression coefficient is therefore

$$\boldsymbol{c} = (A'A)^{-1}A'\boldsymbol{\psi},$$

   the least-squares coefficient of $\boldsymbol{\psi}$ on the columns of A. The regression looks for the combination of the controls' loadings closest to the instrument's loading. With collinear controls, a generalised inverse replaces $(A'A)^{-1}$. The coefficient is then not unique, but the fitted value and $P = A(A'A)^{-1}A'$ are.

3. *Loadings of the orthogonalised instrument.* By steps 1 and 2,

$$\phi^\perp = \phi - \mathbb{E}[\varepsilon_t^p \mathbf{W}'_t]\boldsymbol{c} = \phi, \qquad \boldsymbol{\psi}^\perp = \boldsymbol{\psi} - \mathbb{E}[\boldsymbol{\varepsilon}_t^{-p}\mathbf{W}'_t]\boldsymbol{c} = \boldsymbol{\psi} - A\boldsymbol{c} = (I - P)\boldsymbol{\psi}.$$

   The rest of $z_t^\perp$, ω_t, is uncorrelated with the shocks.

The controls end before the announcement, and have no covariance with the news that arrives from the announcement on. The rows of A for those shocks are zero, and so are the corresponding rows of P. That part of $\boldsymbol{\psi}$ is untouched.

(ii) The predictable part is $\hat{z}_t = \boldsymbol{c}'\mathbf{W}_t = (A\boldsymbol{c})'\boldsymbol{\varepsilon}_t^{-p} = (P\boldsymbol{\psi})'\boldsymbol{\varepsilon}_t^{-p}$. The shocks have unit variance, and $\mathrm{Var}(\hat{z}_t) = \|P\boldsymbol{\psi}\|^2$. P is a projection, and $\|\boldsymbol{\psi}\|^2 = \|P\boldsymbol{\psi}\|^2 + \|(I-P)\boldsymbol{\psi}\|^2$.

(iii) $(I - P)\boldsymbol{\psi} = \mathbf{0}$ if and only if $\boldsymbol{\psi}$ lies in the span of A. If $\boldsymbol{\psi} = A\boldsymbol{w}$, then $\boldsymbol{c} = (A'A)^{-1}A'A\boldsymbol{w} = \boldsymbol{w}$, and the regression finds the weights under which the controls span the contamination. □

When every contamination cancels. P has rank equal to the number of linearly independent controls. $P = I$ if and only if there are as many independent controls as non-policy shocks, and every contamination then cancels, with $\boldsymbol{c} = A^{-1}\boldsymbol{\psi}$. Any non-policy news that arrives from the announcement on gives A a zero row and rules this out. With fewer controls than the shocks they must separate, cancellation requires $\boldsymbol{\psi}$ to lie in the span of A. Otherwise the remainder $(I - P)\boldsymbol{\psi}$ is orthogonal to the span of the controls, not to the shocks the instrument loads on, and it can load on shocks the instrument did not carry. Its size falls, by part (ii), and its direction can change.

A.3 Magnification

Proof of Proposition 2 (magnification). (i) By step 3 of the proof of Proposition 1, the relevance is unchanged, and $\pi - \pi^{\perp} = b'_{p,-p}(\boldsymbol{\psi} - \boldsymbol{\psi}^{\perp}) = b'_{p,-p}P\boldsymbol{\psi}$. Dividing by π gives the second and third expressions for δ_π. The predictable part is $\hat{z}_t = (P\boldsymbol{\psi})'\boldsymbol{\varepsilon}_t^{-p}$, and $\operatorname{Cov}(\hat{z}_t, u_t^p) = (P\boldsymbol{\psi})'\,\mathbb{E}[\boldsymbol{\varepsilon}_t^{-p}\boldsymbol{\varepsilon}_t^{-p\prime}]b_{p,-p} = b'_{p,-p}P\boldsymbol{\psi}$. Dividing by $\pi = \operatorname{Cov}(z_t, u_t^p)$ gives the first.

(ii) By Appendix A.1, $d = M\boldsymbol{\psi}$ and $d^{\perp} = M\boldsymbol{\psi}^{\perp} = M(I - P)\boldsymbol{\psi}$. Since norms are non-negative, $\delta_d \leq 1$.

(iii) Since $\beta = b_p/b_{pp}$, the contamination direction can be written $d = B\big(-b'_{p,-p}\boldsymbol{\psi}/b_{pp},\ \boldsymbol{\psi}'\big)'$. The impact matrix B is invertible, and hence $d \neq \mathbf{0}$ whenever $\boldsymbol{\psi} \neq \mathbf{0}$. By equation (7), $\tilde{\beta} - \beta = d/\pi$ and $\tilde{\beta}^{\perp} - \beta = d^{\perp}/\pi^{\perp}$. Because $\pi, \pi^{\perp} > 0$, the norms factor as

$$\varrho = \frac{\|d^{\perp}\|/\pi^{\perp}}{\|d\|/\pi} = \frac{\|d^{\perp}\|/\|d\|}{\pi^{\perp}/\pi} = \frac{1 - \delta_d}{1 - \delta_\pi}.$$

Positivity of $\pi^{\perp}/\pi$ gives $\delta_\pi < 1$. The denominator $1 - \delta_\pi$ is positive, and $\varrho > 1$ if and only if $1 - \delta_d > 1 - \delta_\pi$, that is $\delta_d < \delta_\pi$. If $\delta_d \geq 0$, the numerator is at most one, and $\varrho \leq 1/(1 - \delta_\pi)$, with equality if and only if $\delta_d = 0$. □

When the bound holds. The loading always shrinks, by Proposition 1(ii). The contamination direction need not. M weighs the shocks by their effect on the economy, and a remainder that moves to shocks with larger effects lengthens d. If the non-policy shocks move the economy by the same amount in orthogonal directions, $M'M = \mu^2 I$, then $\|Mx\| = \mu\|x\|$ for every x, and $1 - \delta_d = \|(I - P)\boldsymbol{\psi}\|/\|\boldsymbol{\psi}\| \leq 1$. The bound $1/(1 - \delta_\pi)$ then holds. In general, $1 - \delta_d \leq \kappa(M)\,\|(I - P)\boldsymbol{\psi}\|/\|\boldsymbol{\psi}\|$, with $\kappa(M)$ the ratio of the largest to the smallest singular value of M, which is positive since B is invertible. Unequal effects of the shocks can more than offset the shrinking loading.

B The loss of first-stage strength and the orthogonalised instruments

This appendix derives equation (16) and the related identities, and shows how Table 1 computes the loss of first-stage strength δ_π from reported statistics. Let $R^2 := \operatorname{Var}(\hat{z}_t)/\operatorname{Var}(z_t)$ be the R^2 of the orthogonalisation regression, with ρ and $\hat{\rho}$ as in equation (16). The fitted value and the residual of the regression are orthogonal, and hence $\operatorname{Var}(\hat{z}_t) = R^2 \operatorname{Var}(z_t)$ and $\operatorname{Var}(z_t^{\perp}) = (1 - R^2)\operatorname{Var}(z_t)$.

The loss of first-stage strength. Equation (16) of the main text is

$$\delta_\pi = \sqrt{R^2}\,\frac{\hat{\rho}}{\rho}. \tag{16}$$

Derivation. By Proposition 2(i), $\delta_\pi = \mathrm{Cov}(\hat{z}_t, u_t^p)\,/\,\mathrm{Cov}(z_t, u_t^p)$. Writing each covariance as a correlation times the two standard deviations,

$$\delta_\pi = \frac{\hat{\rho}\,\sqrt{\mathrm{Var}(\hat{z}_t)}\,\sqrt{\mathrm{Var}(u_t^p)}}{\rho\,\sqrt{\mathrm{Var}(z_t)}\,\sqrt{\mathrm{Var}(u_t^p)}} = \frac{\hat{\rho}}{\rho}\sqrt{\frac{\mathrm{Var}(\hat{z}_t)}{\mathrm{Var}(z_t)}} = \sqrt{R^2}\,\frac{\hat{\rho}}{\rho}.$$

The loss is the size of the predictable part, $\sqrt{R^2}$, times how closely it moves with the policy rate relative to the instrument as a whole, $\hat{\rho}/\rho$.

What lowers $\hat{\rho}/\rho$. The predictable part is $\hat{z}_t = (P\boldsymbol{\psi})'\boldsymbol{\varepsilon}_t^{-p}$ (Appendix A.2), and $\mathrm{Cov}(\hat{z}_t, u_t^p) = b'_{p,-p}P\boldsymbol{\psi}$. Hence

$$\hat{\rho} = \frac{b'_{p,-p}P\boldsymbol{\psi}}{\|P\boldsymbol{\psi}\|\ \mathrm{sd}(u_t^p)}. \tag{33}$$

$\hat{\rho}$ measures how closely the policy rate responds to the news the controls predict. Predictable news to which the policy rate does not respond within the month lowers $\hat{\rho}/\rho$, and with it δ_π.

The correlation after orthogonalisation. Let $\rho^\perp := \mathrm{Corr}(z_t^\perp, u_t^p)$ be the correlation of the orthogonalised instrument with the policy-rate innovation. It is related to the raw correlation by

$$\rho^\perp = \frac{1-\delta_\pi}{\sqrt{1-R^2}}\,\rho. \tag{34}$$

Derivation. By Proposition 2(i), the covariance of the orthogonalised instrument with u_t^p is $1-\delta_\pi$ times that of the raw instrument. Dividing by the standard deviations, and using the variance of $z_t^\perp$ above,

$$\rho^\perp = \frac{(1-\delta_\pi)\,\mathrm{Cov}(z_t, u_t^p)}{\sqrt{1-R^2}\,\sqrt{\mathrm{Var}(z_t)\,\mathrm{Var}(u_t^p)}} = \frac{1-\delta_\pi}{\sqrt{1-R^2}}\,\rho.$$

When strength falls. The correlation falls if and only if

$$\rho^\perp < \rho \iff \delta_\pi > 1-\sqrt{1-R^2} \approx \tfrac{1}{2}R^2. \tag{35}$$

Derivation. Since $\rho > 0$, equation (34) gives $\rho^\perp < \rho$ if and only if $1-\delta_\pi < \sqrt{1-R^2}$. The approximation uses $\sqrt{1-R^2} \approx 1-\frac{1}{2}R^2$. The covariance falls one for one with δ_π, and the standard deviation only with $\sqrt{1-R^2}$. Orthogonalising predictable noise, $\delta_\pi = 0$, would raise strength by $1/\sqrt{1-R^2}$.

From the first-stage F to the loss of strength. The ratio of the orthogonalised to the raw first-stage F is approximately

$$\frac{F^{\perp}}{F} \approx \frac{(1-\delta_{\pi})^2}{1-R^2}.$$

Derivation. For a single instrument, $F = T\rho^2/(1-\rho^2)$, exactly for the homoskedastic statistic. Hence $\rho = \sqrt{F/(F+T)}$, and, when both correlations are small, $F \approx T\rho^2$ and $F^{\perp}/F \approx (\rho^{\perp}/\rho)^2$, which equation (34) turns into the ratio above.

The rule of thumb. Three reported numbers give δ_{π}: the raw and orthogonalised first-stage F and the R^2 of the orthogonalisation regression. The correlations follow from the F, equation (34) gives $\delta_{\pi} = 1 - \sqrt{1-R^2}\,\rho^{\perp}/\rho$, equation (16) gives $\hat{\rho}/\rho = \delta_{\pi}/\sqrt{R^2}$, and equation (13) gives the magnification bound $1/(1-\delta_{\pi})$. Table 1 computes δ_{π} directly from the covariances, with $T = 383$ and the monthly R^2. For the high-frequency surprise, $\delta_{\pi} \approx 0.54$, larger than the threshold $1 - \sqrt{1-R^2} \approx 0.16$ in equation (35), and the bias that survives can be magnified up to 2.15 times when $\delta_d \geq 0$.

B.1 The orthogonalised surprise in the data

The orthogonalised surprise moves the impulse responses furthest. Three pieces of evidence follow, a decomposition of its bias, its magnification variable by variable, and a comparison with weak-instrument bias. Biases are measured against the pooled baseline estimate, which identifies β under block structure.

The decomposition. The raw instrument is the sum of its predictable part and the orthogonalised instrument, and its first stage splits in the same way, with a share δ_{π} from the predictable part. With $\tilde{\beta}_{\hat{z}} := \mathbb{E}[u_t \hat{z}_t]/\,\mathbb{E}[u_t^p \hat{z}_t]$ the estimand with the predictable part as the instrument, the raw estimand is the weighted average

$$\tilde{\beta} = \delta_{\pi}\,\tilde{\beta}_{\hat{z}} + (1-\delta_{\pi})\,\tilde{\beta}^{\perp}, \qquad \tilde{\beta}^{\perp} - \tilde{\beta} = \frac{\delta_{\pi}}{1-\delta_{\pi}}(\tilde{\beta} - \tilde{\beta}_{\hat{z}}).$$

The second expression gives the move of the impulse responses, away from those to the predictable part. Subtracting β from both sides of the first gives the bias of the orthogonalised instrument.

Corollary 1 (Two effects of orthogonalisation)**.** *With* $0 < \delta_{\pi} < 1$,

$$\tilde{\beta}^{\perp} - \beta = \frac{1}{1-\delta_{\pi}}\Big[\underbrace{(\tilde{\beta} - \beta)}_{\text{raw bias}} - \delta_{\pi} \underbrace{(\tilde{\beta}_{\hat{z}} - \beta)}_{\text{bias of the predictable part}}\Big]. \tag{36}$$

Orthogonalisation does two things. It magnifies the raw bias by $1/(1-\delta_{\pi})$, the magnification of Proposition 2. It also subtracts the bias carried by the predictable part, weighted by its share of the first stage. In the reading of Bauer and Swanson (2023b), the predictable part is contamination, and subtracting it is the benefit. In the reading of this paper, the bias that remains is magnified. Orthogonalisation reduces the bias if and only if $\|(\tilde{\beta} - \beta) - \delta_{\pi}(\tilde{\beta}_{\hat{z}} - \beta)\| < (1-\delta_{\pi})\|\tilde{\beta} - \beta\|$. With the predictable part as the instrument, $\tilde{\beta}_{\hat{z}} - \beta = MP\boldsymbol{\psi}/(b'_{p,-p}P\boldsymbol{\psi})$, and substituting it in equation (36) returns $d^{\perp}/\pi^{\perp}$ of Proposition 2.

Magnification by variable and horizon. Proposition 2 compares lengths, which mix units and lose the sign. Orthogonalisation scales the bias of variable k at horizon h by

$$\varrho_{k,h} := \frac{e_k' \Phi_h d^{\perp} / \pi^{\perp}}{e_k' \Phi_h d / \pi} = \frac{e_k' \Phi_h d^{\perp}}{e_k' \Phi_h d} \cdot \frac{1}{1 - \delta_\pi}. \tag{37}$$

It equals ϱ only if $d^{\perp} = (1 - \delta_d)\, d$, and the bound $1/(1 - \delta_\pi)$ does not apply to it. Write the raw contamination as the part the controls span and the remainder, $M\boldsymbol{\psi} = MP\boldsymbol{\psi} + M(I - P)\boldsymbol{\psi}$. $\varrho_{k,h}$ is negative, and the bias of variable k changes sign, when the two parts move variable k in opposite directions and the spanned part dominates. Appendix Table 11 reports the raw and orthogonalised biases and their ratio $\hat{\varrho}_{k,h}$. For the Romer–Romer and Cholesky-based instruments, $\hat{\varrho}_{k,h}$ lies between 0.7 and 1.2, and the sign of the bias is kept. Orthogonalisation barely changes their bias. For the high-frequency surprise, $\hat{\varrho}_{k,h}$ is negative in every cell. The raw surprise points to a milder contraction than the pooled estimate, and the orthogonalised surprise to a deeper one.

The two effects. The decomposition explains this reversal. Appendix Figure 23 plots the decomposition of Corollary 1, horizon by horizon, with the pooled estimate $\hat{\beta}$ in place of β,

$$\underbrace{\Phi_h(\tilde{\beta}^{\perp} - \hat{\beta})}_{\text{orthogonalised bias}} = \underbrace{\frac{1}{1 - \delta_\pi} \Phi_h(\tilde{\beta} - \hat{\beta})}_{\text{raw bias, magnified}} \underbrace{- \frac{\delta_\pi}{1 - \delta_\pi} \Phi_h(\tilde{\beta}_{\hat{z}} - \hat{\beta})}_{\text{predictable part's bias, subtracted}},$$

with the raw bias $\Phi_h(\tilde{\beta} - \hat{\beta})$ in dotted grey. The gap between the grey and orange lines is the magnification, and the blue line is what removing the predictable news does. For the high-frequency surprise, the predictable part carries a large bias of the opposite sign. In the raw surprise, the news the controls span and the remainder offset each other, and the raw bias is small. Orthogonalisation removes one and magnifies the other, and the bias changes sign (Appendix Table 11). For output at twelve months, magnification roughly doubles the raw bias, and the subtraction, with weight $\delta_\pi/(1 - \delta_\pi) = 1.15$, more than offsets it. On impact, the orthogonalised biases of prices and the credit spread come almost entirely from the subtraction. For the Romer–Romer instrument, $\delta_\pi = 0.15$, the subtraction is small, and the two effects nearly offset.

Weak-instrument bias or magnification? With $F = 3.6$, the orthogonalised surprise is weak, and a weak first stage also biases a valid instrument in finite samples. Weak-instrument bias vanishes as the sample grows, and magnification does not. Three exercises separate the two, starting with the one that does not use the pooled estimate.

(A) *The moments before normalisation.* Orthogonalisation removes about half of the surprise's covariance with the funds-rate innovation and 15% of its covariance with output on impact. Its covariances with prices and the credit spread on impact, and with output at twelve months, grow in absolute value. With δ_k the share of the covariance with innovation k removed, the orthogonalised impact response is $\tilde{\beta}_k^{\perp} = \tilde{\beta}_k(1 - \delta_k)/(1 - \delta_\pi)$. If orthogonalisation changed only the strength, $\delta_k = \delta_\pi$ for every variable, up to sampling error, and weak-instrument bias does not move these population shares. The uneven shares show that orthogonalisation changes the contamination. This exercise does not use the pooled estimate.

(B) *Direction.* Weak-instrument bias pulls the estimate toward its least-squares counterpart, the response with the funds rate ordered first in a Cholesky decomposition (Stock and Watson, 2018). On impact, this counterpart lies above the pooled estimate for output and prices and below it for the credit spread. The orthogonalised surprise deviates in the opposite direction for all three variables (Appendix Table 11).

(C) *A clean instrument of the same strength.* The baseline VAR is simulated 1,000 times with the pooled estimate as the true impact vector, and a clean instrument whose first stage matches that of the orthogonalised surprise (median $F = 3.5$) is drawn on the moment window. Its median deviations are small. A deviation at least as large as that of the orthogonalised surprise in Appendix Table 11, and of the same sign, occurs in 9%, 17% and 13% of the draws for prices on impact, output at twelve months and the credit spread on impact, and in 5% for all three together.

Taken together, the moments change unevenly across variables, which a change in strength alone cannot produce. The deviations point away from the least-squares counterpart, and a clean instrument of the same strength rarely produces all three at once. The features of Section 2.2 are therefore mostly a population bias, the contamination that survives orthogonalisation, magnified by a weaker first stage, not finite-sample weak-instrument bias.

C The within estimator: derivations

C.1 Implementation of the within estimator and the pooled instrument

The estimates of Section 6 are computed in seven steps, with sample quantities in place of population ones. Steps 1–3 estimate the direction, steps 4–5 build the pooled instrument, and steps 6–7 give the impulse responses and their uncertainty. The inputs are the VAR residuals $\hat{u}_t$, the Cholesky factor $\hat{L}$ of their covariance matrix, $\hat{\Sigma}_u = \hat{L}\hat{L}'$, the instruments z_{jt}, and the partition $c(\cdot)$ of Section 5.5.

Computation of the pooled instrument. The program in equation (30) involves three unobserved objects, ϕ, Ψ and Σ_ω. At the population moments, equation (22) gives $G^w = q_p\phi' + \mathcal{Q}_{-p}\Psi'$. Since $q_p'\mathcal{Q}_{-p} = \mathbf{0}'$ and $\|q_p\| = 1$, the relevances are $\phi = G^{w\prime}q_p$, and the residual loadings are $R := (I - q_pq_p')G^w = \mathcal{Q}_{-p}\Psi'$. Since $\mathcal{Q}'_{-p}\mathcal{Q}_{-p} = I_{n-1}$, $Rw = \mathbf{0}$ if and only if $\Psi'w = \mathbf{0}$. The cancelling combinations are therefore the weights that set the residual loadings to zero. They can be computed once the direction q_p is known, without knowing Ψ. The steps below replace each population object by its sample counterpart.

1. *Moments and whitening.* For each instrument, $\hat{g}_j = T_j^{-1}\sum_{t\in S_j}\hat{u}_t z_{jt}$, $j = 1,\dots,m$, averaged over the T_j months S_j in which instrument j is observed, and $\hat{G} = [\hat{g}_1,\dots,\hat{g}_m]$. The whitened moments are $\hat{g}_j^w = \hat{L}^{-1}\hat{g}_j$, collected in $\hat{G}^w = \hat{L}^{-1}\hat{G}$. Each moment has a sampling variance $\hat{v}_j$, which gives the precision weight $\varpi_j = 1/\hat{v}_j$ (Appendix C.2).

2. *Demeaning.* With weights $\varpi_j > 0$, $\tilde{\hat{g}}_j^w = \hat{g}_j^w - \sum_{k\in c}\varpi_k\hat{g}_k^w / \sum_{k\in c}\varpi_k$ for $j \in c$, collected in $\widetilde{G}^w = [\tilde{\hat{g}}_1^w,\dots,\tilde{\hat{g}}_m^w]$. Any weights remove τ_c exactly. Demeaning serves only to estimate the direction. Steps 4 and 5 use the moments $\hat{G}^w$ before demeaning.

3. *Direction.* $\hat{q} = \arg\max_{\|q\|=1} q'\widetilde{G}^w\varpi\,\widetilde{G}^{w\prime}q$, with $\varpi = \mathrm{diag}(\varpi_j)$ and the sign fixed by $s_p'\hat{L}\hat{q} > 0$. In the original units, $\hat{b}_p = \hat{L}\hat{q}$ and $\hat{\beta} = \hat{b}_p/(s_p'\hat{b}_p)$. Equal weights give the estimator of Section 5.3.

The application uses the precision weights.

4. *Relevances and residual loadings.* Each whitened moment splits into its projection on $\hat{q}$ and the remainder,
$$\hat{G}^w = \hat{q}\,\hat{\phi}' + \hat{R}, \qquad \hat{\phi}_j := \hat{q}'\hat{g}_j^w, \qquad \hat{R} := (I - \hat{q}\hat{q}')\hat{G}^w. \tag{38}$$
The relevances $\hat{\phi}$ are the projections. The residual loadings $\hat{R}$, the parts of the moments orthogonal to $\hat{q}$, are the sample counterpart of the contamination $\mathcal{Q}_{-p}\Psi'$. The tests and the inference also use the relevance deviations $\hat{\tilde{\phi}}_j = \hat{q}'\tilde{\hat{g}}_j^w$, the whitened within residuals $\hat{r}_j = \tilde{\hat{g}}_j^w - \hat{\tilde{\phi}}_j\hat{q}$, and the family-average contaminations $\hat{\tau}_c^w$ from the residual loadings.

5. *Pooled instrument.* The precision weights replace the noise covariance, $\hat{\Sigma}_\omega := \mathrm{diag}(\hat{v}_j)$. The solution of Appendix C.7, with $\hat{R}$ in place of Ψ', gives the weights
$$\hat{w} \;\propto\; \hat{\Sigma}_\omega^{-1}\hat{\phi}^\perp, \qquad \hat{\phi}^\perp := \hat{\phi} - \hat{R}'(\hat{R}\hat{\Sigma}_\omega^{-1}\hat{R}')^+\hat{R}\hat{\Sigma}_\omega^{-1}\hat{\phi}, \qquad \hat{w}'\hat{\phi} = 1. \tag{39}$$
The solution is exact and in closed form. It is unique once scaled to $\hat{w}'\hat{\phi} = 1$. The pseudo-inverse $(\cdot)^+$ allows $\hat{R}$ to have reduced rank. The pooled instrument is the series $z_t^\star = \sum_j \hat{w}_j\,(T/T_j)\,\tilde{z}_{jt}\mathbf{1}\{t \in S_j\}$, with $\tilde{z}_{jt}$ instrument j demeaned on its support and T the length of the moment window. Its covariance with the VAR residuals is $T^{-1}\sum_t \hat{u}_t z_t^\star = \hat{G}\hat{w}$ exactly. Its first-stage F is the reported strength.

6. *Impulse responses.* $\hat{\Theta}_h = \hat{\Phi}_h\hat{\beta}$. The shock series follows from the proxy-SVAR step with $z_t^\star$.

7. *Uncertainty.* The direction has a two-component variance and a level set (Online Appendix OA.1). The moving-block bootstrap re-runs steps 1–6 on each draw, the weights included, with the partition and the panel of instruments held fixed. The idiosyncratic contamination is not redrawn.

Same impulse responses as the within estimator. The weights of step 5 satisfy $\hat{R}\hat{w} = \mathbf{0}$. Hence,
$$\hat{G}^w\hat{w} = \hat{q}\,(\hat{\phi}'\hat{w}) + \hat{R}\hat{w} = (\hat{\phi}'\hat{w})\,\hat{q}, \qquad \hat{G}\hat{w} = (\hat{\phi}'\hat{w})\,\hat{L}\hat{q}.$$
The moment of the pooled instrument is proportional to $\hat{b}_p = \hat{L}\hat{q}$. After normalisation on the policy rate, the pooled instrument and the within estimator give the same impact vector and the same impulse responses at every horizon, in the sample and in every bootstrap draw that uses the same $\hat{\Sigma}_u$ and moments. Because $\hat{R}$ is built from $\hat{q}$, the pooled instrument inherits any error in the estimated direction.

C.2 The eigenvector step with general weights

The criterion. With a positive definite weight W on the variables, the fit is
$$\sum_c \sum_{j\in c} \varpi_j\,(\hat{g}_j - \phi_j b - \tau_c)'\,W\,(\hat{g}_j - \phi_j b - \tau_c).$$
It is a weighted least-squares fit of the moments. The weight W acts on the variables within each moment, and the weights ϖ_j act on the instruments. The whitened form is the case $W = \Sigma_u^{-1}$. The

application uses the precision weights $\varpi_j = 1/\hat{v}_j$, with $\hat{v}_j := T_j^{-2}\sum_{t\in S_j}\tilde{z}_{jt}^2$ the sampling variance of the whitened moment and $\tilde{z}_{jt}$ the instrument demeaned on its support. The weights also make the direction invariant to rescaling a whole family. Let $\bar{g}_c^{\varpi}$ and $\bar{\phi}_c^{\varpi}$ be the precision-weighted family means of the moments and the relevances.

Concentrating out the family effects and the relevance deviations. The first-order condition for τ_c is $\sum_{j\in c}\varpi_j W(\hat{g}_j - \phi_j b - \tau_c) = \mathbf{0}$. It gives the estimated common contamination of family c, for given b and ϕ,

$$\hat{\tau}_c = \bar{g}_c^{\varpi} - \bar{\phi}_c^{\varpi}\, b.$$

Substituting it back replaces each moment and relevance by its deviation from the weighted family mean, $\tilde{\hat{g}}_j := \hat{g}_j - \bar{g}_c^{\varpi}$ and $\tilde{\phi}_j := \phi_j - \bar{\phi}_c^{\varpi}$. The criterion becomes $\sum_j \varpi_j(\tilde{\hat{g}}_j - \tilde{\phi}_j b)'W(\tilde{\hat{g}}_j - \tilde{\phi}_j b)$. Each term involves only its own $\tilde{\phi}_j$. The derivative of term j with respect to $\tilde{\phi}_j$ is $-2\varpi_j\, b'W(\tilde{\hat{g}}_j - \tilde{\phi}_j b)$. Setting it to zero gives

$$\hat{\tilde{\phi}}_j = \frac{b'W\tilde{\hat{g}}_j}{b'Wb},$$

in which ϖ_j drops out. Substituting back, the criterion is $\sum_j \varpi_j\, \tilde{\hat{g}}_j'W\tilde{\hat{g}}_j - f(b)$, with

$$f(b) = \frac{b'W\widetilde{G}\varpi\widetilde{G}'Wb}{b'Wb}, \qquad \widetilde{G} := [\tilde{\hat{g}}_1, \ldots, \tilde{\hat{g}}_m], \quad \varpi := \operatorname{diag}(\varpi_1, \ldots, \varpi_m).$$

The first term does not depend on b. The direction maximises f.

Eigenvector. With $v := W^{1/2}b$, where $W = W^{1/2\prime}W^{1/2}$, $f = v'Sv/v'v$ with $S := W^{1/2}\widetilde{G}\varpi\widetilde{G}'W^{1/2\prime}$. For a unit vector v, $f = \sum_k \lambda_k (v'v_k)^2$, a weighted average of the eigenvalues of S with weights that sum to one. It is maximised at the leading eigenvector v_1, uniquely up to sign when λ_1 is simple. Hence $\hat{b} \propto W^{-1/2}v_1$, or equivalently $\widetilde{G}\varpi\widetilde{G}'W\,\hat{b} = \lambda_1\,\hat{b}$. With $W = \Sigma_u^{-1}$, $v = q$ and $S = \widetilde{G}^{w}\varpi\widetilde{G}^{w\prime}$, this is step 3 of Appendix C.1.

C.3 Identification

This appendix proves Proposition 3. The lemma shows that the moments determine b_p only up to a relevance-proportional shift.

Lemma C.1 (Observational equivalence)**.** *Hold B_{-p} fixed, let $B = [\, b_p \;\; B_{-p}\,]$ be nonsingular and $\phi \neq \mathbf{0}$. For every $\alpha \in \mathbb{R}^{n-1}$, the pair $(b_p + B_{-p}\alpha,\ \Psi - \phi\alpha')$ generates the same G as (b_p, Ψ) in equation (18). Conversely, if $(\tilde{b}_p, \tilde{\phi}, \tilde{\Psi})$ generates the same G, then for some $\kappa \neq 0$ and $\alpha \in \mathbb{R}^{n-1}$*

$$\tilde{b}_p = \kappa\,(b_p + B_{-p}\alpha), \qquad \tilde{\phi} = \phi/\kappa, \qquad \tilde{\Psi} = \Psi - \phi\,\alpha'.$$

Hence b_p is identified up to scale if and only if the maintained restriction on the contamination excludes $\Psi - \phi\alpha'$ for every $\alpha \neq \mathbf{0}$.

Proof. The first part follows by substitution, $(b_p + B_{-p}\alpha)\phi' + B_{-p}(\Psi - \phi\alpha')' = b_p\phi' + B_{-p}\Psi' = G$. For the converse, because B is nonsingular, $\tilde{b}_p = \kappa\, b_p + B_{-p}a$ for unique $\kappa \in \mathbb{R}$ and $a \in \mathbb{R}^{n-1}$.

Equating the two expressions for G gives $b_p\,(\kappa\tilde{\phi}-\phi)' + B_{-p}\big(a\tilde{\phi}' + \tilde{\Psi}' - \Psi'\big) = \mathbf{0}$. Both brackets therefore vanish. Hence $\phi = \kappa\tilde{\phi}$, which implies $\kappa \neq 0$, and with $\alpha := a/\kappa$, $\tilde{\Psi} = \Psi - \phi\alpha'$. □

Proof of Proposition 3. By Lemma C.1, every alternative is a shift α. Writing $\phi_j = \bar{\phi}_c + \tilde{\phi}_j$, the shift changes the two parts of each loading,

$$\bar{\boldsymbol{\psi}}_c \mapsto \bar{\boldsymbol{\psi}}_c - \bar{\phi}_c\,\alpha, \qquad \boldsymbol{\nu}_j \mapsto \boldsymbol{\nu}_j - \tilde{\phi}_j\,\alpha.$$

The family mean is unrestricted and absorbs the first change. Identification therefore turns on which shifts of the idiosyncratic contamination, $-\tilde{\phi}_j\alpha$, the restriction allows.

(i) Under condition (b), $\mathbb{E}[\boldsymbol{\nu}_j - \tilde{\phi}_j\alpha \mid \boldsymbol{\phi}_c, c] = \mathbb{E}[\boldsymbol{\nu}_j \mid c] - \tilde{\phi}_j\alpha$. Since $\sigma_\phi^2 > 0$, some $\tilde{\phi}_j \neq 0$, and the right-hand side depends on $\boldsymbol{\phi}_c$ unless $\alpha = \mathbf{0}$. Only $\alpha = \mathbf{0}$ is admissible, and b_p is identified up to scale.

(ii) Without condition (b), $\mathbb{E}[\boldsymbol{\nu}_j - \tilde{\phi}_j\alpha \mid \boldsymbol{\phi}_c, c] = \mathbb{E}[\boldsymbol{\nu}_j \mid \boldsymbol{\phi}_c, c] - \tilde{\phi}_j\alpha$ may depend on $\boldsymbol{\phi}_c$ for any α. Only condition (a) restricts the deviations, through their family average. Since the $\tilde{\phi}_j$ sum to zero within each family, the shift leaves that average unchanged, $|c|^{-1}\sum_{j\in c}(\boldsymbol{\nu}_j - \tilde{\phi}_j\alpha) = \bar{\nu}_c$. Every α is admissible, and b_p is not identified. □

C.4 Recovery by the leading eigenvector

This appendix gives the formal version of Proposition 4. It needs a regularity condition on the idiosyncratic contamination, since Assumption 1 restricts only its conditional mean.

Assumption C.1 (Regularity of the idiosyncratic contamination)**.** *Conditionally on the relevances, the $\boldsymbol{\nu}_j$ are independent across instruments, with a common conditional variance $\Sigma_\nu := \mathrm{Var}(\boldsymbol{\nu}_j \mid \boldsymbol{\phi}_c, c)$ in every family, and finite fourth moments.*

The proposition describes the population within scatter $S_0 := \widetilde{G}^w\widetilde{G}^{w\prime}$, the matrix whose leading eigenvector the estimator computes when the moments are known exactly. Each demeaned moment is $\tilde{g}_j^w = \tilde{\phi}_j q_p + \mathcal{Q}_{-p}\tilde{\boldsymbol{\nu}}_j$. Besides the dispersion of relevance σ_ϕ^2, two terms of the idiosyncratic contamination enter S_0. The cross-moment $\boldsymbol{\zeta} := \sum_j \tilde{\phi}_j\boldsymbol{\nu}_j$ measures how much it lines up with the relevance deviations, and it tilts the target axis. The dispersion of the idiosyncratic contamination $N := \sum_j \tilde{\boldsymbol{\nu}}_j\tilde{\boldsymbol{\nu}}_j'$ adds rival axes. Part (i) has no idiosyncratic contamination, part (ii) lets the number of instruments grow, and part (iii) holds in the finite panel at hand.

Proposition C.1 (Recovery by the leading eigenvector)**.** *Let $\sigma_\phi^2 > 0$.*

(i) (Exact block structure.) *If $\boldsymbol{\nu}_j \equiv \mathbf{0}$, the demeaned moments before whitening satisfy $\widetilde{G}\widetilde{G}' = \sigma_\phi^2\, b_p b_p'$, of rank one. For any positive definite weight W on the variables, the leading eigenvector of $\widetilde{G}\widetilde{G}'W$ is b_p. With $W = \Sigma_u^{-1}$, $S_0 = \sigma_\phi^2\, q_p q_p'$, its unique leading eigenvector is q_p, and $b_p = \Sigma_u^{1/2} q_p$.*

(ii) (Idiosyncratic contamination, $m \to \infty$.) *Under Assumptions 1 and C.1, as $m \to \infty$ with $\sigma_\phi^2/m \to \bar{\sigma}_\phi^2$ and the share of instruments left after demeaning $(m-C)/m \to \bar{s} \in (0,1]$, $m^{-1}S_0 \to \bar{\sigma}_\phi^2\, q_p q_p' + \bar{s}\,\mathcal{Q}_{-p}\Sigma_\nu\mathcal{Q}_{-p}'$. Its unique leading eigenvector is q_p if and only if*

$$\bar{\sigma}_\phi^2 > \bar{s}\,\lambda_{\max}(\Sigma_\nu) \qquad \textit{(the ordering condition)}, \tag{40}$$

in which case the leading eigenvector of S_0 converges to q_p.

(iii) (Idiosyncratic contamination, finite m.) *For any realised* $\{\boldsymbol{\nu}_j\}$*, identically,*

$$S_0 = \sigma_\phi^2\, q_p q_p' + q_p(\mathcal{Q}_{-p}\boldsymbol{\zeta})' + (\mathcal{Q}_{-p}\boldsymbol{\zeta})\, q_p' + \mathcal{Q}_{-p} N \mathcal{Q}_{-p}'. \tag{41}$$

If $\sigma_\phi^2 > \lambda_{\max}(N)$ *and* $\|\boldsymbol{\zeta}\|$ *is small relative to the gap* $\varsigma := \sigma_\phi^2 - \lambda_{\max}(N)$*, the leading eigenvector of* S_0 *is*

$$q_p + \mathcal{Q}_{-p}\big(\sigma_\phi^2 I - N\big)^{-1}\boldsymbol{\zeta} + O\big(\|\boldsymbol{\zeta}\|^2/\varsigma^2\big). \tag{42}$$

The second term is the tilt. Under Assumptions 1 and C.1, the tilt has mean zero over the draw of the $\{\boldsymbol{\nu}_j\}$ *to first order. It is of order* $\|\Sigma_\nu\|^{1/2}/\sigma_\phi$*, hence* $O(m^{-1/2})$ *when* σ_ϕ^2 *grows in proportion to* m*.*

Equation (40) defines the per-instrument limit of the ordering condition, equation (27), with $N/m \to \bar{s}\,\Sigma_\nu$.

Proof of Proposition C.1 (recovery by the leading eigenvector). Setup. Work in the orthonormal basis $\mathcal{Q} = [\, q_p \;\; \mathcal{Q}_{-p} \,]$ of equation (22), in which the target is the first axis and the contamination space the remaining $n-1$. Since $\mathcal{Q}' q_p = (1, \mathbf{0}')'$ and $\mathcal{Q}'\mathcal{Q}_{-p} = (\mathbf{0}, I_{n-1})'$, each demeaned moment $\tilde{g}_j^w = \tilde{\phi}_j\, q_p + \mathcal{Q}_{-p}\tilde{\boldsymbol{\nu}}_j$ has coordinates $(\tilde{\phi}_j, \tilde{\boldsymbol{\nu}}_j')'$. Hence,

$$\mathcal{Q}' S_0\, \mathcal{Q} = \sum_j \begin{pmatrix} \tilde{\phi}_j \\ \tilde{\boldsymbol{\nu}}_j \end{pmatrix} \begin{pmatrix} \tilde{\phi}_j & \tilde{\boldsymbol{\nu}}_j' \end{pmatrix} = \begin{pmatrix} \sum_j \tilde{\phi}_j^2 & \sum_j \tilde{\phi}_j \tilde{\boldsymbol{\nu}}_j' \\ \sum_j \tilde{\phi}_j \tilde{\boldsymbol{\nu}}_j & \sum_j \tilde{\boldsymbol{\nu}}_j \tilde{\boldsymbol{\nu}}_j' \end{pmatrix} = \begin{pmatrix} \sigma_\phi^2 & \boldsymbol{\zeta}' \\ \boldsymbol{\zeta} & N \end{pmatrix},$$

where σ_ϕ^2 is the dispersion of relevance within families, N the dispersion of the idiosyncratic contamination, and $\boldsymbol{\zeta}$ the cross-moment between the relevance deviations and the idiosyncratic contamination. Demeaning the contamination does not change the cross-moment, since the $\tilde{\phi}_j$ sum to zero within each family,

$$\sum_j \tilde{\phi}_j \tilde{\boldsymbol{\nu}}_j = \sum_j \tilde{\phi}_j \boldsymbol{\nu}_j - \sum_c \bar{\boldsymbol{\nu}}_c \sum_{j \in c} \tilde{\phi}_j = \sum_j \tilde{\phi}_j \boldsymbol{\nu}_j = \boldsymbol{\zeta}.$$

The three parts of the proposition are three cases of this matrix.

(i) Exact block structure. With $\boldsymbol{\nu}_j \equiv \mathbf{0}$, both $\boldsymbol{\zeta}$ and N vanish, and

$$\mathcal{Q}' S_0\, \mathcal{Q} = \begin{pmatrix} \sigma_\phi^2 & \mathbf{0}' \\ \mathbf{0} & \mathbf{0} \end{pmatrix}.$$

Hence, $S_0 = \sigma_\phi^2\, q_p q_p'$, of rank one, with unique leading eigenvector q_p. Before whitening, $\tilde{g}_j = \tilde{\phi}_j b_p$, and $\widetilde{G}\widetilde{G}' = \sigma_\phi^2\, b_p b_p'$. For any $W \succ 0$, $\widetilde{G}\widetilde{G}' W b = \sigma_\phi^2 (b_p' W b)\, b_p$, whose only eigenvector with a nonzero eigenvalue is b_p.

(ii) Idiosyncratic contamination, $m \to \infty$. The three terms σ_ϕ^2, $\boldsymbol{\zeta}$ and N are sums over instruments, and they grow as instruments are added. Dividing by m gives a matrix with a finite limit and the same eigenvectors,

$$\frac{1}{m}\, \mathcal{Q}' S_0\, \mathcal{Q} = \begin{pmatrix} \sigma_\phi^2/m & \boldsymbol{\zeta}'/m \\ \boldsymbol{\zeta}/m & N/m \end{pmatrix}.$$

The dispersion of relevance converges by assumption, $\sigma_\phi^2/m \to \bar{\sigma}_\phi^2$. The cross-moment $\boldsymbol{\zeta} =$

$\sum_j \tilde{\phi}_j \boldsymbol{\nu}_j$ is a sum of independent terms with mean zero, by Assumptions 1 and C.1. Its variance is $\mathrm{Var}(\boldsymbol{\zeta} \mid \{\phi_j\}) = \sum_j \tilde{\phi}_j^2 \Sigma_\nu = \sigma_\phi^2 \Sigma_\nu$. Its size is therefore of order σ_ϕ, growing at rate $m^{1/2}$. Hence, $\boldsymbol{\zeta}/m = O_p(m^{-1/2})$, which vanishes as $m \to \infty$. The dispersion of the idiosyncratic contamination $N = \sum_j \tilde{\boldsymbol{\nu}}_j \tilde{\boldsymbol{\nu}}_j'$ has terms with expectation $\mathbb{E}[\tilde{\boldsymbol{\nu}}_j \tilde{\boldsymbol{\nu}}_j' \mid c] = (1 - |c|^{-1})\Sigma_\nu$, since demeaning within a family of $|c|$ members removes a share $1/|c|$ of the variance. Summing over j,

$$\frac{1}{m}\,\mathbb{E}[N] = \frac{1}{m}\sum_c (|c| - 1)\,\Sigma_\nu = \frac{m - C}{m}\,\Sigma_\nu \;\longrightarrow\; \bar{s}\,\Sigma_\nu.$$

By the law of large numbers across instruments, $N/m \to \bar{s}\,\Sigma_\nu$, the same limit, since the terms are independent across families, depend on each other within a family only through the family mean, and have finite variance by the finite fourth moments of Assumption C.1. Hence,

$$\frac{1}{m}\,\mathcal{Q}' S_0\,\mathcal{Q} \;\longrightarrow\; \begin{pmatrix} \bar{\sigma}_\phi^2 & \mathbf{0}' \\ \mathbf{0} & \bar{s}\,\Sigma_\nu \end{pmatrix}.$$

As in part (i), the cross-moment vanishes, but the dispersion of the idiosyncratic contamination does not. The eigenvalues of the limit are $\bar{\sigma}_\phi^2$ on the target axis and $\bar{s}\lambda_k(\Sigma_\nu)$ in the contamination space. The target is the unique leading eigenvector if and only if $\bar{\sigma}_\phi^2 > \bar{s}\lambda_{\max}(\Sigma_\nu)$. The leading eigenvector of S_0 converges to q_p.

(iii) Idiosyncratic contamination, finite m. In a finite panel, all three terms stay in the matrix,

$$\mathcal{Q}' S_0\,\mathcal{Q} = \begin{pmatrix} \sigma_\phi^2 & \boldsymbol{\zeta}' \\ \boldsymbol{\zeta} & N \end{pmatrix}.$$

Unlike in parts (i) and (ii), the cross-moment does not vanish. It tilts the leading eigenvector away from q_p, and the eigenvector equation gives the tilt.

Let λ_1 be the largest eigenvalue. Evaluating the Rayleigh quotient at the first axis gives $\lambda_1 \geq \sigma_\phi^2 > \lambda_{\max}(N)$. Write the leading eigenvector in coordinates as $(1, \boldsymbol{x}')'$. The two block rows of the eigenvector equation read $\sigma_\phi^2 + \boldsymbol{\zeta}'\boldsymbol{x} = \lambda_1$ and $\boldsymbol{\zeta} + N\boldsymbol{x} = \lambda_1 \boldsymbol{x}$. Hence,

$$\boldsymbol{x} = (\lambda_1 I - N)^{-1}\boldsymbol{\zeta}, \tag{43}$$

$$\lambda_1 = \sigma_\phi^2 + \boldsymbol{\zeta}'(\lambda_1 I - N)^{-1}\boldsymbol{\zeta}. \tag{44}$$

The solution is unique, and λ_1 is simple. Rotating $(1, \boldsymbol{x}')'$ back with $\mathcal{Q}$ gives $q_p \cdot 1 + \mathcal{Q}_{-p}\boldsymbol{x}$, and the leading eigenvector is, exactly,

$$q_p + \mathcal{Q}_{-p}(\lambda_1 I - N)^{-1}\boldsymbol{\zeta}.$$

From the exact eigenvector to equation (42). The exact eigenvector depends on λ_1, which is itself unknown, since it solves equation (44). Equation (42) replaces λ_1 by the known σ_ϕ^2. The cost of this replacement depends on the gap between the dispersion of relevance and the largest dispersion of the idiosyncratic contamination,

$$\varsigma := \sigma_\phi^2 - \lambda_{\max}(N) > 0.$$

Step 1: λ_1 is close to σ_ϕ^2. Since $\lambda_1 \geq \sigma_\phi^2$, every eigenvalue of $\lambda_1 I - N$ is at least $\lambda_1 - \lambda_{\max}(N) \geq \varsigma$.

Hence, $\|(\lambda_1 I - N)^{-1}\| \le 1/\varsigma$, and equation (44) gives

$$0 \;\le\; \lambda_1 - \sigma_\phi^2 \;=\; \boldsymbol{\zeta}'(\lambda_1 I - N)^{-1}\boldsymbol{\zeta} \;\le\; \frac{\|\boldsymbol{\zeta}\|^2}{\varsigma}.$$

Step 2: replacing λ_1 by σ_ϕ^2. By equation (43), the tilt is $\mathcal{Q}_{-p}\boldsymbol{x}$, with $\boldsymbol{x} = (\lambda_1 I - N)^{-1}\boldsymbol{\zeta}$. Replacing λ_1 by σ_ϕ^2 gives $\boldsymbol{x}_0 := (\sigma_\phi^2 I - N)^{-1}\boldsymbol{\zeta}$. For two invertible matrices, $A^{-1} - B^{-1} = A^{-1}(B - A)B^{-1}$. With $A = \lambda_1 I - N$ and $B = \sigma_\phi^2 I - N$,

$$\boldsymbol{x} - \boldsymbol{x}_0 = -(\lambda_1 - \sigma_\phi^2)\,(\lambda_1 I - N)^{-1}(\sigma_\phi^2 I - N)^{-1}\boldsymbol{\zeta}.$$

Each inverse has norm at most $1/\varsigma$, and Step 1 bounds $\lambda_1 - \sigma_\phi^2$. Hence,

$$\|\boldsymbol{x} - \boldsymbol{x}_0\| \;\le\; \frac{\|\boldsymbol{\zeta}\|^2}{\varsigma}\cdot\frac{1}{\varsigma^2}\cdot\|\boldsymbol{\zeta}\| \;=\; \frac{\|\boldsymbol{\zeta}\|^3}{\varsigma^3}.$$

Step 3: unit length. Eigenvectors are reported with unit length, as q_p is. The vector $(1, \boldsymbol{x}')'$ has length $(1 + \|\boldsymbol{x}\|^2)^{1/2}$, with $\|\boldsymbol{x}\| \le \|\boldsymbol{\zeta}\|/\varsigma$ by equation (43). The unit eigenvector is therefore

$$\frac{q_p + \mathcal{Q}_{-p}\boldsymbol{x}}{(1 + \|\boldsymbol{x}\|^2)^{1/2}}, \qquad \frac{1}{(1 + \|\boldsymbol{x}\|^2)^{1/2}} = 1 + O\big(\|\boldsymbol{\zeta}\|^2/\varsigma^2\big).$$

Step 4: Recovering equation (42). Combining Steps 2 and 3, $\boldsymbol{x} = \boldsymbol{x}_0 + O(\|\boldsymbol{\zeta}\|^3/\varsigma^3)$ and the normalisation adds $O(\|\boldsymbol{\zeta}\|^2/\varsigma^2)$. Hence, the unit leading eigenvector is

$$q_p + \mathcal{Q}_{-p}\boldsymbol{x}_0 + O\big(\|\boldsymbol{\zeta}\|^2/\varsigma^2\big) = q_p + \mathcal{Q}_{-p}(\sigma_\phi^2 I - N)^{-1}\boldsymbol{\zeta} + O\big(\|\boldsymbol{\zeta}\|^2/\varsigma^2\big).$$

First order. Write $(\sigma_\phi^2 I - N)^{-1} = \sigma_\phi^{-2}(I - N/\sigma_\phi^2)^{-1}$. Since $\|N\| < \sigma_\phi^2$, the inverse expands as a geometric series,

$$(\sigma_\phi^2 I - N)^{-1} = \sigma_\phi^{-2}\Big(I + \frac{N}{\sigma_\phi^2} + \frac{N^2}{\sigma_\phi^4} + \cdots\Big) = \sigma_\phi^{-2} I + O\big(\|N\|/\sigma_\phi^4\big).$$

Keeping the leading term, the tilt is $\sigma_\phi^{-2}\mathcal{Q}_{-p}\boldsymbol{\zeta}$. Other terms, such as $\sigma_\phi^{-4}\mathcal{Q}_{-p}N\boldsymbol{\zeta}$, involve products of three or more $\boldsymbol{\nu}_j$, and are smaller when the idiosyncratic contamination is small relative to the dispersion of relevance.

Mean of the tilt. By condition (b) and then condition (a) of Assumption 1,

$$\mathbb{E}[\boldsymbol{\zeta} \mid \{\phi_j\}] = \sum_j \tilde{\phi}_j\, \mathbb{E}[\boldsymbol{\nu}_j \mid \boldsymbol{\phi}_c, c] = \sum_j \tilde{\phi}_j\, \mathbb{E}[\boldsymbol{\nu}_j \mid c] = \mathbf{0}.$$

The first-order tilt has mean zero.

Size of the tilt. As in part (ii), $\mathrm{Var}(\boldsymbol{\zeta} \mid \{\phi_j\}) = \sigma_\phi^2\Sigma_\nu$, and the size of $\boldsymbol{\zeta}$ is of order $\sigma_\phi\|\Sigma_\nu\|^{1/2}$. Since $\|(\sigma_\phi^2 I - N)^{-1}\| \le 1/\varsigma$, the tilt is at most $\|\boldsymbol{\zeta}\|/\varsigma$. With ς of the order of σ_ϕ^2, the tilt is of order $\|\Sigma_\nu\|^{1/2}/\sigma_\phi$, which is $O(m^{-1/2})$ when σ_ϕ^2 grows in proportion to m. Therefore, in a finite panel, the leading eigenvector differs from the target by a tilt that is centred on zero and shrinks as

instruments are added. □

C.5 The two tests of the partition

Section 5.5 states the two tests and their null hypotheses. This appendix gives the construction of the test of no common direction and the asymptotic distribution of the overidentification test.

The test of no common direction: construction. Under $H_0^{\mathrm{d}} : \sigma_\phi^2 = 0$, the demeaned whitened moments contain only sampling noise. The statistic is the leading eigenvalue $\hat{\lambda}_1$ of the weighted within scatter $\widetilde{G}^w \omega \widetilde{G}^{w\prime}$. Its distribution under H_0^{d} depends on the number of instruments, the partition, the weights and the correlation of the sampling errors across instruments. It has no closed form and is simulated in three steps.

1. Draw the m moments in every whitened coordinate from $N(\mathbf{0}, \hat{V})$, with $\hat{V}$ the estimated sampling covariance of the moments, including their correlation across instruments.
2. Demean the draws within the families with the precision weights, and compute the leading eigenvalue of their weighted within scatter.
3. Repeat, and take as the p-value the share of draws in which the leading eigenvalue exceeds $\hat{\lambda}_1$.

The simulation details and the results are in Online Appendix OA.1.3.

The overidentification test: asymptotic distribution. Under H_0^{o}, the whitened moment of $j \in c$ is $g_j^w = \phi_j q_p + \tau_c^w$, a multiple of the direction and the family's common contamination. Stacked over instruments,

$$G^w = q_p \phi' + \sum_c \tau_c^w \iota_c',$$

with ι_c the indicator of the members of family c. The statistic J_T of equation (29) is built in three steps.

Removing the direction. $\hat{X} := \hat{G}^{w\prime}(I_n - \hat{q}\hat{q}')$ is an $m \times n$ matrix. Row j is the whitened moment of instrument j with its component along $\hat{q}$ removed. Under H_0^{o}, at $\hat{q} = q_p$, row j is $\tau_c^{w\prime}(I_n - q_p q_p')$, the same for every member of family c.

Coordinates. Let $\hat{h}_1, \ldots, \hat{h}_{n-1}$ be an orthonormal basis of the directions orthogonal to $\hat{q}$, and $\hat{x}_i := \hat{X}\hat{h}_i$. The m-vector $\hat{x}_i$ collects the coordinates of the m moments along $\hat{h}_i$. Since $\hat{X}\hat{q} = \mathbf{0}$, the $n-1$ vectors $\hat{x}_i$ contain all the information in $\hat{X}$.

Fitting the family structure. Each $\hat{x}_i$ is fitted by generalised least squares, with weight $\hat{V}^{-1}$, on the design $K_R := [\iota_1, \ldots, \iota_C, \hat{\phi}]$, of rank $C+1$. The indicators fit one value per family, the family's common contamination along $\hat{h}_i$. The relevances $\hat{\phi}$ fit what is left of the direction, since $\hat{q}$ differs from q_p in the sample. With

$$\hat{P}_R := \hat{V}^{-1} - \hat{V}^{-1} K_R (K_R' \hat{V}^{-1} K_R)^{-1} K_R' \hat{V}^{-1},$$

which gives the weighted squared residuals of a fit on K_R, the statistic is

$$J_T = \sum_{i=1}^{n-1} \hat{x}_i' \, \hat{P}_R \, \hat{x}_i.$$

It matches equation (29), as the trace splits over the columns of $\hat{X}$ in the basis $(\hat{q}, \hat{h}_1, \ldots, \hat{h}_{n-1})$, and the column along $\hat{q}$ is zero. It does not depend on the weights used to demean. The distribution follows under five assumptions.

T1. The partition is fixed before the moments are formed, and m and n are fixed as $T \to \infty$.

T2. $\sqrt{T}\,\mathrm{vec}(\hat{G}^w - G^w) \xrightarrow{d} N(\mathbf{0}, \Omega)$ (Online Appendix Assumption OA.1.1).

T3. $\Omega = \bar{V} \otimes I_n$, which conditional homoskedasticity of the innovations ensures.

T4. $T\hat{V} \xrightarrow{p} \bar{V}$, positive definite.

T5. Condition (P1) holds with a distinct leading eigenvalue, and $\hat{q} - q_p = O_p(T^{-1/2})$.

T1 rules out choosing the families after seeing the moments. T2 and T3 give the sampling errors a normal limit, independent across the n whitened coordinates. T4 makes $\hat{V}$ a valid weight. T5 makes the estimated direction consistent, which the first test supports.

Proposition C.2 (The overidentification test). *Under H_0^o and T1–T5, $J_T \xrightarrow{d} \chi^2_{(n-1)(m-C-1)}$.*

Proof. The coordinates under H_0^o. Let $\hat{\mathcal{E}} := \hat{G}^w - G^w$ be the sampling error. Since $\hat{q}'\hat{h}_i = 0$, $\hat{x}_i = \hat{G}^{w\prime}(I_n - \hat{q}\hat{q}')\hat{h}_i = \hat{G}^{w\prime}\hat{h}_i = G^{w\prime}\hat{h}_i + \hat{\mathcal{E}}'\hat{h}_i$. Substituting the stacked moments,

$$\hat{x}_i = \underbrace{\sum_c \iota_c\,(\tau_c^{w\prime}\hat{h}_i)}_{\text{family effects}} + \underbrace{\phi\,(q_p'\hat{h}_i)}_{\text{relevance term}} + \underbrace{\hat{\mathcal{E}}'\hat{h}_i}_{\text{sampling error}} \quad . \tag{45}$$

The family effects. The first term gives every member of family c the same value, $\tau_c^{w\prime}\hat{h}_i$, the family's common contamination along $\hat{h}_i$. It is a combination of the indicators ι_c, the first C columns of K_R, and the fit removes it exactly.

The relevance term. Its coefficient is $q_p'\hat{h}_i$, which is $O_p(T^{-1/2})$ by T5, since $\hat{h}_i$ is orthogonal to $\hat{q}$ and $\hat{q}$ is close to q_p. Writing it with $\hat{\phi} = \phi + O_p(T^{-1/2})$, the last column of K_R, changes it by $O_p(T^{-1})$. Since J_T does not depend on the choice of basis, fix an orthonormal basis $h_1, \ldots, h_{n-1}$ of the directions orthogonal to q_p, and take the $\hat{h}_i$ as their projections onto the directions orthogonal to $\hat{q}$, orthonormalised. By T5, $\hat{h}_i - h_i = O_p(T^{-1/2})$. Replacing $\hat{h}_i$ by h_i in the sampling error changes it by $O_p(T^{-1})$ as well. Hence,

$$\hat{x}_i = K_R\gamma_i + \hat{\mathcal{E}}'h_i + O_p(T^{-1}), \qquad \gamma_i := \left(\tau_1^{w\prime}\hat{h}_i, \ldots, \tau_C^{w\prime}\hat{h}_i,\ q_p'\hat{h}_i\right)'.$$

The first C entries of γ_i are the family effects, one per indicator ι_c. The last entry is the coefficient of the relevance term, now on $\hat{\phi}$. The two systematic terms of equation (45) are thus one combination of the columns of K_R.

From $\hat{x}_i$ to J_T. The two systematic terms are unknown. Removing them leaves only the sampling error, whose distribution is known. Multiplying out,

$$\hat{P}_R K_R = \hat{V}^{-1}K_R - \hat{V}^{-1}K_R(K_R'\hat{V}^{-1}K_R)^{-1}K_R'\hat{V}^{-1}K_R = \mathbf{0}.$$

A vector in the span of the columns of K_R is fitted exactly and leaves no residual. Multiplying $\hat{x}_i$ by $\hat{P}_R$, the term $K_R\gamma_i$ drops out whatever γ_i is, and leaves $\hat{P}_R\hat{x}_i = \hat{P}_R\left(\hat{\mathcal{E}}'h_i + O_p(T^{-1})\right)$. Only the

sampling error of equation (45) remains, with h_i in place of $\hat{h}_i$. By T4, $\hat{V}^{-1} = T\bar{V}^{-1}(1 + o_p(1))$, and $\hat{P}_R = TP_R(1 + o_p(1))$, with P_R the same matrix with $\bar{V}^{-1}$ in place of $\hat{V}^{-1}$ and ϕ in place of $\hat{\phi}$. Hence,

$$J_T = \sum_{i=1}^{n-1} \hat{x}_i' \hat{P}_R \hat{x}_i = \sum_{i=1}^{n-1} \left(\sqrt{T}\hat{\mathcal{E}}' h_i\right)' P_R \left(\sqrt{T}\hat{\mathcal{E}}' h_i\right) + o_p(1).$$

The distribution. By T2 and T3, the vectors $\sqrt{T}\hat{\mathcal{E}}' h_i$ are asymptotically independent $N(\mathbf{0}, \bar{V})$ across i, since $\Omega = \bar{V} \otimes I_n$ makes the sampling errors independent across the whitened coordinates with a common covariance $\bar{V}$, and the h_i are orthonormal. Each quadratic form is χ^2 with $\operatorname{rank}(P_R) = m - C - 1$ degrees of freedom, since P_R removes the $C + 1$ columns of K_R from the m dimensions, and $\bar{V}^{1/2} P_R \bar{V}^{1/2}$ is idempotent. The $n - 1$ forms add. □

C.6 What averaging removes, and what it leaves

The simple average (Section 5.6) shrinks the idiosyncratic contamination at rate $m^{-1/2}$ and leaves the common component untouched. Write each loading as a panel-wide mean plus a deviation, $\boldsymbol{\psi}_j = \bar{\boldsymbol{\psi}} + \boldsymbol{\xi}_j$, with the deviations $\boldsymbol{\xi}_j$ independent across instruments, of mean zero and variance Σ_ψ. A combination $w' z_t$ then carries the contamination

$$\Psi' w = \Big(\sum_j w_j\Big)\bar{\boldsymbol{\psi}} + \sum_j w_j \boldsymbol{\xi}_j. \tag{46}$$

The first term enters with the sum of the weights, which is one for any average. The averaged instrument therefore identifies the direction of $b_p + B_{-p}\bar{\boldsymbol{\psi}}/\bar{\phi}$, with $\bar{\phi}$ the mean relevance. No number of instruments reduces this tilt.

The idiosyncratic component shrinks. The equal-weight average, $w_j = 1/m$, has contamination

$$\Psi' w = \bar{\boldsymbol{\psi}} + \bar{\boldsymbol{\xi}}_m, \qquad \bar{\boldsymbol{\xi}}_m := \frac{1}{m}\sum_j \boldsymbol{\xi}_j,$$

and, since the deviations are independent across instruments, with mean zero and finite variance,

$$\mathbb{E}[\bar{\boldsymbol{\xi}}_m] = \mathbf{0}, \qquad \operatorname{Var}(\bar{\boldsymbol{\xi}}_m) = \frac{1}{m^2}\sum_j \operatorname{Var}(\boldsymbol{\xi}_j) = \frac{\Sigma_\psi}{m}.$$

The average of m independent deviations has a $1/m$ share of the variance of one of them. Chebyshev's inequality turns this into a rate: for every $M > 0$,

$$\Pr\left(\sqrt{m}\,\|\bar{\boldsymbol{\xi}}_m\| > M\right) \le \frac{m\ \mathbb{E}\,\|\bar{\boldsymbol{\xi}}_m\|^2}{M^2} = \frac{m\ \operatorname{tr}\operatorname{Var}(\bar{\boldsymbol{\xi}}_m)}{M^2} = \frac{\operatorname{tr}\Sigma_\psi}{M^2}.$$

The bound does not depend on m. Hence, $\bar{\boldsymbol{\xi}}_m = O_p(m^{-1/2})$. Only a finite second moment is needed.

The common component does not. With $\bar{\phi} := m^{-1}\sum_j \phi_j$ the mean relevance, the moment of the averaged instrument is

$$\frac{1}{m}\sum_j g_j = \bar{\phi}\, b_p + B_{-p}\bar{\boldsymbol{\psi}} + B_{-p}\bar{\boldsymbol{\xi}}_m = \bar{\phi}\Big(b_p + \frac{B_{-p}\bar{\boldsymbol{\psi}}}{\bar{\phi}}\Big) + O_p(m^{-1/2}). \tag{47}$$

Provided $\bar{\phi}$ stays bounded away from zero, the proxy-SVAR estimand of the averaged instrument converges to the normalised $b_p + B_{-p}\bar{\boldsymbol{\psi}}/\bar{\phi}$. This is the contaminated impact vector equation (6) of a single instrument carrying the panel's mean loading. Adding instruments measures the wrong target more precisely.

C.7 The constrained-optimal instrument

The weights of the pooled instrument solve the program in equation (30). They maximise the concentration parameter, relevance squared over noise, subject to cancelling the contamination, $\Psi' w = \mathbf{0}$. Standardise the noise, with $v := \Sigma_\omega^{1/2} w$, $\tilde{a} := \Sigma_\omega^{-1/2}\phi$ and $\tilde{\Psi} := \Sigma_\omega^{-1/2}\Psi$. The program in equation (30) becomes $\max_v (v'\tilde{a})^2/v'v$ subject to $\tilde{\Psi}' v = \mathbf{0}$. Let $\tilde{P}_\perp := I - \tilde{\Psi}(\tilde{\Psi}'\tilde{\Psi})^{+}\tilde{\Psi}'$ be the projection onto the orthogonal complement of the columns of $\tilde{\Psi}$. A vector is feasible if and only if $\tilde{P}_\perp v = v$. Hence, $v'\tilde{a} = v'\tilde{P}_\perp\tilde{a}$. By Cauchy–Schwarz the maximum, $\|\tilde{P}_\perp\tilde{a}\|^2$, is attained at $v^\star \propto \tilde{P}_\perp\tilde{a}$, which is feasible since $\tilde{\Psi}'\tilde{P}_\perp = \mathbf{0}$. Returning to $w = \Sigma_\omega^{-1/2} v$ and normalising $w'\phi = 1$,

$$w^\star = \frac{\Sigma_\omega^{-1}\phi^\perp}{\phi^{\perp\prime}\Sigma_\omega^{-1}\phi^\perp}, \qquad \phi^\perp := \phi - \Psi(\Psi'\Sigma_\omega^{-1}\Psi)^{+}\Psi'\Sigma_\omega^{-1}\phi, \qquad \mu^2(w^\star) = T\,\phi^{\perp\prime}\Sigma_\omega^{-1}\phi^\perp.$$

Here $\phi^\perp$ is the residual of ϕ on the columns of Ψ in the Σ_ω^{-1} metric. The Moore–Penrose inverse means no rank condition on Ψ is needed. With $\Sigma_\omega = \sigma_\omega^2 I$, $\tilde{P}_\perp = P_\perp$ and $\phi^\perp = P_\perp\phi$, which gives equation (31). Since $\|\tilde{P}_\perp\tilde{a}\|^2 = \|\tilde{a}\|^2 - \|(I-\tilde{P}_\perp)\tilde{a}\|^2$, the constraint costs the relevance collinear with the contamination.

C.8 Unconstrained and constrained pooling

Unconstrained pooling only adds strength. Constrained pooling adds strength and cancels the contamination, at the cost of the relevance that lines up with it. The three cases below use the solution of Appendix C.7, with independent homoskedastic noise, $\Sigma_\omega = \sigma_\omega^2 I$, as in Appendix C.9.

(i) The simple average. With equal weights, $w_j = 1/m$,

$$w'\phi = \frac{1}{m}\sum_j \phi_j, \qquad \mathrm{Var}(w'\omega_t) = \frac{\sigma_\omega^2}{m}, \qquad \mu^2 = \frac{T}{\sigma_\omega^2}\frac{1}{m}\Big(\sum_j \phi_j\Big)^2.$$

The average uses only the mean relevance across all instruments. Its strength grows in proportion to the number of instruments at a given mean relevance. Its contamination is $\bar{\boldsymbol{\psi}} + \bar{\boldsymbol{\xi}}_m$ (Appendix C.6). The idiosyncratic part shrinks as instruments are added, and the common part does not.

(ii) Unconstrained pooling. Without the restriction $\Psi' w = \mathbf{0}$, the weights maximise strength alone.

The maximum is attained at $w \propto \phi$, the weights proportional to the relevances, and

$$\mu^2 = \frac{T}{\sigma_\omega^2}\|\phi\|^2 = \sum_j \mu_j^2 = \frac{T}{\sigma_\omega^2}\Big[\,\sigma_\phi^2 + \sum_c |c|\,\bar{\phi}_c^2\Big].$$

The strengths of the instruments add. The pool is at least as strong as the strongest instrument, which is the combination $w = e_j$, and at least as strong as the simple average. The squared relevance splits into the dispersion within families and the family levels, and unconstrained pooling uses both terms in full. Its contamination is $\Psi' w$, which the weights do not restrict. The common contamination therefore remains.

(iii) Constrained pooling. The strongest cancelling combination keeps the dispersion of relevance within families in full, and the family levels only in the combinations whose contaminations offset (Appendix C.9),

$$\mu^2(w^\star) = \frac{T}{\sigma_\omega^2}\Big[\,\sigma_\phi^2 + \|P_F\,\phi\|^2\Big].$$

Relative to unconstrained pooling, it gives up the part of the family levels that the family contaminations can reproduce. This is the cost of cancelling the contamination.

C.9 The strength of the strongest cancelling combination

The strength of the strongest cancelling combination splits into two sources, the dispersion of relevance within families and the relevance of the family levels. The derivation splits the weights in the same way.

Contamination depends only on family totals. Under exact block structure, every member of family c carries the family's contamination $\bar{\boldsymbol{\psi}}_c$. The contamination of a combination is therefore

$$\Psi' w = \sum_j w_j \bar{\boldsymbol{\psi}}_{c(j)} = \sum_c a_c \bar{\boldsymbol{\psi}}_c, \qquad a_c := \sum_{j \in c} w_j.$$

Moving weight between members of the same family leaves it unchanged. Only the total weight on each family matters.

Splitting the weights. Any weight vector splits into a family part, which sets the family totals, and a within part, which moves weight inside families,

$$w = w_F + w_W, \qquad (w_F)_j = \frac{a_c}{|c|}, \qquad (w_W)_j = w_j - \frac{a_c}{|c|} \qquad (j \in c).$$

The family part spreads each family total equally over its members and is constant within each family. The within part sums to zero within every family.

What each part cancels. The within part has family totals of zero. It therefore always cancels the contamination, $\Psi' w_W = \mathbf{0}$. These combinations, the within-family contrasts, span $m - C$ dimensions. The family part carries all the contamination, $\Psi' w = \Psi' w_F = \sum_c a_c \bar{\boldsymbol{\psi}}_c$. It cancels only if the family totals satisfy $\sum_c a_c \bar{\boldsymbol{\psi}}_c = \mathbf{0}$. The C family contaminations span r dimensions. The

condition imposes r restrictions on C totals, and the cancelling family-level combinations span $C - r$ dimensions.

What each part captures of relevance. The relevance of a combination splits in the same way. The family part gives $w_F'\phi = \sum_c (a_c/|c|) \sum_{j \in c} \phi_j = \sum_c a_c \bar{\phi}_c$. Since the within part sums to zero within each family, subtracting the family mean from ϕ_j leaves $w_W'\phi$ unchanged, $w_W'\phi = \sum_j (w_W)_j \phi_j - \sum_c \bar{\phi}_c \sum_{j \in c} (w_W)_j = \sum_j (w_W)_j (\phi_j - \bar{\phi}_c)$ with $\sum_{j \in c} (w_W)_j = 0$. Hence,

$$w'\phi = w_F'\phi + w_W'\phi = \underbrace{\sum_c a_c \, \bar{\phi}_c}_{\text{family part}} \; + \; \underbrace{\sum_j (w_W)_j \, (\phi_j - \bar{\phi}_c)}_{\text{within part}} .$$

The family part captures only the mean relevance of each family, as the family averages would. The within part captures only the relevance deviations, as contrasts between members of a family would.

The two parts are orthogonal. Within each family, the two components are orthogonal,

$$w_F' w_W = \sum_c \frac{a_c}{|c|} \sum_{j \in c} (w_W)_j = 0,$$

since the within weights sum to zero in every family, $\sum_{j \in c} (w_W)_j = 0$. The cancelling combinations are therefore the orthogonal sum of the within-family contrasts and the cancelling family-level combinations. The projection on them splits as $P_\perp = P_W + P_F$, with P_W the projection on the within-family contrasts and P_F the projection on the cancelling family-level combinations.

The strength. With homoskedastic noise, the strength of the strongest cancelling combination is $\mu^2(w^\star) = T\|P_\perp \phi\|^2 / \sigma_\omega^2$ (Section 5.6). The factor T/σ_ω^2 converts squared relevance into concentration. Since the two projections are orthogonal, $\|P_\perp \phi\|^2 = \|P_W \phi\|^2 + \|P_F \phi\|^2$. Projecting ϕ on the within-family contrasts subtracts each family's mean relevance. Its entries are the relevance deviations $\phi_j - \bar{\phi}_c$, and the sum of their squares is σ_ϕ^2. Hence,

$$\|P_\perp \phi\|^2 = \sigma_\phi^2 + \|P_F \, \phi\|^2. \tag{48}$$

The first term is the dispersion of relevance within families. It comes from the within-family contrasts alone, and it is the strength of the pooled contrast instrument $\sum_j \tilde{\phi}_j z_{jt}$. The second term is the relevance of the family levels that survives the cancellation. It is nonzero for generic ϕ if and only if $C > r$, which is partition condition (P2) of Section 5.5.

A closed form for the family-level term. Let the C families have equal size k, such that $m = Ck$, and stack the family means, $\bar{\phi}_F := (\bar{\phi}_1, \ldots, \bar{\phi}_C)'$ and $\bar{\Psi} := [\bar{\boldsymbol{\psi}}_1, \ldots, \bar{\boldsymbol{\psi}}_C]$, of rank r. A family-level combination with totals a has squared length $\|w_F\|^2 = \|a\|^2/k$ and relevance $w_F'\phi = a'\bar{\phi}_F$. It cancels the contamination if and only if $a \in \ker \bar{\Psi}$. The projection of ϕ on the cancelling family-level

combinations is therefore

$$\|P_F\,\phi\|^2 = k\,\bar{\phi}_F'\big(I_C - H\big)\bar{\phi}_F, \qquad H := \bar{\Psi}'(\bar{\Psi}\bar{\Psi}')^{+}\bar{\Psi},$$

with $I_C - H$ the projection on $\ker\bar{\Psi}$. Equivalently, $\|P_F\,\phi\|^2$ is k times the residual sum of squares of a regression across families of the mean relevances on the family contaminations,

$$\|P_F\,\phi\|^2 = k\,\big\|\bar{\phi}_F - \bar{\Psi}'\hat{\gamma}\big\|^2, \qquad \hat{\gamma} = (\bar{\Psi}\bar{\Psi}')^{+}\bar{\Psi}\bar{\phi}_F,$$

since the fitted values are $\bar{\Psi}'\hat{\gamma} = H\bar{\phi}_F$. The term counts the part of the family levels of relevance that the family contaminations cannot reproduce. Let $a_1^\star, \ldots, a_{C-r}^\star$ be an orthonormal basis of $\ker\bar{\Psi}$, the cancelling family totals. Then $I_C - H = \sum_i a_i^\star a_i^{\star\prime}$, and

$$\|P_F\,\phi\|^2 = k\sum_{i=1}^{C-r}\big(a_i^{\star\prime}\bar{\phi}_F\big)^2.$$

Each independent family-level combination whose contaminations offset adds the squared relevance it picks up. With $C \le r$, there is no such combination, $H = I_C$, and the term is zero. In the baseline, $C = 4$ and $r = 3$. One combination of family totals, unique up to scale, makes the family contaminations offset, and $\|P_F\,\phi\|^2 = k(a^{\star\prime}\bar{\phi}_F)^2$.

The concentration parameter of the pooled instrument. Combining the two terms, the strongest cancelling combination has a closed-form concentration parameter,

$$\mu^2(w^\star) = \frac{T}{\sigma_\omega^2}\Big[\,\sigma_\phi^2 + k\sum_{i=1}^{C-r}\big(a_i^{\star\prime}\bar{\phi}_F\big)^2\Big].$$

The first term comes from the within-family contrasts and needs only condition (P1). The second comes from the family levels and needs condition (P2). With unequal family sizes, the regression across families is weighted by the family sizes $|c|$.

D Simulated impulse responses

Figure 3 shows in two dimensions how demeaning within families recovers the direction of the policy shock. This appendix shows what that does to the impulse responses, in a simulated three-variable VAR. Figure 3 uses its own simulated points, with the same structure of the moments.

Design. The VAR has lag order one, $y_t = A\,y_{t-1} + u_t$, with the policy variable third, and

$$A = \begin{pmatrix} 0.90 & 0 & -0.08 \\ 0.10 & 0.85 & -0.05 \\ 0.05 & 0.05 & 0.80 \end{pmatrix}, \qquad b_p = (-0.45,\ -0.30,\ 1)'.$$

Two families of six instruments have moments $g_j = \phi_j\,b_p + \tau_c + \nu_j$, with relevance ϕ_j spread evenly from 0.4 to 2.6 within each family, with a small perturbation, and family contaminations $\tau_1 = (-0.5,\ 0.9,\ 0.35)'$ and $\tau_2 = (0.8,\ -0.6,\ -0.25)'$. The idiosyncratic contamination is $\nu_j \sim N(\mathbf{0}, 0.05^2 I)$. Within each family, its part orthogonal to b_p is made uncorrelated with relevance and with its part

along b_p. This imposes in the sample what Assumption 1(b) imposes in population. Hence, the figure shows identification rather than sampling error. Four directions are computed. Two are the mean moments of the two families, each family's average instrument used as the only instrument. The third is the first principal component of all twelve moments, without demeaning. The fourth is the first principal component after demeaning within families. Each is normalised to $+1$ on the policy variable on impact, with impulse responses $A^h b$. The whitening step is omitted. Hence, the example isolates the within transform.

Results. With its average instrument as the only instrument, each family identifies the true direction shifted by its contamination direction, magnified by its first stage, as the bias identity in equation (7) predicts, and the two families err in opposite directions. Pooling without demeaning does not average the errors away. Its first principal component is pulled toward the contamination and lies further from the truth than either family alone. After demeaning, the first principal component is the true direction, and its impulse responses coincide with the true ones at every horizon (Figure 7).

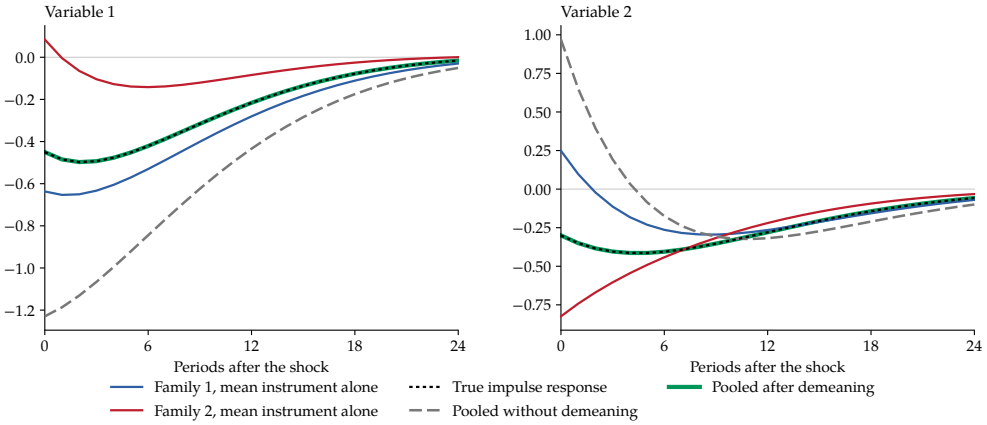


Figure 7: **Simulated impulse responses: demeaning within families recovers the true response.** Variables 1 and 2, after a policy shock that raises the policy variable by one unit on impact. Family 1 and 2: each family's average instrument as the only instrument. Pooled: the first principal component of the moments, without and after demeaning within families. True: $A^h b_p$.

E Supplementary analyses for the empirical application

E.1 The family-common component: derivation

Section 6.4 compares each family's average instrument with the pooled estimate. This appendix derives equation (32).

The family's average instrument. With the precision weights $\varpi_j = 1/\hat{v}_j$ of the baseline, the average instrument of family c is $\bar{z}_{ct} := \sum_{j\in c} \varpi_j z_{jt} / \sum_{j\in c} \varpi_j$, on the supports of its members as in step 5 of Appendix C.1. Its moment is the precision-weighted mean moment, $\bar{g}_c = \mathrm{Cov}(u_t, \bar{z}_{ct}) = \sum_{j\in c} \varpi_j g_j / \sum_{j\in c} \varpi_j$, with sample analogue $\hat{\bar{g}}_c$. Under Assumption 1, every member has $g_j = \phi_j b_p + \tau_c + B_{-p}\nu_j$, and hence

$$\bar{g}_c = \bar{\phi}_c\, b_p + \bar{\tau}_c, \qquad \bar{\tau}_c := \tau_c + B_{-p}\bar{\nu}_c,$$

with $\bar{\phi}_c$ and $\bar{\nu}_c$ the precision-weighted means of the relevances and of the idiosyncratic deviations. The family's mean contamination $\bar{\tau}_c$ equals τ_c under exact block structure.

The shift in the sample. Let $\hat{\pi}_c := s_p' \hat{\bar{g}}_c$ be the covariance of the average instrument with the funds-rate innovation, the family's first stage. As the only instrument, the average instrument $\bar{z}_{ct}$, the precision-weighted mean of the family's members defined above, gives the impact vector normalised on the funds rate, $\tilde{\beta}_c = \hat{\bar{g}}_c / \hat{\pi}_c$. Subtracting the pooled estimate $\hat{\beta}$, written as $\hat{\pi}_c\hat{\beta}/\hat{\pi}_c$,

$$\tilde{\beta}_c - \hat{\beta} = \frac{\hat{\bar{g}}_c - \hat{\pi}_c\,\hat{\beta}}{\hat{\pi}_c}.$$

The numerator is the family's mean moment net of the part the pooled estimate reads as a policy shock. Each bar of Figure 5 plots the impulse response $\Phi_h(\tilde{\beta}_c - \hat{\beta})$ at horizon h.

The limit. As $T \to \infty$, $\hat{\bar{g}}_c \xrightarrow{p} \bar{g}_c$ and $\hat{\pi}_c \xrightarrow{p} s_p'\bar{g}_c = \bar{\phi}_c\, b_{pp} + \bar{\tau}_{c,p} = \bar{\phi}_c^p + \bar{\tau}_{c,p}$, with $\bar{\phi}_c^p := \bar{\phi}_c\, b_{pp}$ the family's mean relevance in units of the policy rate. When the pooled estimate is consistent, $\hat{\beta} \xrightarrow{p} \beta = b_p / b_{pp}$, as under exact block structure (Proposition 3), the numerator converges to

$$\bar{g}_c - (\bar{\phi}_c^p + \bar{\tau}_{c,p})\beta = \bar{\phi}_c\, b_p + \bar{\tau}_c - \bar{\phi}_c\, b_{pp}\,\beta - \bar{\tau}_{c,p}\,\beta = \bar{\tau}_c - \bar{\tau}_{c,p}\,\beta,$$

since $b_{pp}\beta = b_p$. The relevance drops out. Dividing by the limit of $\hat{\pi}_c$ gives the limit in equation (32). Each bar therefore measures the family's common contamination together with the mean idiosyncratic contamination of its members.

Contamination and first stage. The bars are in units of impulse responses, free of the units of the instruments, since $\tilde{\beta}_c$ is normalised on the funds rate. Their numerator and denominator, taken separately, are not. Each is in the units of the family's instruments, which differ across families. Dividing both by $\mathrm{sd}(\hat{u}_t^p)\,\mathrm{sd}(\bar{z}_{ct})$ gives

$$\tilde{\beta}_c - \hat{\beta} = \frac{\hat{\kappa}_c}{\hat{\rho}_c}, \qquad \hat{\kappa}_c := \frac{\hat{\bar{g}}_c - \hat{\pi}_c\,\hat{\beta}}{\mathrm{sd}(\hat{u}_t^p)\,\mathrm{sd}(\bar{z}_{ct})}, \qquad \hat{\rho}_c := \mathrm{corr}(\hat{u}_t^p, \bar{z}_{ct}).$$

The first-stage correlation $\hat{\rho}_c$ is unit-free, and it determines the first-stage F, which is approximately $T\hat{\rho}_c^2/(1-\hat{\rho}_c^2)$. The standardised contamination $\hat{\kappa}_c$ is in units of impulse responses, like the bars. Each bar is the family's standardised contamination magnified by $1/\hat{\rho}_c$. The division changes neither the bars nor their ranking. It only splits each bar into its two parts, which Appendix Table 7 reports.

E.2 Decomposing a change of VAR

Adding a variable to the VAR changes the impulse responses for two reasons. The VAR changes, and the pooled instrument is re-estimated in the new VAR. Holding the baseline pooled instrument fixed in the larger VAR separates the two. Section 7.2 reports the decomposition.

Three estimates. Let VAR A be the baseline and VAR B a five-variable system. The baseline pooled instrument is $z^\star_{A,t} := \sum_j w_{A,j} z_{jt}$, with the weights w_A estimated in VAR A. The re-pooled instrument $z^\star_{B,t}$ has weights w_B estimated in VAR B. The three estimates are

(1) VAR A with the instrument $z^\star_{A,t}$: $\Theta^{(1)}_h = \Phi^A_h \tilde{\beta}^{(1)}$, the baseline;

(2) VAR B with the same instrument $z^\star_{A,t}$: $\Theta^{(2)}_h = \Phi^B_h \tilde{\beta}^{(2)}$;

(3) VAR B with the re-pooled instrument $z^\star_{B,t}$: $\Theta^{(3)}_h = \Phi^B_h \tilde{\beta}^{(3)}$, the estimate of Figure 6.

Φ^A_h and Φ^B_h are the moving-average coefficients of the two VARs.

Change of VAR and change of instrument. For the variables common to both VARs,

$$\underbrace{\Theta^{(3)}_h - \Theta^{(1)}_h}_{\text{total change}} = \underbrace{\Theta^{(2)}_h - \Theta^{(1)}_h}_{\text{change of VAR, same instrument}} + \underbrace{\Theta^{(3)}_h - \Theta^{(2)}_h}_{\text{change of instrument, same VAR}} .$$

The decomposition is sequential. It changes the VAR first and the instrument second.

What each term contains. VAR B has the non-policy shocks of A and one new shock ε^k_t, with impact vector b^B_k and funds-rate entry b^B_{pk}. Both VARs are assumed to recover the same policy shock, with normalised impact vectors β_A and β_B. The weights w_A satisfy $\Psi'_A w_A = \mathbf{0}$. The baseline instrument therefore carries no contamination from the non-policy shocks of A. Its loading on the new shock, $\psi^\star_k := \sum_j w_{A,j}\psi_{jk}$, with ψ_{jk} the loading of instrument j on ε^k_t, is not restricted. Its moment with the innovations of B is $(w'_A\phi)\, b^B_p + b^B_k\, \psi^\star_k$. By equation (7),

$$\tilde{\beta}^{(2)} = \beta_B + \frac{d^\star_k}{\pi^\star}, \qquad d^\star_k := \left(b^B_k - \beta_B\, b^B_{pk}\right) \psi^\star_k, \qquad \pi^\star := (w'_A\phi)\, b^B_{pp} + b^B_{pk}\, \psi^\star_k.$$

The baseline instrument identifies the impact vector of B and the contamination direction of the new shock, magnified by its first stage. Re-pooled in B, the within transform removes each family's common contamination, from the new shock as well, for any number of families. Under exact block

structure, $\tilde{\beta}^{(3)} = \beta_B$, and $\tilde{\beta}^{(1)} = \beta_A$. The two terms of the decomposition are therefore

$$\underbrace{\Theta_h^{(2)} - \Theta_h^{(1)}}_{\text{change of VAR}} = \left(\Phi_h^B \beta_B - \Phi_h^A \beta_A\right) + \Phi_h^B \frac{d_k^\star}{\pi^\star}, \qquad \underbrace{\Theta_h^{(3)} - \Theta_h^{(2)}}_{\text{change of instrument}} = -\Phi_h^B \frac{d_k^\star}{\pi^\star}.$$

The change of instrument isolates the contamination that the baseline instrument carries from the new shock, with its sign reversed. The change of VAR contains the change in the impulse responses of the policy shock, $\Phi_h^B \beta_B - \Phi_h^A \beta_A$, and the same contamination. It measures the change in the impulse responses alone only when the baseline instrument does not load on the new shock, $\psi_k^\star = 0$. The derivation assumes that the larger VAR recovers the same policy shock. If the baseline VAR is not informationally sufficient, its policy shock is a mix of shocks, and the change of VAR also reflects that.

E.3 The internal-instrument VAR and local projections

Internal-instrument VAR. The pooled instrument is stacked above the variables of the baseline, $w_t = (z_t^\star, y_t')'$ with y_t in (IP, CPI, EBP, Fed Funds), and the system

$$w_t = \tilde{c} + \sum_{\ell=1}^{6} \tilde{\Pi}_\ell \, w_{t-\ell} + \eta_t, \qquad \text{Var}(\eta_t) = \Sigma_\eta = PP', \tag{49}$$

is estimated with six lags and a constant, with P the lower-triangular Cholesky factor of Σ_η. With $z_t^\star$ ordered first, the first recursive shock $e_1' P^{-1} \eta_t$ is the part of the instrument's innovation not predicted by the lags of the system, and its impact on the other variables is unrestricted. With $\tilde{\Phi}_h$ the moving-average coefficients of equation (49) and s_p the coordinate of the funds rate in w_t, the response to a shock that raises the funds rate by one point on impact is the y-block of

$$\Theta_h^{\text{int}} = \frac{\tilde{\Phi}_h \, Pe_1}{s_p' Pe_1}, \qquad h = 0, \ldots, 36. \tag{50}$$

This is the normalisation in equation (4) of the external-instrument estimator. On impact Pe_1 is proportional to $\text{Cov}(\eta_t, \eta_{z,t})$, the external-instrument moment with $z_t^\star$ replaced by its innovation. The two estimators differ at later horizons because the lags of $z_t^\star$ enter the equations of y_t. They coincide when the instrument carries no information about future y_t beyond the lags of y_t, which holds when the shock is invertible and the instrument's noise is serially uncorrelated (Plagborg-Møller and Wolf, 2021).

Local projections. For each variable k of y_t and each horizon $h = 0, \ldots, 36$,

$$y_{k,t+h} = \alpha_{k,h} + \theta_{k,h} \, i_t + \sum_{\ell=1}^{6} \gamma_{k,h,\ell}' \, w_{t-\ell} + e_{k,t+h}, \tag{51}$$

with the funds rate i_t instrumented by $z_t^\star$ (Stock and Watson, 2018). The controls are those of equation (49). Hence, the two estimators condition on the same information. With $\tilde{x}_t$ a variable residualised on the controls, $\theta_{k,h} = \text{Cov}(\tilde{y}_{k,t+h}, \tilde{z}_t^\star) / \text{Cov}(\tilde{i}_t, \tilde{z}_t^\star)$ is the response to a one-point rise

in the funds rate, the normalisation of the baseline. In a finite sample, local projections avoid the VAR's extrapolation of the dynamics, at the cost of higher variance. (Plagborg-Møller and Wolf, 2021) The leads are restricted to December 2019. Hence, the sample shortens by one month per horizon. Confidence bands are 68% intervals from the moving-block bootstrap of the baseline, with the pooling weights held fixed.

Assumptions. Both estimators use the pooled instrument $z_t^\star$ of Section 5.6. They inherit its identifying assumptions, namely the family partition, Assumption 1 and the constraint that removes the contamination the families share. They add no restriction on the families. Each adds only the conditions of its own estimator. Both require $z_t^\star$ to be uncorrelated with past and future shocks once the controls are included (Stock and Watson, 2018). Ordering $z_t^\star$ first in the internal-instrument VAR drops the invertibility requirement of the external-instrument VAR (Plagborg-Møller and Wolf, 2021). The pooling weights are estimated once and held fixed in both estimators.

F Appendix tables

Table 3: Predictability of the raw instruments and the pooled instrument from pre-announcement information. Each is regressed on a constant and the six Bauer and Swanson (2023b) controls of Section 2.1; HC1 standard errors in parentheses; *, **, ***: 10, 5 and 1%. Romer–Romer: the instrument before orthogonalisation. MAR: the Miranda-Agrippino and Ricco (2021) surprise. Cholesky-based: the Cholesky-based instrument. Pooled $z^{\star}$: the baseline pooled instrument. The first three at meeting frequency, the last two monthly (months with an FOMC decision). Joint F: HC1 test that all six coefficients are zero.

	Monetary policy surprises	Romer–Romer	MAR	Cholesky	Pooled $z^{\star}$
NFP surprise	0.093**	0.113	0.101**	0.276**	0.502
	(0.039)	(0.101)	(0.044)	(0.128)	(3.362)
NFP 12M	0.005**	−0.002	−0.000	0.001	−0.050
	(0.002)	(0.009)	(0.002)	(0.012)	(0.219)
S&P 500 3M	0.079	−0.156	0.033	−0.056	0.461
	(0.059)	(0.184)	(0.056)	(0.209)	(5.892)
Slope 3M	−0.011	−0.097***	−0.006	−0.076***	−0.223
	(0.007)	(0.028)	(0.009)	(0.025)	(0.924)
BCOM 3M	0.122**	0.022	0.034	−0.041	0.314
	(0.052)	(0.147)	(0.050)	(0.179)	(5.365)
Treasury skew	0.032***	0.117***	0.011	0.078*	0.310
	(0.011)	(0.038)	(0.010)	(0.041)	(1.069)
R^2	0.165	0.105	0.066	0.060	0.001
Joint F (p)	5.22 (0.000)	3.59 (0.002)	1.11 (0.356)	2.64 (0.017)	0.03 (1.000)
N	316 meetings	256 meetings	219 meetings	267 months	267 months
Sample	1988m2–2019m12	1988m2–2019m12	1994m2–2019m12	1988m2–2019m12	1988m2–2019m12

Table 4: First-stage F (HC0) by instrument, orthogonalisation and policy rate. Each instrument is used alone, in the VAR in (IP, CPI, EBP, policy rate) of Section 6.1, on the 383 months within 1988m1–2019m12 on which all its versions are observed. The raw Romer–Romer and Cholesky-based instruments are re-estimated on the window of their orthogonalised versions. **Bold**: $F > 10$ (Stock and Yogo, 2005). PTA: purge-then-aggregate. ATP: aggregate-then-purge. FWL: orthogonalised jointly with the instrument's own first stage.

Instrument (statement window)	Fed funds	2-year	1-year	Shadow
Monetary policy surprises (MPS)				
Raw	**14.5**	9.4	**18.6**	8.0
Orthogonalised (ATP)	7.1	2.8	6.4	5.7
Orthogonalised (PTA)	3.6	0.6	2.7	2.4
Romer–Romer				
Raw	**37.3**	5.7	**12.9**	**25.1**
Orthogonalised (ATP)	**17.5**	0.9	1.8	**10.4**
Orthogonalised (PTA, FWL)	**29.6**	2.9	5.8	**21.8**
Cholesky-based				
Raw	**965.7**	**39.6**	**92.3**	**573.3**
Orthogonalised (ATP, FWL)	**533.5**	**22.7**	**41.6**	**311.1**

Table 5: The instrument panel. Twelve orthogonalised instruments in four families. Support is the span of non-missing months in the moment window (January 1988 to December 2019); T_j its length. PTA: purge-then-aggregate; ATP: aggregate-then-purge; FWL: orthogonalised jointly with the instrument's own first stage.

Family	Instrument	Support	T_j
High-frequency surprises	FF4, orthogonalised (PTA)	1994m02–2019m12	311
	ED4, orthogonalised (PTA)	1994m02–2019m12	311
	Bauer–Swanson MPS, orthogonalised (PTA)	1988m02–2019m12	383
Swanson factors	Swanson (2024) federal funds factor, orthogonalised (PTA)	1988m02–2019m12	383
	Swanson (2024) forward guidance factor, orthogonalised (PTA)	1988m02–2019m12	383
	Swanson (2024) asset purchase factor, tightening sign, orthogonalised (PTA)	1988m02–2019m12	383
Romer–Romer	Romer–Romer, orthogonalised (PTA)	1988m02–2019m12	383
	Romer–Romer, orthogonalised (PTA, FWL)	1988m02–2019m12	383
	Romer–Romer, orthogonalised (ATP)	1988m02–2019m12	383
Cholesky-based	Cholesky, federal funds rate, orthogonalised (ATP, FWL)	1988m02–2019m12	383
	Cholesky, 1-year Treasury, orthogonalised (ATP, FWL)	1988m02–2019m12	383
	Cholesky, 2-year Treasury, orthogonalised (ATP, FWL)	1988m02–2019m12	383

Table 6: Single instruments and pools on the baseline VAR and moment window. F: HC0 first stage against the funds-rate innovation. Impulse responses to a $+1$ percentage-point funds-rate shock; the peak decline of output is its largest absolute response, with its horizon. Strong instruments only: the instruments with a single-instrument $F > 10$ (footnote 11).

Identification	F	IP_0	IP peak decline (h)	CPI_0	CPI_{24}	EBP_0
Bauer–Swanson surprise, raw	14.5	−0.28	−1.93 (20)	−0.15	−0.11	+0.25
Bauer–Swanson surprise, orthogonalised	3.6	−0.52	−6.58 (13)	−0.83	−3.00	+1.97
Swanson funds-rate factor, PTA	10.6	−1.61	−5.88 (13)	−0.33	−1.56	+1.09
Romer–Romer, PTA	22.3	−0.13	−1.69 (23)	−0.11	+0.05	+0.19
MAR information-robust, PTA	6.3	−2.20	−8.35 (13)	−0.46	−2.53	+1.69
Cholesky funds rate, ATP-FWL	533.5	+0.01	−1.40 (30)	−0.03	+0.34	+0.05
HF and Swanson families only	5.1	−1.16	−6.56 (13)	−0.62	−2.45	+1.61
All 12 instruments pooled, no within transform	146.9	−0.19	−2.04 (19)	−0.14	−0.16	+0.32
Strong instruments only ($F > 10$), pooled estimator	49.8	+0.11	−2.07 (20)	+0.07	+0.18	+0.34
Strong instruments only ($F > 10$), no within transform	284.3	−0.08	−1.47 (28)	−0.07	+0.25	+0.07
Pooled estimate (baseline)	32.6	−0.48	−3.97 (15)	−0.22	−0.94	+0.88

Table 7: Each bar of Figure 5 split into the family's contamination and its first stage. F: HC0 first-stage F of the family's average instrument. $\hat{\rho}_c$: its first-stage correlation with the funds-rate innovation. Standardised contamination: $\Phi_h \hat{\kappa}_c$, free of the units of the instrument. Bar: $\Phi_h \hat{\kappa}_c / \hat{\rho}_c$, the values plotted in Figure 5. Both in percent (pp for EBP). Subscripts: horizon in months.

				Standardised contamination $\Phi_h \hat{\kappa}_c$			Bar $\Phi_h \hat{\kappa}_c / \hat{\rho}_c$		
Family	F	$\hat{\rho}_c$	$1/\hat{\rho}_c$	IP_{12}	CPI_{12}	EBP_0	IP_{12}	CPI_{12}	EBP_0
High-frequency surprises	4.8	0.11	8.8	−0.50	−0.21	+0.16	−4.38	−1.86	+1.39
Swanson factors	1.7	0.09	11.2	−0.03	+0.02	−0.02	−0.37	+0.18	−0.23
Romer–Romer	24.4	0.26	3.8	+0.65	+0.18	−0.17	+2.52	+0.69	−0.66
Cholesky-based	202.2	0.58	1.7	+1.89	+0.52	−0.53	+3.26	+0.89	−0.92

Table 8: Variations of the panel. The baseline, the baseline with the information-robust family (IR) added as a fifth family, the five-family panel with each family dropped in turn, and the repartitioned families (Section 7.1). m instruments, C families. p, J_T: test of exact block structure. Eigengap: $\hat{\lambda}_1/\hat{\lambda}_2$ of the weighted within scatter. Impulse responses to a $+1$ percentage-point funds-rate shock.

Panel	m	C	F	p, J_T	eigengap	IP_0	IP peak decline (h)	CPI_{24}	EBP_0
Baseline: four families of three	12	4	32.6	0.848	3.7	−0.48	−3.97 (15)	−0.94	+0.88
Adding a family									
Adding the information-robust family (IR)	15	5	26.8	0.725	3.6	−0.57	−4.29 (15)	−1.04	+0.95
Leaving one family out of the five-family panel									
Without the high-frequency family	12	4	22.4	0.565	4.3	−0.52	−4.19 (15)	−1.04	+0.94
Without the Swanson family	12	4	29.2	0.467	4.0	−0.04	−2.43 (18)	+0.12	+0.40
Without the Romer–Romer family	12	4	25.2	0.557	3.7	−0.58	−4.30 (15)	−1.04	+0.95
Without the Cholesky family	12	4	7.5	0.858	2.9	−1.40	−7.45 (13)	−2.71	+1.80
Repartitioned families									
Each Swanson factor in the family of its horizon	18	5	30.7	0.641	3.9	−0.61	−3.90 (15)	−0.79	+0.78
Path factor with the Treasury curve	18	5	30.5	0.646	3.9	−0.60	−3.90 (15)	−0.80	+0.79
Without the Swanson factors	15	5	61.4	0.338	4.3	−0.20	−2.56 (18)	−0.02	+0.41

Table 9: Further five-variable systems. Each row is a separate VAR in the baseline four variables plus the variable listed, with the four baseline families and the IR family as a fifth, and the baseline weights and windows. F: first stage of the pooled instrument. J_T: test of exact block structure. x_0 and x peak: impulse response of the added variable on impact and at its largest absolute value, with its horizon. IP_0, IP peak decline and CPI_{24}: impulse responses of output and prices in the same system. Impulse responses to a $+1$ percentage-point funds-rate shock.

Added variable	F	J_T (df)	p	x_0	x peak (h)	IP_0	IP peak decline	CPI_{24}
In the figure: demand and the labour market								
Real consumption (%)	21.9	38.0 (36)	0.377	−0.21	−1.18 (16)	−0.71	−4.71	−0.81
Retail sales (%)	15.9	39.8 (36)	0.303	−3.16	−4.71 (12)	−1.04	−5.83	−1.48
Real personal income (%)	26.1	27.4 (36)	0.848	−0.12	−1.38 (21)	−0.23	−3.59	−0.77
Payroll employment (%)	25.8	27.6 (36)	0.840	+0.03	−1.56 (22)	−0.64	−4.37	−0.96
In the figure: credit and financial conditions								
Business loans (%)	29.8	35.5 (36)	0.493	−0.00	−8.38 (29)	−0.23	−3.14	−0.80
Consumer credit, non-revolving (%)	25.1	27.1 (36)	0.858	−0.31	−2.89 (25)	−0.44	−4.37	−1.14
Mortgage spread (pp)	14.7	35.7 (36)	0.484	+0.76	+0.95 (1)	−1.39	−6.40	−1.69
VIX (index points)	16.9	26.2 (36)	0.886	+12.56	+13.40 (1)	−0.75	−5.03	−1.79
Other candidates, not plotted								
Core CPI (%)	21.7	24.8 (36)	0.920	−0.02	−0.86 (36)	−0.38	−3.90	−1.28
PCE price index (%)	19.7	29.9 (36)	0.755	−0.27	−1.28 (36)	−0.86	−4.53	−1.48
Building permits (%)	22.4	37.2 (36)	0.414	−13.38	−20.47 (5)	−0.41	−4.67	−1.19
Capacity utilisation, manufacturing (pp)	23.4	32.8 (36)	0.623	−0.08	−2.91 (12)	−0.74	−5.12	−0.90
Crude oil price (%)	22.5	35.6 (36)	0.486	+6.45	−7.83 (36)	−0.58	−4.09	−1.14
Trade-weighted dollar (%)	20.1	42.2 (36)	0.220	+0.24	+1.50 (5)	−0.37	−3.77	−1.26
Yen per dollar (%)	30.2	33.2 (36)	0.601	−0.49	−2.41 (36)	−0.44	−3.83	−0.76
Canadian dollars per dollar (%)	17.6	40.2 (36)	0.288	+2.43	+4.87 (5)	−0.48	−4.41	−1.50
10-year Treasury yield (pp)	9.9	55.4 (36)	0.020	−2.60	−3.55 (1)	−1.02	−3.97	+1.25
2-year Treasury yield (pp)	7.4	58.9 (36)	0.009	−3.33	−4.74 (1)	−0.51	−2.92	+1.19
BAA–10-year spread (pp)	7.9	36.0 (36)	0.470	+0.99	+1.56 (2)	−1.74	−7.90	−2.60
Real M2 (%)	17.0	48.3 (36)	0.082	+0.71	+2.37 (22)	−0.72	−4.13	−0.84

Table 10: Diversifying the families. Same VAR, windows, precision weights and estimator as the baseline; families other than those listed as in the baseline. IR: information-robust family of three distinct series; NR (narrative family): Romer–Romer and two Aruoba–Drechsel shocks; ladder: two-, five- and ten-year Treasury surprises added to the high-frequency family; minutes: Swanson factors on minutes releases as a sixth family. m instruments, C families. Swanson share: the family's share of the precision-weighted within-family variation along the estimated direction. Angles between normalised impact vectors, in the Σ_u^{-1} metric: to the baseline in the left block, to the same partition with all families in the right block. J_T p: test of exact block structure.

					angle		IP peak decline		without the Swanson family			
Partition	m	C	F	Swanson share	(°)	J_T p	(h)	CPI_{24}	angle (°)	F	IP peak decline (h)	CPI_{24}
Baseline	12	4	32.6	0.71	0.0	0.848	−3.97 (15)	−0.94	34.0	38.5	−2.17 (19)	+0.18
IR + NR	15	5	33.2	0.62	4.0	0.540	−3.79 (15)	−0.81	27.6	72.2	−2.22 (19)	+0.11
Maturity ladder in the high-frequency family	15	4	23.4	0.56	10.6	0.849	−4.87 (14)	−1.25	16.3	43.0	−3.59 (15)	−0.39
Ladder + IR + NR	18	5	22.9	0.51	10.8	0.644	−4.61 (14)	−1.10	15.0	42.9	−3.42 (16)	−0.34
Ladder + IR + NR + Swanson factors on minutes	21	6	13.1	0.37	15.6	0.768	−6.41 (14)	−1.66	7.1	8.2	−6.50 (14)	−1.27

Table 11: Magnification by variable and horizon. Biases are measured against the pooled baseline estimate. $\hat{\varrho}_{k,h}$: orthogonalised bias over raw bias, equation (37); a negative value means that orthogonalisation reverses the sign of the bias. Output and prices in percent, the credit spread (excess bond premium) in percentage points, for a +1 percentage-point funds-rate shock. Instruments, orthogonalisations and months as in Table 1.

	Output		Prices		Credit spread
	impact	12 months	impact	12 months	impact
High-frequency surprise ($\delta_\pi = 0.54$, bound 2.15)					
Raw bias	+0.20	+2.16	+0.07	+0.54	−0.63
Orthogonalised bias	−0.04	−2.66	−0.61	−1.71	+1.08
$\hat{\varrho}_{k,h}$	−0.21	−1.23	−8.93	−3.16	−1.71
Romer–Romer instrument ($\delta_\pi = 0.15$, bound 1.18)					
Raw bias	+0.53	+3.09	+0.21	+0.95	−0.85
Orthogonalised bias	+0.63	+2.55	+0.18	+0.75	−0.62
$\hat{\varrho}_{k,h}$	+1.19	+0.82	+0.83	+0.79	+0.73
Cholesky-based instrument ($\delta_\pi = 0.36$, bound 1.55)					
Raw bias	+0.49	+3.15	+0.20	+0.95	−0.88
Orthogonalised bias	+0.49	+3.01	+0.19	+0.89	−0.83
$\hat{\varrho}_{k,h}$	+0.99	+0.96	+0.93	+0.94	+0.95

G Appendix figures

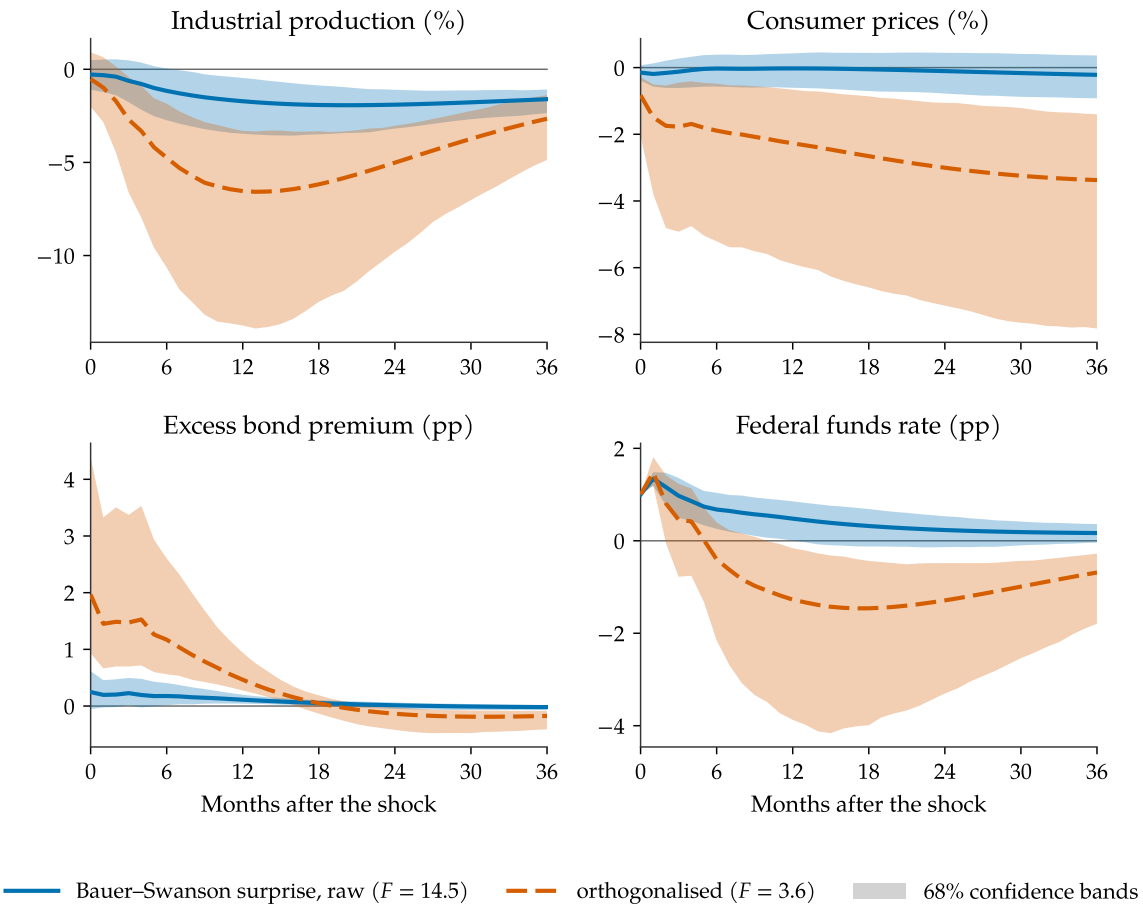


Figure 8: **Orthogonalising the surprise weakens it and enlarges its impulse responses.** Single-instrument impulse responses to a +1 percentage-point funds-rate shock in the monthly VAR of Section 6.1, identified with the Bauer and Swanson (2023b) surprise, raw (solid, $F = 14.5$) and orthogonalised announcement by announcement and then aggregated (PTA; dashed, $F = 3.6$), each with its 68% moving-block-bootstrap confidence band.

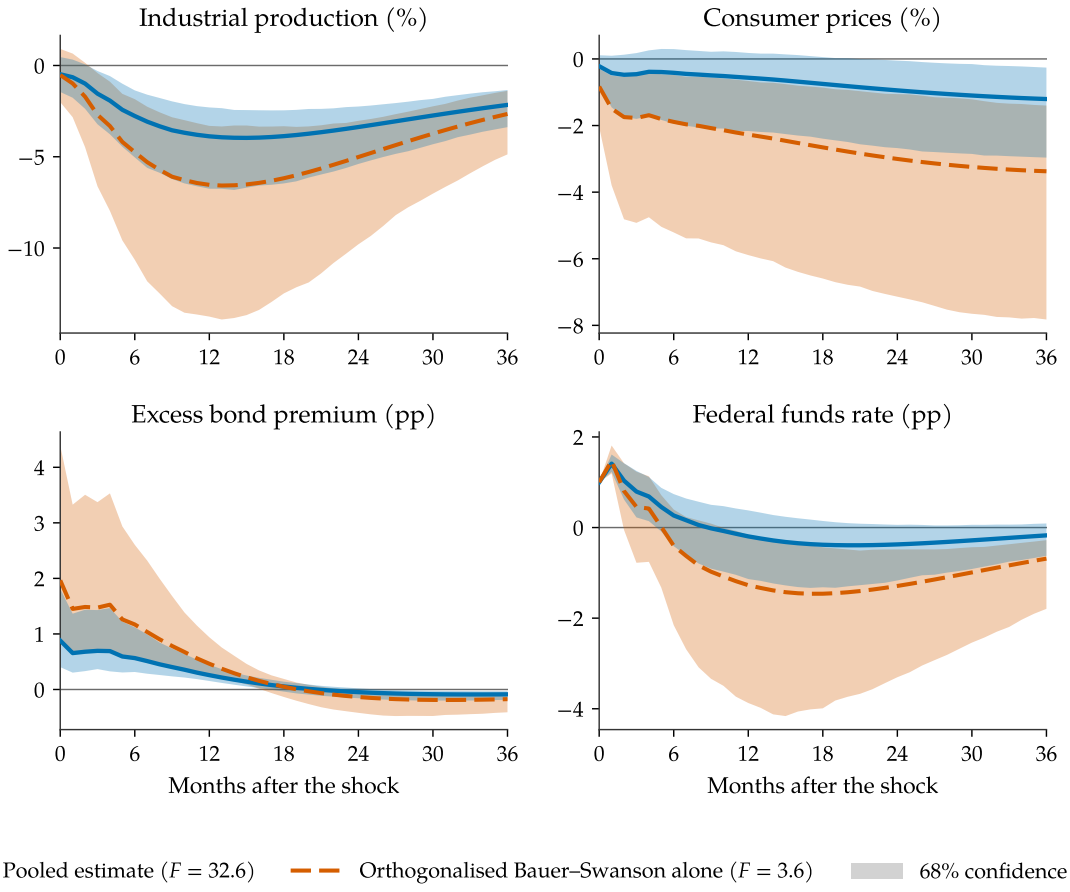


Figure 9: **The pooled monetary policy shock.** Impulse responses to a +1 percentage-point shock to the federal funds rate. Solid: the baseline pooled estimate from twelve orthogonalised instruments in four families ($F = 32.6$). Dashed: the orthogonalised Bauer and Swanson (2023b) surprise alone ($F = 3.6$). Shaded: 68% median-recentred percentile confidence bands from the moving-block bootstrap, 1,000 draws, with the VAR, the moments and the weights re-estimated in each draw.

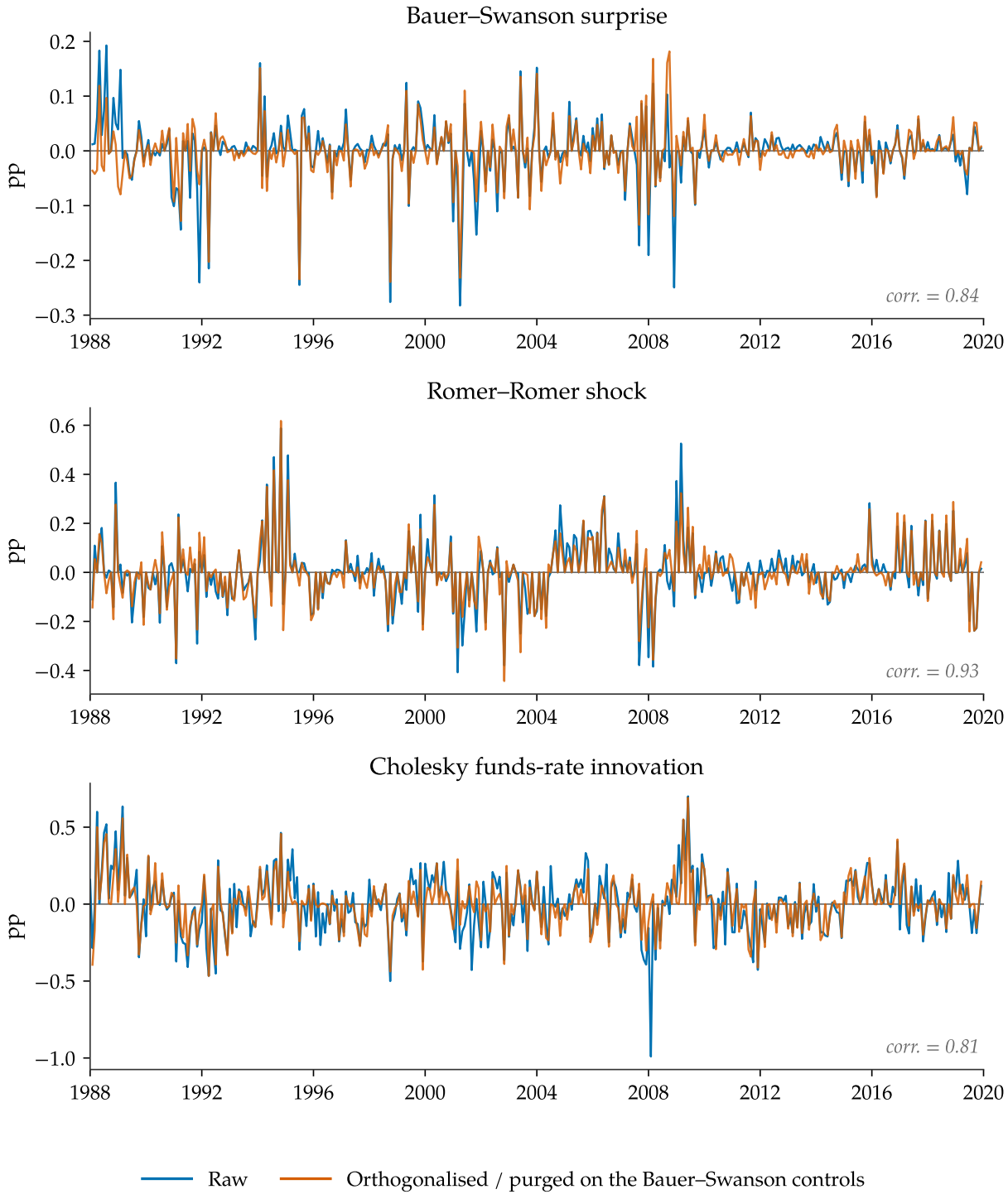


Figure 10: **The raw and orthogonalised instruments, January 1988 to December 2019.** Monthly series in percentage points. Top: the Bauer and Swanson (2023b) surprise (PTA). Middle: the Romer–Romer instrument (PTA-FWL). Bottom: the Cholesky-based instrument (ATP-FWL), zero in months without an FOMC announcement. The raw wide-window series are re-estimated on the window of the orthogonalised ones. Each panel reports the correlation of the two series.

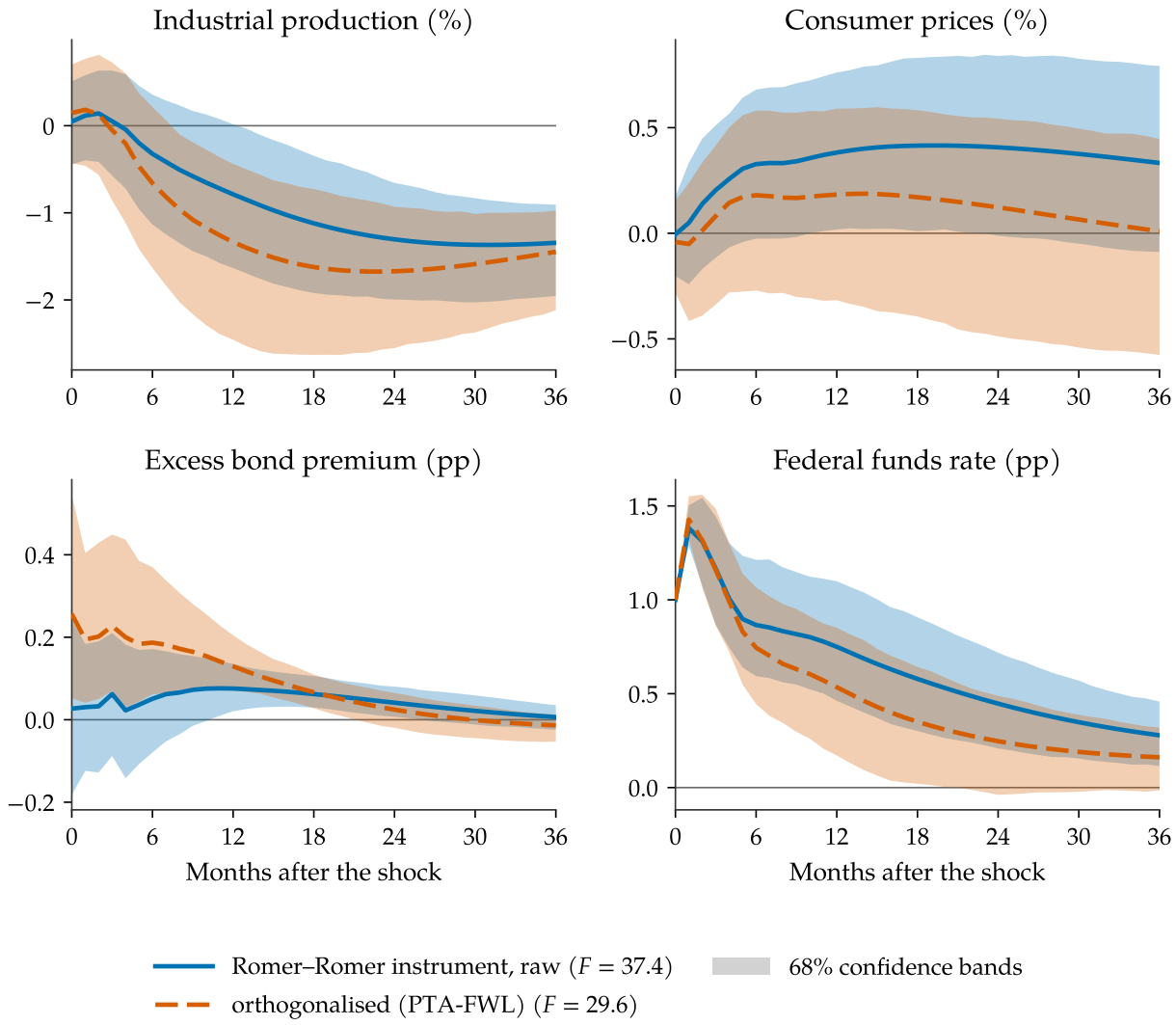


Figure 11: **Romer–Romer instrument: raw versus orthogonalised.** Single-instrument impulse responses to a +1 percentage-point funds-rate shock identified with the Romer and Romer (2004) instrument, raw (solid, $F = 37.4$) and orthogonalised on the six Bauer–Swanson controls at the announcement level, jointly with its first stage (PTA-FWL; dashed, $F = 29.6$). The raw instrument is the Romer and Romer (2004) residual, estimated on the same 256 meetings, February 1988 to December 2019, as the orthogonalised one, such that the two differ only by the controls, as in the raw row of Table 4. Baseline VAR; moments from January 1988 to December 2019. Shaded: 68% median-recentred percentile confidence bands from the moving-block bootstrap, 1,000 draws. The orthogonalisation costs about a fifth of the first-stage strength and enlarges the output and credit-spread impulse responses slightly.

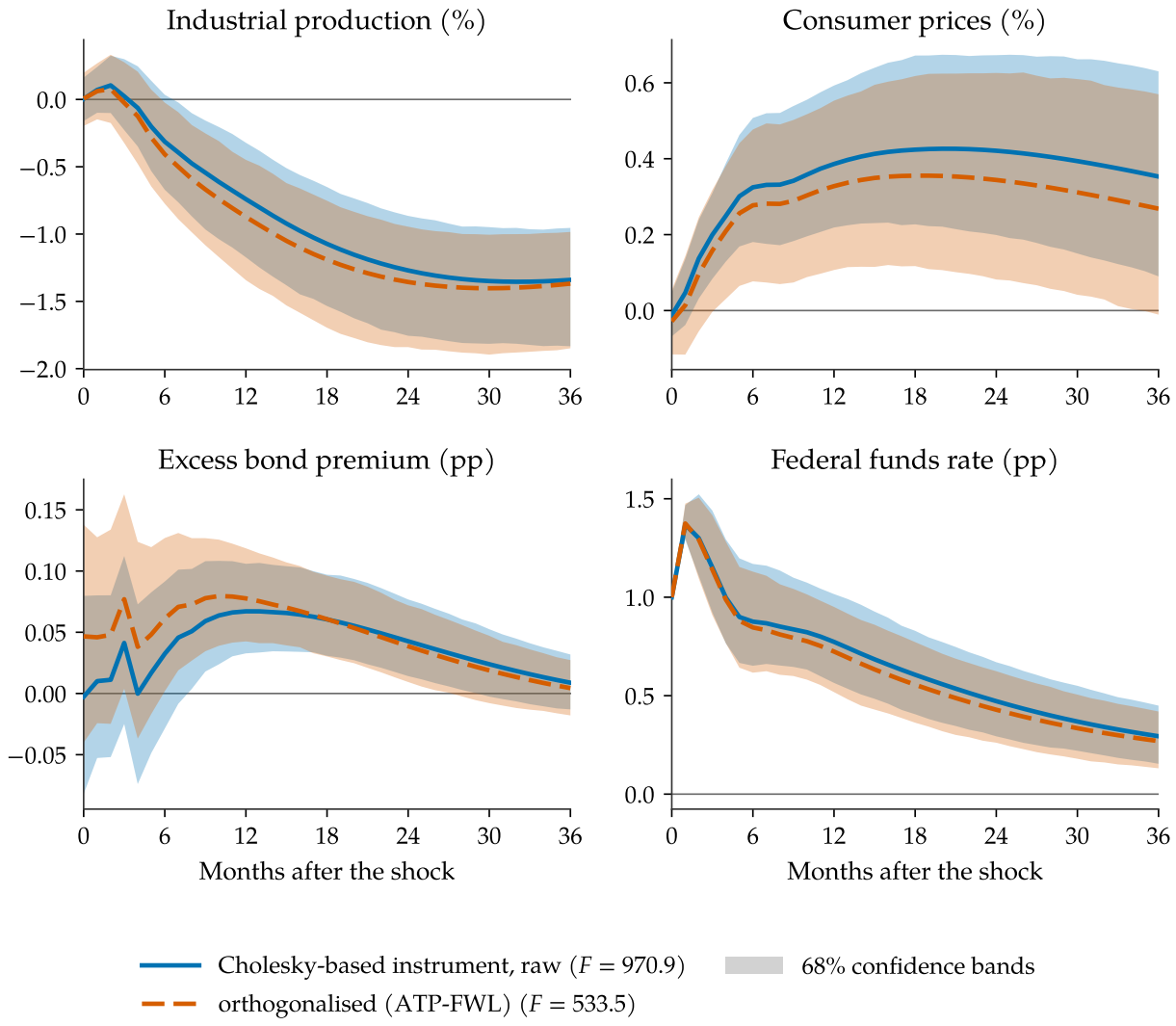


Figure 12: **Cholesky-based instrument: raw versus orthogonalised.** Single-instrument impulse responses to a +1 percentage-point funds-rate shock identified with the recursive funds-rate innovation of the instrument database, raw (solid, $F = 971$) and orthogonalised on the six Bauer–Swanson controls after monthly aggregation, jointly with its first stage (ATP-FWL; dashed, $F = 534$). The raw innovation is the database VAR's funds-rate innovation orthogonalised on the output and price innovations from January 1988 to December 2019, the window of the orthogonalised one, such that the two differ only by the controls, as in the raw row of Table 4. Baseline VAR; moments from January 1988 to December 2019. Shaded: 68% median-recentred percentile confidence bands from the moving-block bootstrap, 1,000 draws. The orthogonalisation removes close to half of the first-stage F and leaves the impulse responses essentially unchanged.

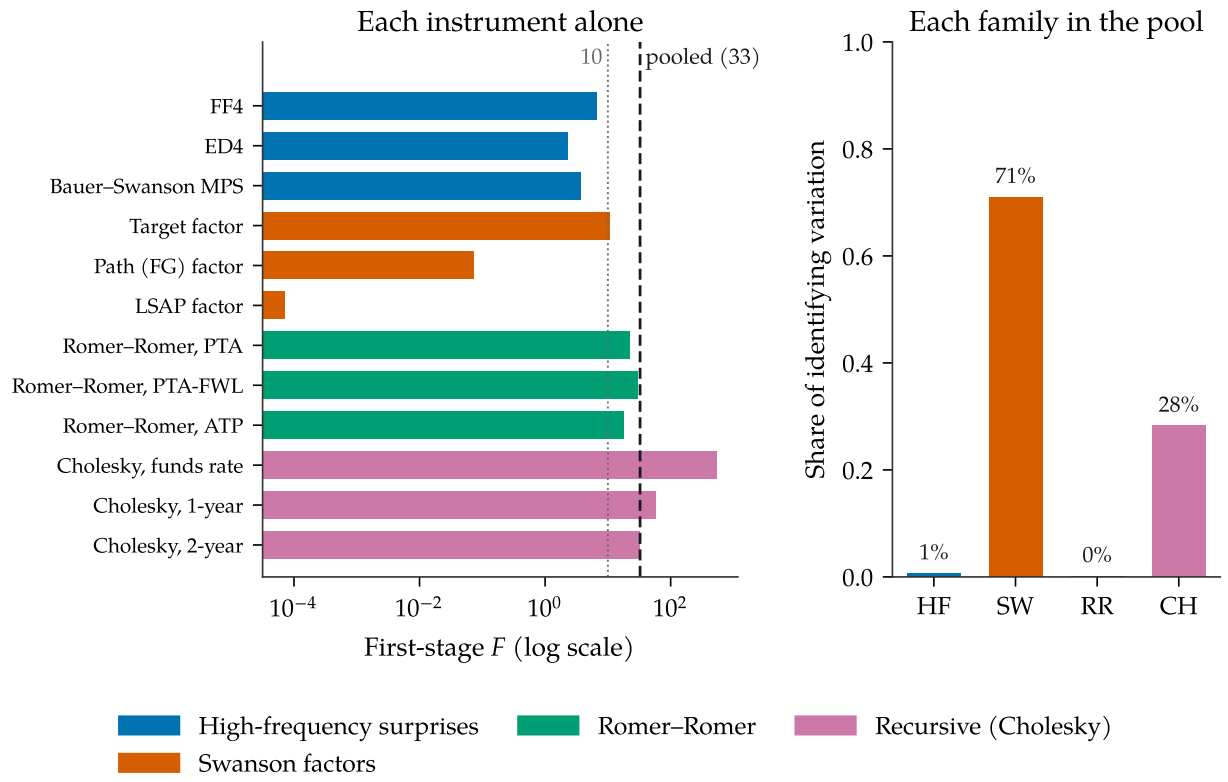


Figure 13: **What identifies the shock.** Left: first-stage F of each instrument entered alone against the funds-rate innovation on the moment window (log scale); the dotted line marks $F = 10$, the dashed line the pooled instrument. Right: each family's share of the identifying variation $\sum_{j \in c} \hat{\phi}_j^2 / \hat{v}_j$ in the precision-weighted within scatter, with the share in percent above each bar.

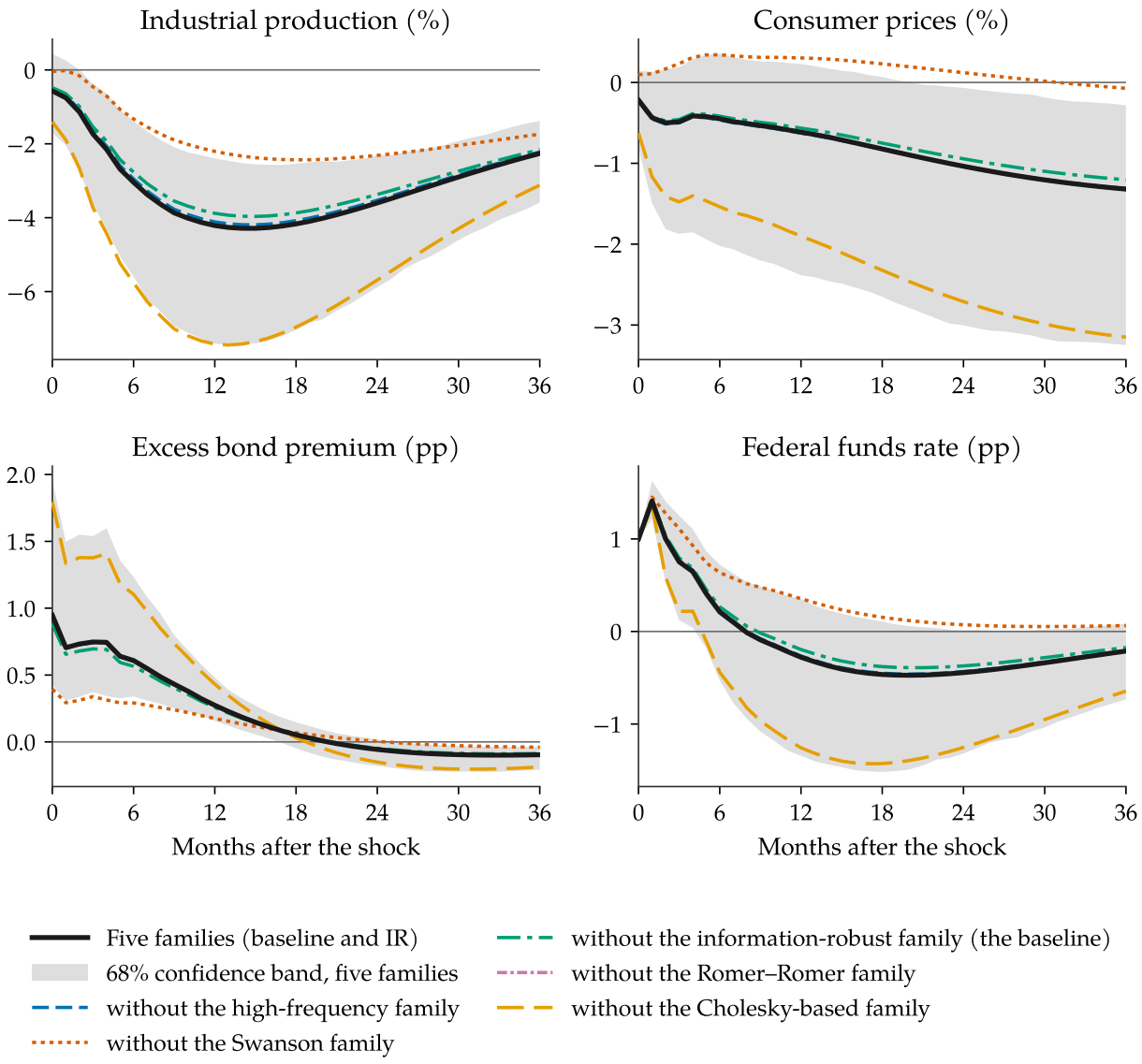


Figure 14: **Leaving one family out.** The five-family panel, the baseline families and the information-robust family (solid black, with its 68% confidence band), and the estimate with each family dropped in turn. With four families left, partition condition (P2) of Section 5.5 holds in every case. Dropping the information-robust family returns the baseline. Same VAR, windows and precision weights.

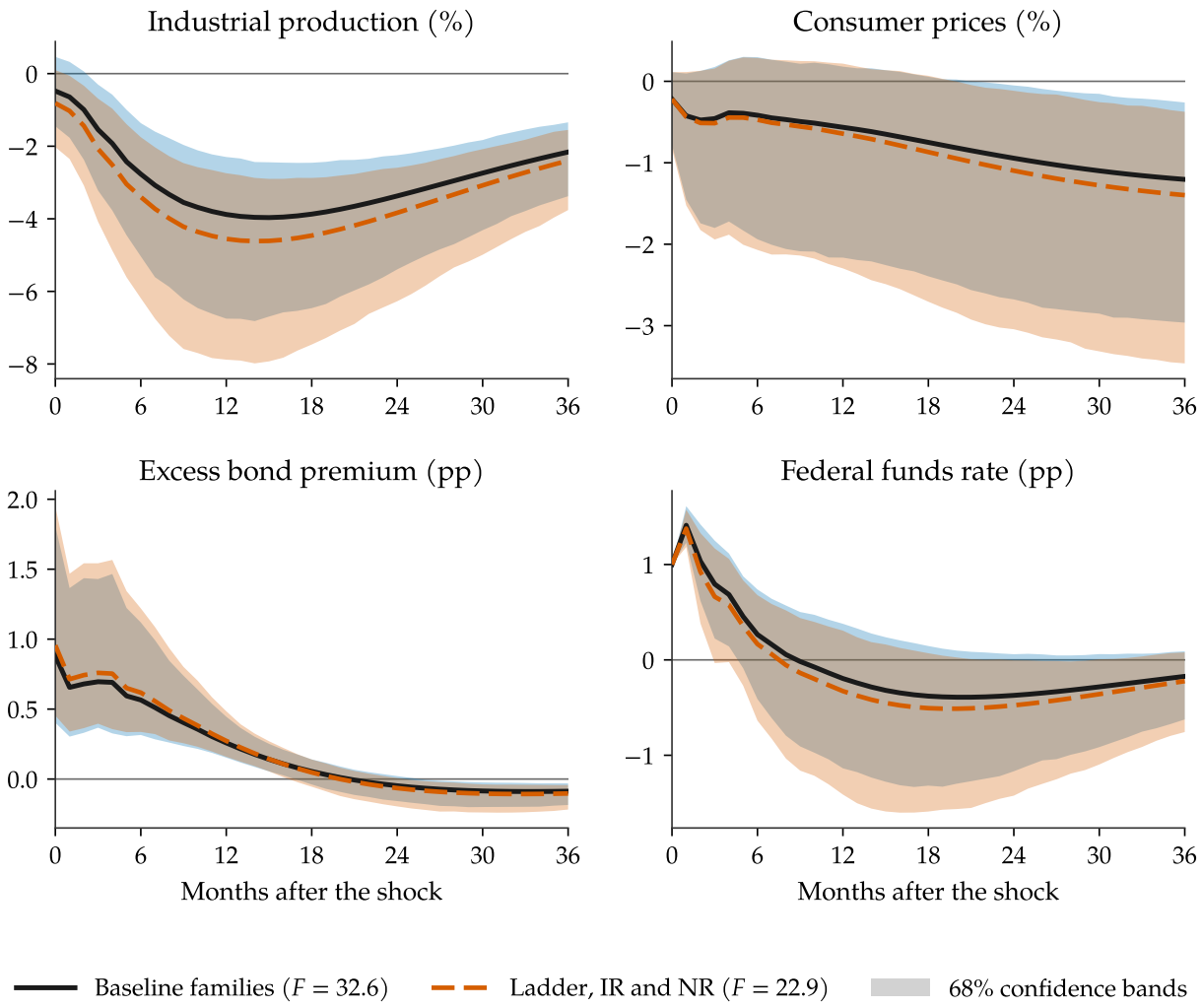


Figure 15: **Diversified families.** Baseline (solid, $F = 32.6$) against the partition with the maturity ladder in the high-frequency family, the information-robust family (IR) and the Romer–Romer family diversified with two Aruoba and Drechsel (2024b) shocks (NR) (dashed, $F = 22.9$). Other families, VAR, windows, weights and bootstrap as in the baseline. Each with its 68% confidence band.

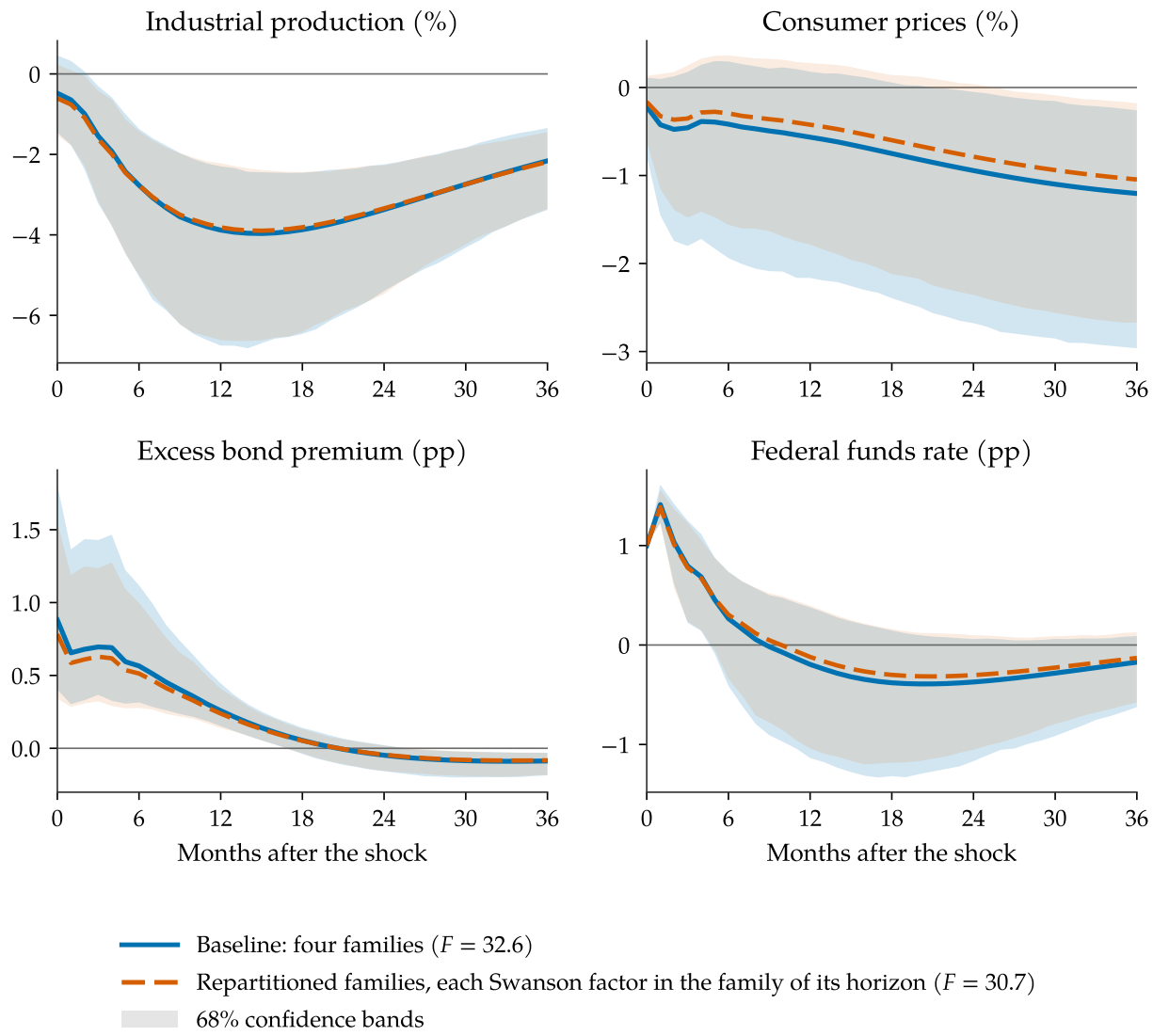


Figure 16: **Repartitioned families.** Impulse responses to a +1 percentage-point funds-rate shock. Solid: the baseline, four families. Dashed: five families formed by window and horizon (Section 7.1), the target and path factors with the near-dated futures and the asset-purchase factor with the Treasury curve. Same VAR, moment window, precision weights and bootstrap; 68% confidence bands.

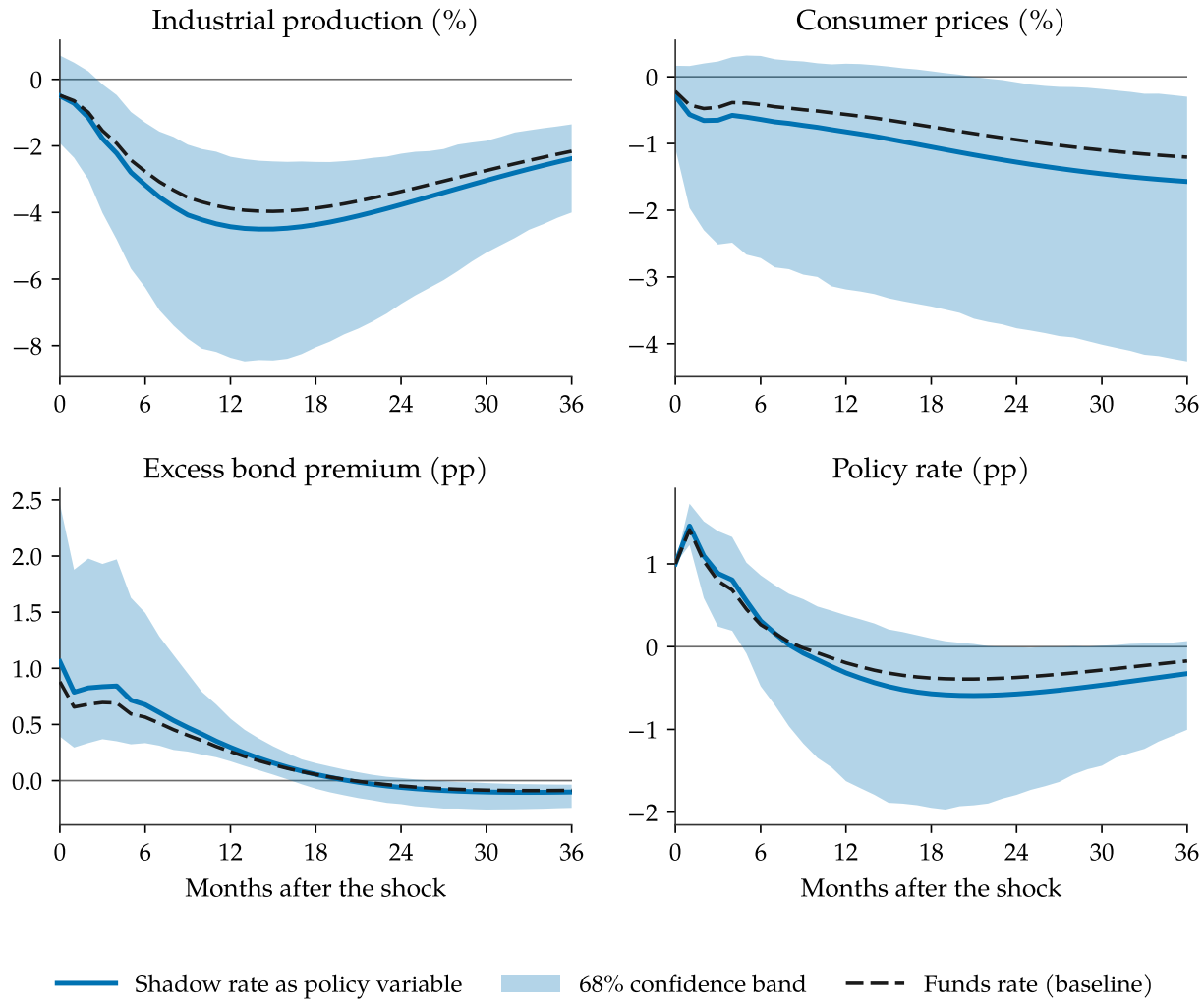


Figure 17: **The shadow rate as policy indicator.** Impulse responses to a +1 percentage-point shock to the shadow rate (the funds rate away from the zero lower bound, the Wu and Xia (2016) shadow rate at it), with 68% confidence bands; the funds-rate baseline is dashed. The bottom-right panel is the response of each specification's own policy rate.

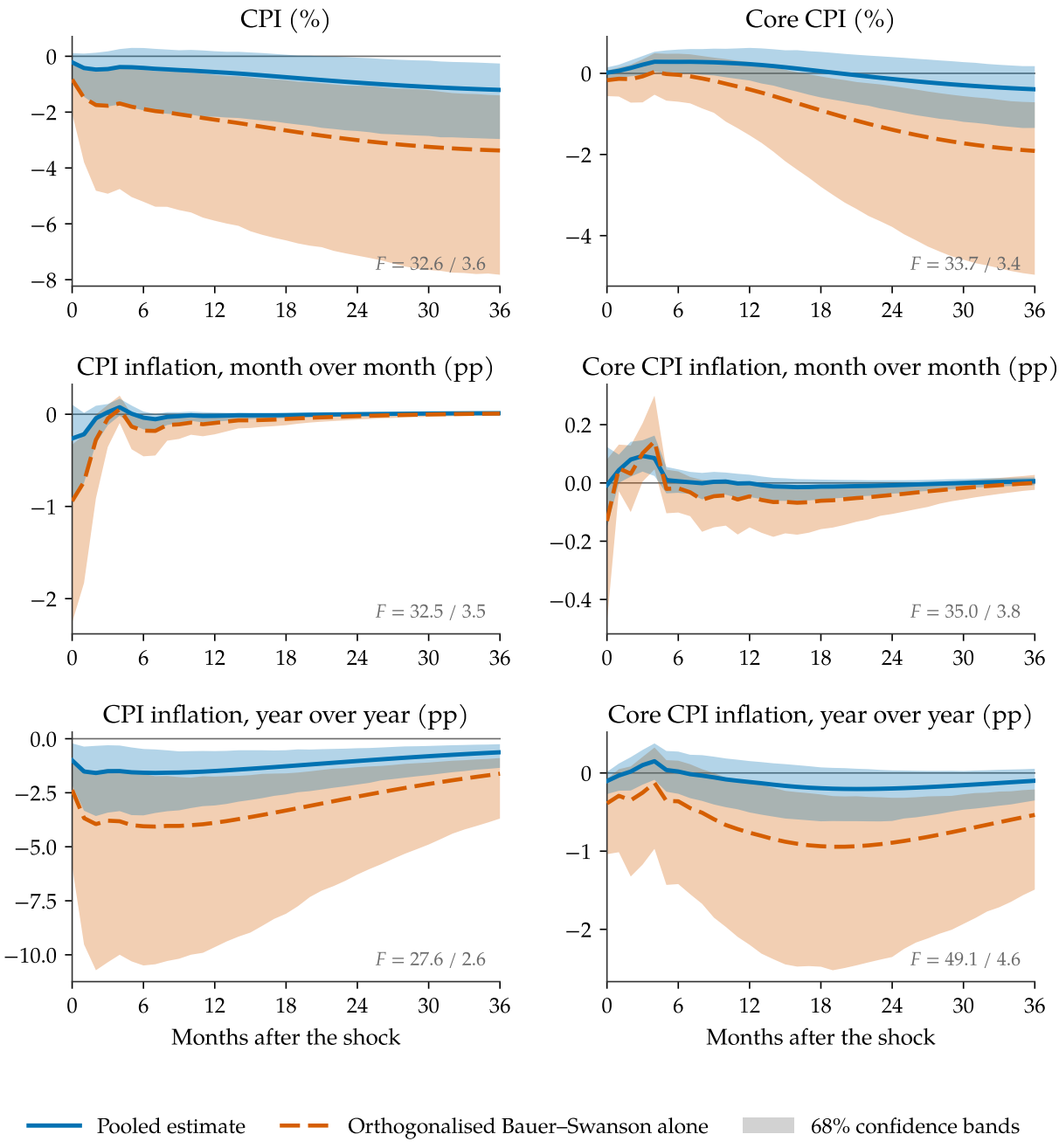


Figure 18: **The price measure: levels, month-over-month and year-over-year inflation.** Impulse responses to a +1 percentage-point funds-rate shock of the CPI and core CPI: top, 100×log level; middle, month-over-month inflation, $100\Delta \log P_t$; bottom, year-over-year inflation, $100(\log P_t - \log P_{t-12})$, in percentage points. Each panel is from a separate four-variable VAR in which the price variable is the one shown. Pooled estimate (solid) and the orthogonalised Bauer and Swanson (2023b) surprise alone (dashed), each with its 68% confidence band. Same families, precision weights, windows and bootstrap as the baseline. F: first stage of the pooled instrument / of the orthogonalised surprise.

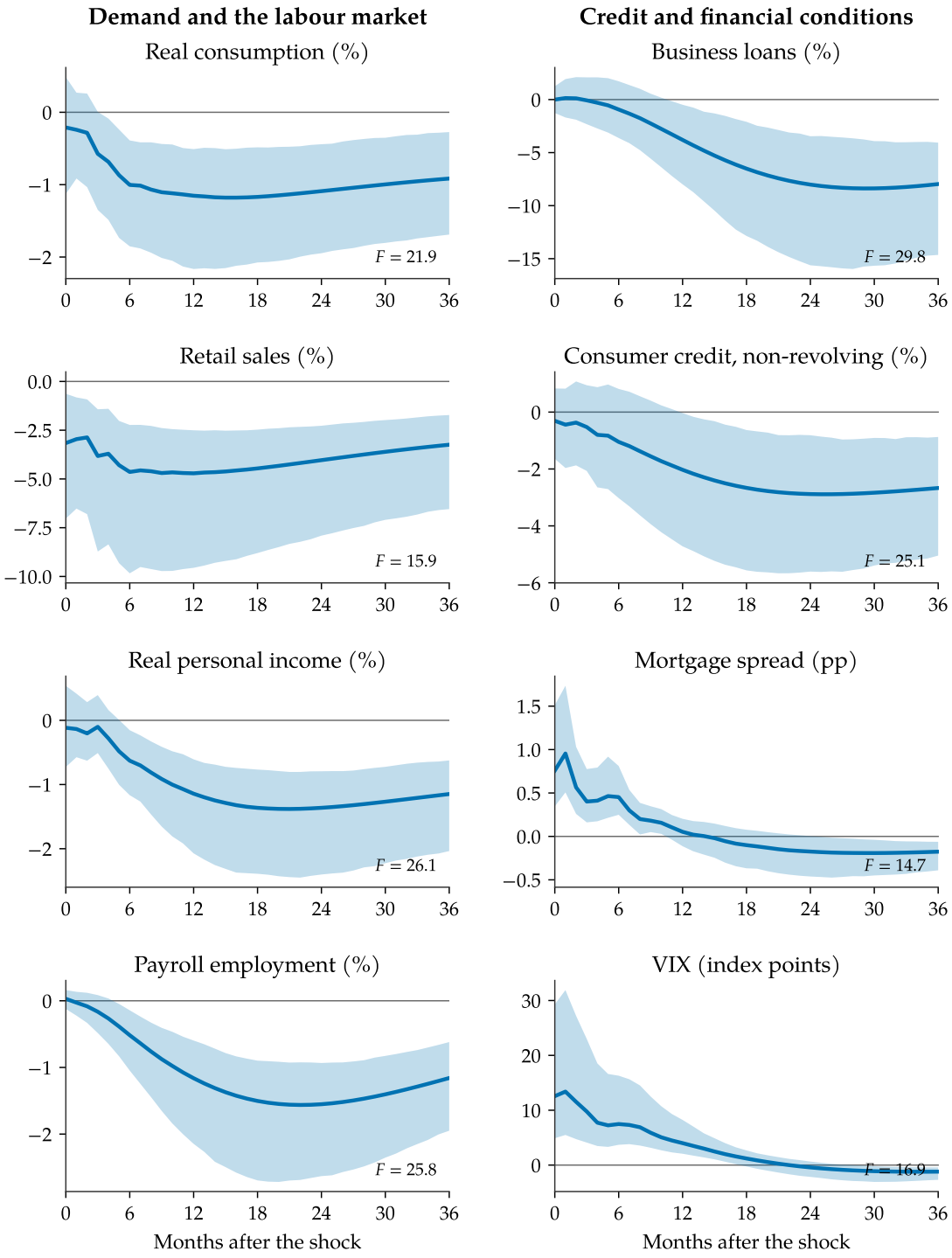


Figure 19: **Further variables, one at a time.** As in Figure 6: each panel is a separate five-variable VAR, and plots the impulse response of the added variable with its 68% confidence band. Left column: demand and the labour market. Right column: credit and financial conditions.

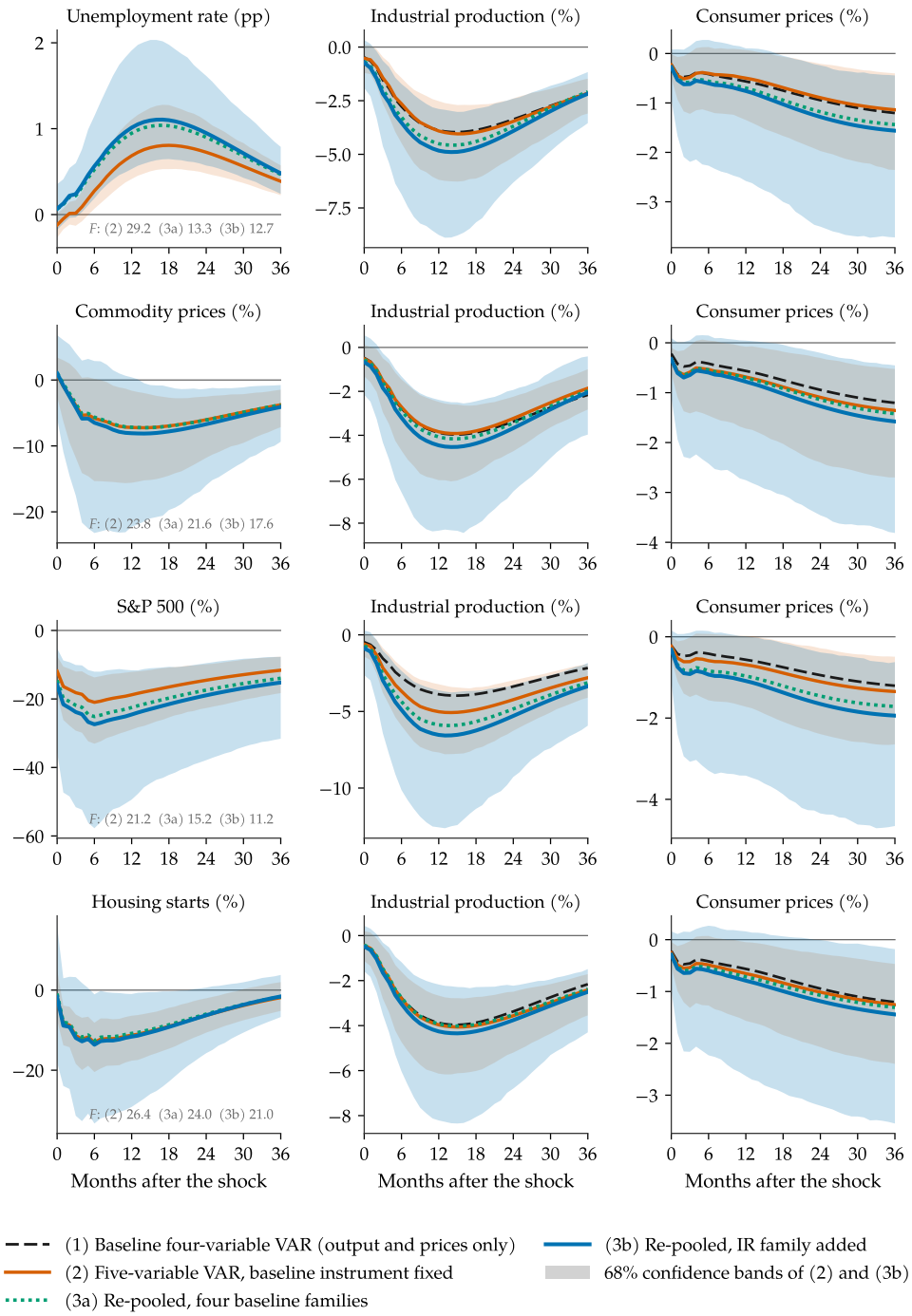


Figure 20: **Decomposing a change of VAR.** One row per five-variable system: the added variable, output and prices. Impulse responses to a +1 percentage-point funds-rate shock. (1) Baseline VAR and pooled instrument. (2) Five-variable VAR with the baseline pooled instrument held fixed. Its gap to (1) is the change of VAR. (3a) Re-pooled with the four baseline families. (3b) Re-pooled with the IR family added, as in Figure 6. Its gap to (2) is the change of instrument. First panel of each row: first-stage F of each step. 68% confidence bands for (2) and (3b), moving-block bootstrap with the same resampled months for every step, and the weights fixed in (2). Details are in Appendix E.2.

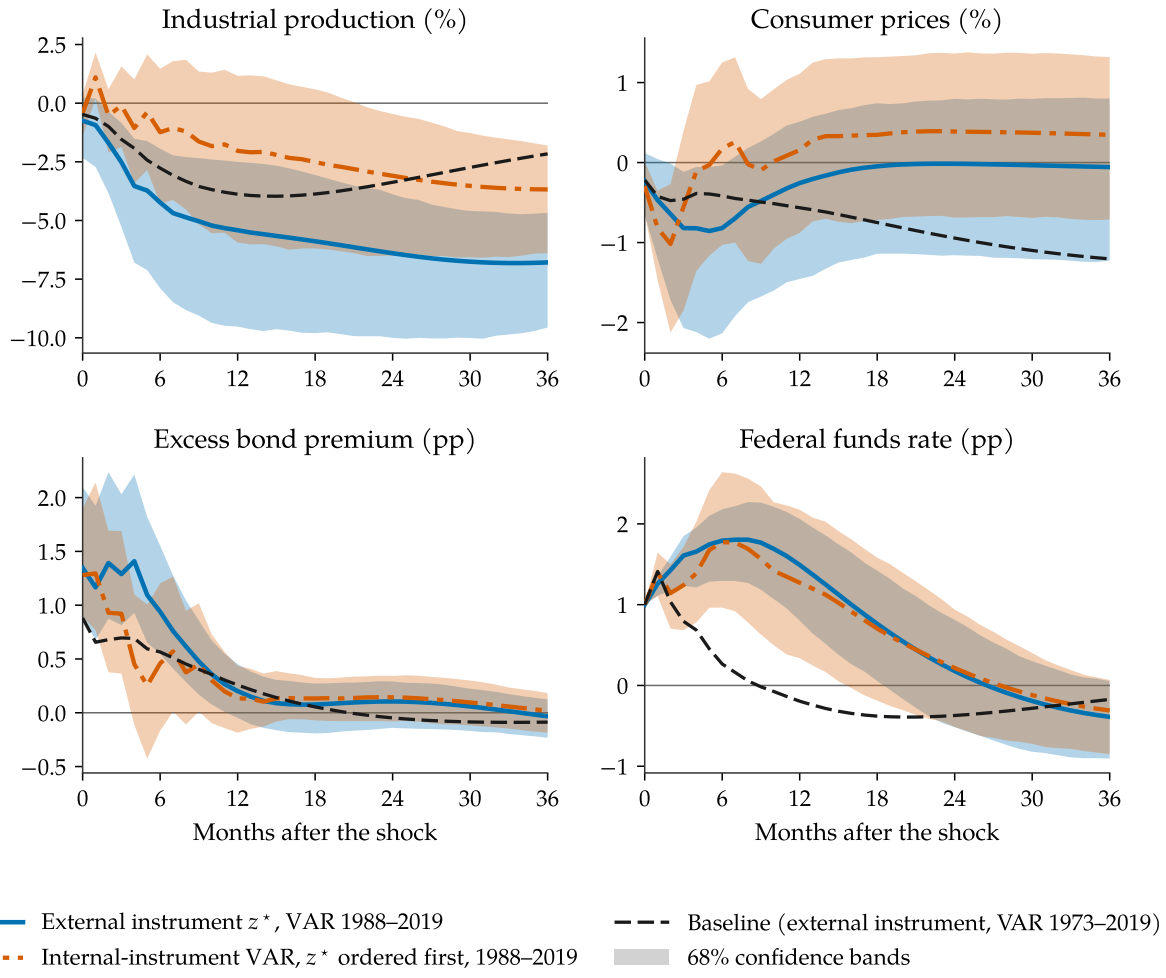


Figure 21: **The pooled instrument in an internal-instrument VAR.** Impulse responses to a +1 percentage-point funds-rate shock. Solid, with 68% confidence band: $z_t^\star$ as an external instrument on a VAR in (IP, CPI, EBP, Fed Funds) estimated from January 1988 to December 2019. Dash-dotted, with 68% confidence band: the internal-instrument VAR of Appendix E.3, with $z_t^\star$ ordered first, on the same sample. Dashed: the baseline (external instrument, VAR from January 1973 to December 2019; Appendix Figure 9). Six lags and a constant throughout. Confidence bands: median-recentred percentile intervals from the moving-block bootstrap, 1,000 draws, with the pooling weights held at their estimates.

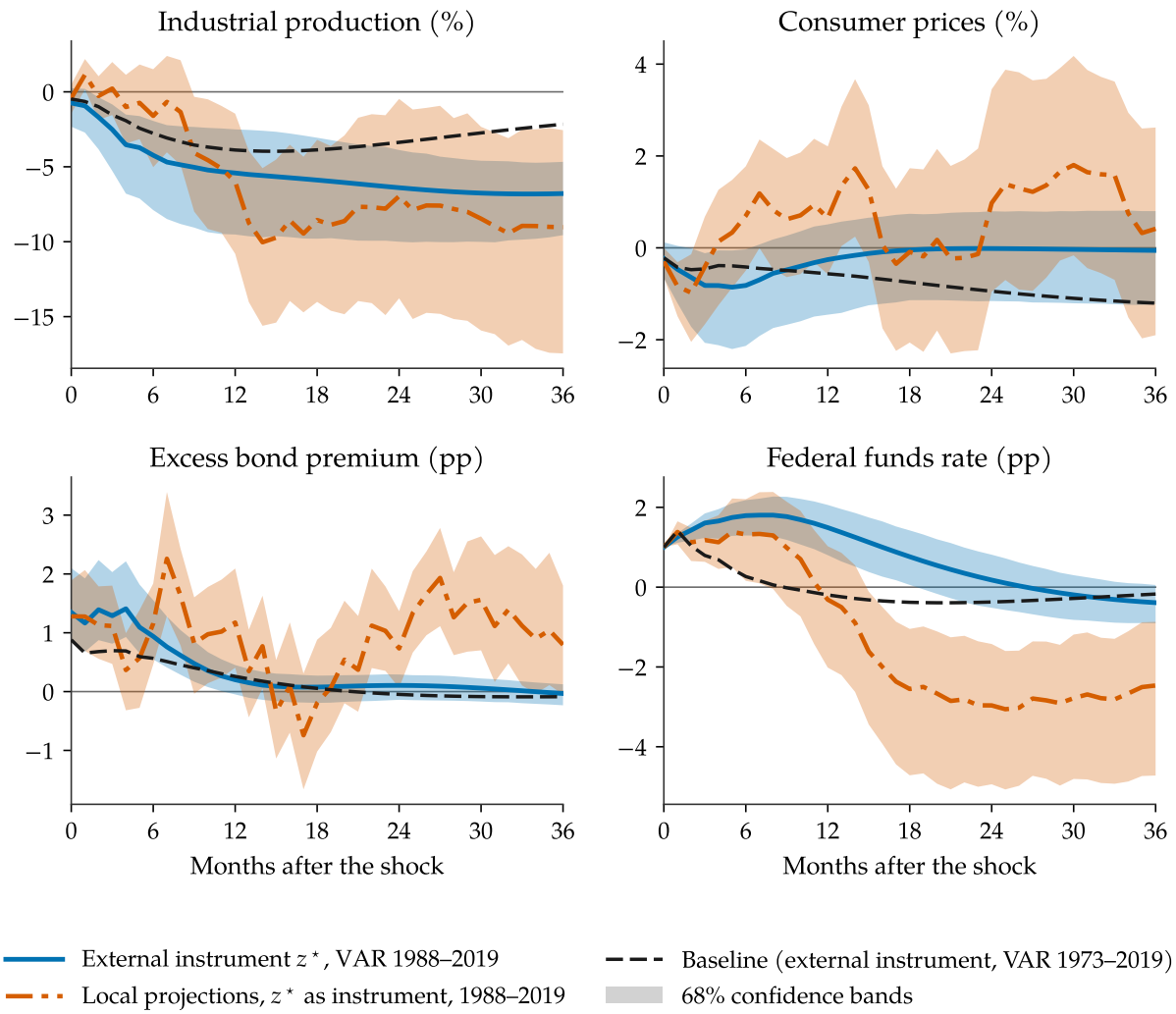


Figure 22: **The pooled instrument in local projections.** Impulse responses to a +1 percentage-point funds-rate shock. Solid, with 68% confidence band: $z_t^\star$ as an external instrument on a VAR estimated from January 1988 to December 2019, as in Figure 21. Dash-dotted, with 68% confidence band: the local projections of Appendix E.3, with the funds rate instrumented by $z_t^\star$, leads up to December 2019. Dashed: the baseline. Confidence bands: median-recentred percentile intervals from a moving-block bootstrap of the time index, 1,000 draws, with the pooling weights held at their estimates.

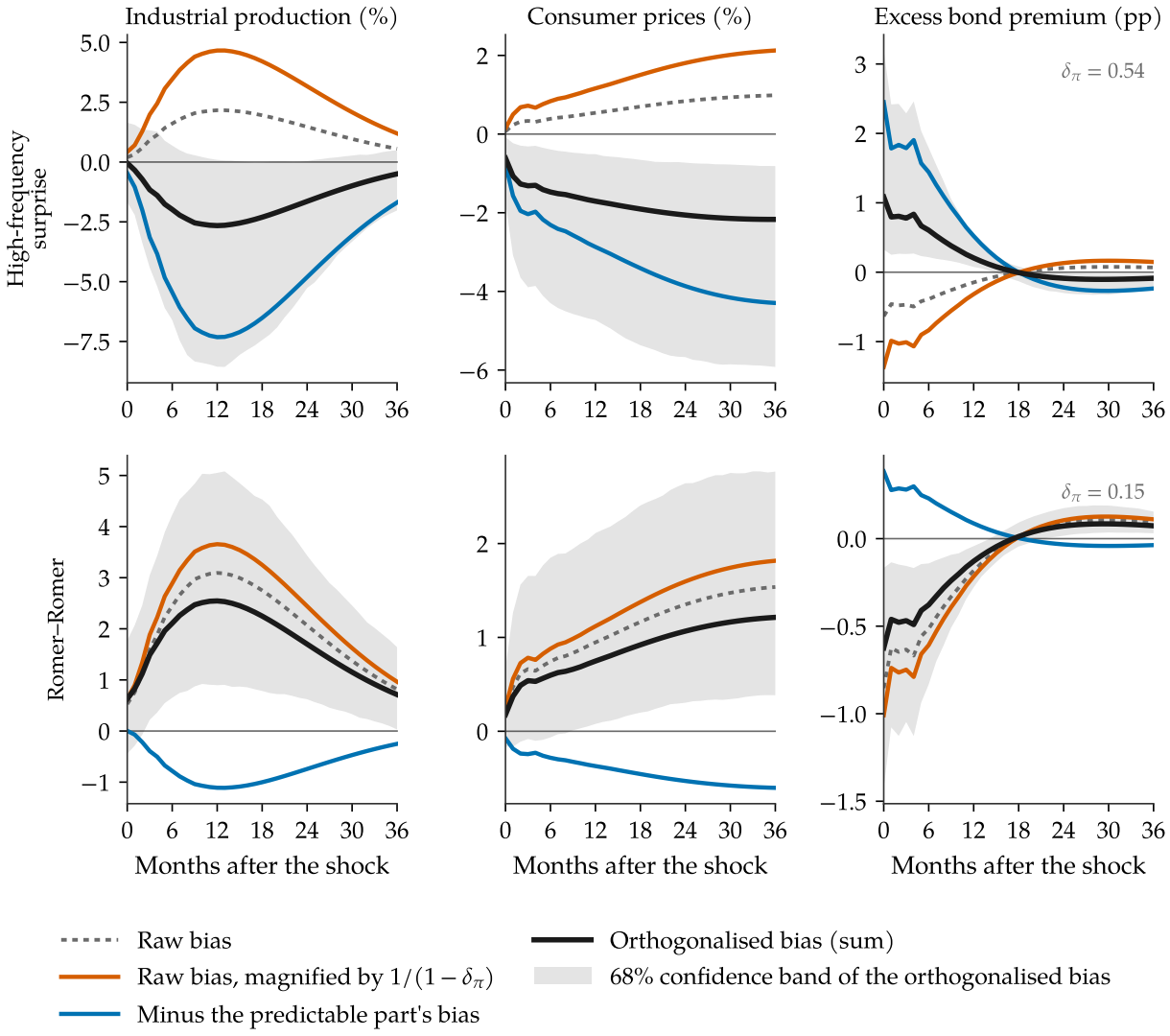


Figure 23: **Two effects of orthogonalisation.** The orthogonalised bias (black) is the raw bias magnified by $1/(1-\delta_\pi)$ (orange) minus the bias of the predictable part weighted by $\delta_\pi/(1-\delta_\pi)$ (blue), equation (36). Grey dotted: the raw bias. Biases are measured against the pooled baseline estimate, for a +1 percentage-point funds-rate shock. Top row: high-frequency surprise; bottom row: Romer–Romer instrument. Shaded: 68% median-recentred moving-block bootstrap confidence band of the orthogonalised bias, with the VAR, the moments and the pooled direction re-estimated in each of 1,000 draws.